TU WIEN Informatics

Increasing User Satisfaction, Engagement and Retention in Life-Tracking Apps for Students

A Case Study of Methods on the App Habits Network

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Wien, 30. November 2025

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A Case Study of Methods on the App Habits Network

DIPLOMA THESIS

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by

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to the Faculty of Informatics

at the TU Wien

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Vienna, November 30, 2025

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Kurzfassung

Verhaltensänderungsmodelle wie das Hook Model [EH14] und das Fogg Behavior Model [Fog09] bieten theoretische Grundlagen für die Gestaltung persuasiver Systeme, doch es gibt kaum empirische Untersuchungen darüber, wie diese Modelle in der Praxis bei Langzeit-Self-Tracking-Anwendungen, die mehr als einen Lebensbereich abdecken, funktionieren. Diese Arbeit untersucht persuasive Designstrategien und Visualisierungen in *Habits Network*, einer Self-Tracking-App, die für diese Arbeit entwickelt wurde und die das Tracken von Aktivitäten und sozialen Interaktionen kombiniert. Mithilfe einer Mixed-Methods-Studie wird untersucht, wie sich theoretische Konzepte in einer Self-Tracking-Anwendung umsetzen lassen und welche Elemente das Engagement und die Bindung der Nutzenden im Laufe der Zeit fördern. Dafür wurden semistrukturierte Interviews mit zehn Nutzenden (mit einer Nutzungsdauer von fünf bis 17 Monaten), vier Interviews mit Experten von österreichischen Universitäten aus den Bereichen Visualisierung und Verhaltensänderung sowie eine Verhaltensanalyse anhand von Nutzungsdaten und eine System Usability Scale (SUS) Befragung durchgeführt und ausgewertet. Die SUS ergab einen Mittelwert von 77,00 (SD: 9,47), was auf eine gute Benutzungsfreundlichkeit hindeutet, obwohl die Komplexität sich als häufiges Problem herausgestellt hat. Die Interviews mit Nutzenden und die Nutzungsdaten zeigen, dass einfache Visualisierungen wie Heatmaps und Liniendiagramme am häufigsten genutzt wurden, während komplexeren Visualisierungen, die genauere Einblicke in das Leben der Nutzenden bieten könnten, weniger Beachtung geschenkt wurde. Die Interviews mit Nutzenden zeigen, dass die App erfolgreich interne Trigger für die Nutzung etablieren konnte. Die Ergebnisse zeigen, dass das Schaffen von Bewusstsein und ein Gefühl des Überblicks über das eigene Leben der Haupteffekt der App auf die Nutzenden ist. Diese Arbeit liefert empirische Belege dafür, welche persuasiven Designelemente und Visualisierungen die Zufriedenheit, das Engagement und die Bindung im Kontext des langzeitigen Self-Trackings verbessern.

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Abstract

While behavior change models like the Hook Model and Fogg Behavior Model provide theoretical foundations for persuasive system design, little empirical evidence exists on how these models perform when applied to long-term, multi-faceted self-tracking applications in practice. This thesis investigates the effectiveness of persuasive design strategies and visualizations in *Habits Network*, a self-tracking app developed for this thesis that combines activity and social interaction tracking. Using a mixed-methods approach, the study examines how theoretical concepts can be translated into a self-tracking application and what factors drive engagement and retention over time. The evaluation employs the System Usability Scale (SUS), semi-structured interviews with ten users (ranging from five to 17 months of usage), four expert interviews with professionals in visualization and behavior change from Austrian universities, and behavioral analysis using app analytics data. The SUS yields a mean score of 77.00 (SD: 9.47), indicating good usability, though complexity emerges as a primary concern across all evaluation methods. Behavioral data reveal that simple visualizations, such as heatmaps and line charts, generate sustained engagement, while complex visualizations that could provide deeper insights into users' lives receive less attention. User interviews reveal that the app successfully established internal triggers. The findings reveal that self-awareness and a sense of control over one's life are the primary effects of the app on users. This work contributes empirical evidence on which persuasive design elements and visualizations effectively enhance satisfaction, engagement, and retention in long-term self-tracking contexts.

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CHAPTER 1

Introduction

Self-tracking has entered the mainstream in recent years through wearables and smartphone applications that monitor physical activity, sleep patterns, heart rate, and other physiological metrics. Millions of users worldwide have adopted these technologies with the goal of improving their health, productivity, and overall well-being. While automated tracking through sensors has achieved widespread adoption, manually tracking aspects of life that cannot be captured automatically presents a fundamentally different challenge. Activities such as social interactions, emotional states, creative pursuits, or personal achievements require conscious effort to document, placing the burden of sustained engagement entirely on the user.

Despite initial enthusiasm, most users abandon their tracking practices within three to six months [LDF10]. This high dropout rate highlights a critical challenge for self-tracking applications: maintaining user engagement over time. The problem intensifies when tracking extends beyond single metrics to encompass multiple facets of life, where users must manually document diverse activities and relationships.

Persuasive technology offers a systematic approach to addressing engagement challenges by applying behavior change principles to system design. Persuasive systems aim to influence user attitudes and behaviors without coercion or deception [Fog02; OH09]. By encoding motivation mechanisms, reducing barriers to use, and providing meaningful feedback, persuasive design strategies can potentially help users maintain tracking practices long enough to achieve their goals and experience the benefits of increased self-awareness.

1.1 Problem Statement

While behavior change models like the Hook Model [EH14] and the Fogg Behavior Model [Fog02; Fog09] provide theoretical foundations for persuasive system design, little empirical evidence exists on how these models perform when applied to long-

term, multifaceted self-tracking applications in practice. Existing research has identified barriers throughout the personal informatics lifecycle [LDF10]. Still, it has not adequately examined which specific persuasive design strategies effectively enhance user satisfaction, engagement, and retention over extended periods of real-world use. Long-term tracking differs fundamentally from short-term tracking, requiring different intervention concepts that current research has not adequately addressed [Mey+20]. Short-term visualization approaches do not scale well to unstructured, heterogeneous, and large long-term data. While there is an opportunity to provide visualizations that show relationships between different facets of life, open questions remain about what such visualizations should look like and how they should be designed so that non-experts can gain insights and make sense of their data [Mey+20; LDF10].

1.2 Aim, Context and Scope of this Thesis

This thesis investigates the effectiveness of persuasive design in *Habits Network*, a multifaceted life-tracking app that combines activity and social interaction tracking. The application demonstrates notable user retention, with ten users maintaining engagement over periods ranging from five to seventeen months, creating entries at rates between 0.8 and 3.8 per day since registration.

The research aims to identify which persuasive design elements and visualizations drive sustained app usage and provide meaningful insights to users. Following the theoretical review of persuasive technology principles, these concepts were implemented as concrete features in *Habits Network*. The evaluation examines whether these implementations function as intended through four complementary approaches: the System Usability Scale to establish comparable usability metrics; user perspectives on what motivates their continued use; expert assessment of the theoretical implementation quality; and behavioral data analysis revealing actual feature usage patterns.

A specific focus is on the effectiveness of visualization for life insights. Two advanced visualizations were developed to reveal relationships in users' lives: a bubble chart showing activity co-occurrence patterns and an ego network graph displaying social interaction structures. The research investigates which visualizations users engage with to gain desired insights, what types of insights users actually seek from their tracking data, and how visualization complexity affects comprehension and sustained usage.

1.3 Research Questions

This thesis aims to answer the following research questions:

- **RQ1:** How does considering the Hook Model and Fogg Behavior Model enhance user satisfaction, engagement, and retention in a self-tracking app for students?

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- **RQ2:** Which visualization methods best convey Insights from tracked life domains, measured through user comprehension and engagement?

- **RQ3:** To what extent can well-being metrics, such as productivity and motivation, be improved through persuasive design strategies in a self-tracking app?

1.4 Methods

This research employed a mixed-methods approach to evaluate both the effectiveness of persuasive design strategies and the comprehensibility and engagement of different visualization methods in *Habits Network*. The study combined quantitative usability metrics, qualitative insights from interviews, and behavioral data analysis to triangulate findings and provide comprehensive answers to the research questions.

Ten users who had been actively using the application for periods ranging from five to seventeen months participated in the study. Before conducting semi-structured interviews, all participants completed the System Usability Scale questionnaire to establish standardized usability metrics comparable to other applications. The interviews, conducted online via video conferencing, lasted between 40 and 80 minutes and explored their motivations, usage patterns, experiences with different features, and comprehension of various visualizations, including heatmaps, line charts, bubble charts, and ego network graphs. Interviews were transcribed for thematic analysis.

To complement user perspectives, four expert interviews were conducted with professionals specializing in visualization, UI/UX design, and behavior change. These in-person interviews, lasting 65-95 minutes, provided a critical external assessment of both the theoretical implementation of persuasive design elements and the effectiveness of visualization choices for conveying life insights to non-expert users.

Behavioral data were collected through app analytics tracking user interactions with features and visualizations over their entire usage periods. This quantitative data revealed actual feature usage patterns and visualization engagement, providing an objective counterpoint to self-reported experiences from interviews and enabling analysis of which persuasive design elements and visualization methods drove sustained engagement over time.

1.5 Thesis Structure

This thesis is structured to progress from theoretical foundations through empirical investigation to interpretation and synthesis.

Chapter 1 introduces the research context, formulates the problem statement, defines the research questions, and outlines the methodology.

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Chapter 2 presents the theoretical background, examining persuasive technology principles including the Hook Model, the Fogg Behavior Model, gamification, behavior change theory, and persuasive system design principles.

Chapter 3 describes the *Habits Network* application, detailing its design philosophy, use case, technical implementation, and features. Each feature is explicitly connected to the theoretical principles from Chapter 2, demonstrating how behavior change models were translated into concrete design decisions.

Chapter 4 outlines the evaluation methods employed in this research: behavioral analysis through app analytics, expert interviews with professionals in visualization and behavior change, the System Usability Scale questionnaire, and semi-structured user interviews.

Chapter 5 presents the findings from each evaluation method without interpretation. The chapter begins with quantitative usage data revealing which features were used and how users interacted with the app over time. This is followed by qualitative findings from expert interviews identifying central themes and recommendations. The System Usability Scale results are then presented, providing standardized usability metrics. Finally, findings from user interviews are presented, capturing user perspectives and experiences.

Chapter 6 discusses these findings in relation to the theoretical frameworks and research questions. The discussion interprets how the empirical evidence informs understanding of persuasive design effectiveness in long-term self-tracking contexts, synthesizing insights across all evaluation methods.

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CHAPTER 2

Related Work

2.1 Persuasive Technology

Persuasion has traditionally been defined as human communication intended to shape or influence the independent judgments and behaviors of others [OH09]. In the context of technology, some authors refer to persuasion as attempts to influence people's behavior, not attitudes [Fog09], whereas others also include attitude. According to the theory of cognitive consistency, people strive to align their behavior, attitudes, and beliefs so that changes in behavior will also lead to changes in attitudes and beliefs.

Persuasive systems may be defined as "*computerized software or information systems designed to reinforce, change, or shape attitudes or behaviors or both without using coercion or deception*" [OH08, p. 202].

There are three potential successful outcomes of a persuasive system:

- **Reinforcement** – making the user continue with the behavior and more resilient to change,
- **Shaping** – creating new behaviors, and
- **Changing** – changing existing behaviors.

All of those must be done voluntarily and without the use of coercion or deception [OH09].

2.2 The Hook Model

The Hook Model, developed by Nir Eyal (2014) [EH14], provides a theoretical framework for understanding how digital products create user habits through repeated cycles of

interaction. Drawing on behavioral psychology and persuasive design principles, the model outlines a four-phase process comprising *trigger*, *action*, *variable reward*, and *investment*. Through continuous repetition of these stages, users gradually internalize behavioral patterns, transitioning from externally prompted actions to self-initiated and context-dependent habits.

Rather than relying solely on external incentives, habit-forming systems aim to resonate with users' internal drives and emotional states, encouraging repeated engagement that satisfies psychological needs. In this sense, the model extends earlier theories of persuasive technology, particularly Fogg's Behavior Model (FBM) [Fog02], which posits that behavior arises from the simultaneous convergence of motivation, ability, and a trigger.

The model offers a structured lens for analyzing and designing experiences that transform fleeting user actions into persistent routines. The following sections examine each phase of the Hook Model in detail, beginning with the concept of the *trigger*, which initiates the behavioral loop and catalyzes the formation of habitual use [EH14].

2.2.1 Trigger

The *trigger* phase is the crucial starting point of the Hook Model's behavioral loop, as it initiates user engagement with the product. Triggers act as cues or prompts that signal the user to take a specific action, effectively bridging external stimuli or internal emotions with behavioral responses. These triggers can be external, such as notifications, emails, or advertisements, providing clear and timely calls to action that direct attention toward the product. Alternatively, they can be internal, arising from the user's own thoughts, feelings, or existing habits, such as boredom, curiosity, or loneliness, that unconsciously motivate the use of the product. The effectiveness of a trigger depends on its ability to resonate with the user's current context and emotional state. Well-designed triggers guide users effortlessly into the behavior loop by appearing at psychologically opportune moments, thereby increasing the likelihood of action. Over time, as the Hook cycle repeats, internal triggers gradually replace external cues, promoting habitual self-sustained engagement without the need for constant external prompting. This transition reflects how the Hook Model leverages both environmental stimuli and intrinsic motivators to cultivate durable user habits that align with psychological needs for connection, achievement, and relief [EH14].

External Triggers prompt users to take specific actions by presenting cues in their environment. According to Eyal (2014) [EH14], these can be categorized into four types: *paid triggers*, *earned triggers*, *relationship triggers*, and *owned triggers*. These triggers serve as initial stimuli that draw attention and encourage behavioral engagement.

- **Paid Triggers** are promotional messages that require a financial investment to reach users. Examples include banner ads, sponsored posts, and pay-per-click campaigns. For instance, a mobile game might use a paid Facebook ad to invite users to try it, prompting new users to download it.

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- **Earned Triggers** arise when users or others voluntarily share information about the product, typically through word-of-mouth or organic social sharing. An example is a user posting a positive review or recommendation that motivates others to engage with the product, without incurring direct advertising costs.
- **Relationship Triggers** come from interpersonal interactions, where people in a user's social circle prompt engagement. For example, a notification that a friend tagged you in a photo on Instagram or invited you to join a live stream acts as a relationship trigger, leveraging social connections to drive action.
- **Owned Triggers** reside within the product itself, such as notifications, reminders, icons, or company-sent emails that encourage user return without relying on external platforms. Examples include push notifications from Duolingo reminding users to complete their daily language lessons, and the app icon badge indicating unread messages.

Internal triggers, in contrast, prompt behavior through associations stored in memory. These associations can be linked to emotions, thoughts, or existing routines. When a behavior is repeatedly associated with certain internal states, such as pride, loneliness, or boredom, it may eventually occur automatically in response to these feelings [EH14].

> *Emotions, particularly negative ones, are powerful internal triggers and greatly influence our daily routines. Feelings of boredom, loneliness, frustration, confusion, and indecisiveness often instigate a slight pain or irritation and prompt an almost instantaneous and often mindless action to quell the negative sensation.* [EH14, p. 34]

Negative emotions, therefore, play a central role in triggering habitual actions. Feelings such as lack of productivity, loss of control, or loneliness can drive people to engage in behaviors that provide emotional relief or validation. Actions triggered in this way often serve to "scratch the itch" caused by internal discomfort, functioning as coping mechanisms for regulating affect. In this sense, the Hook Model emphasizes how both positive and negative emotional states can drive repeated engagement and eventually form habits [EH14].

2.2.2 Action

The *Action* phase represents the simplest behavior an individual can perform in anticipation of a reward. According to Eyal's Hook Model [EH14], the goal of this phase is to minimize friction between the trigger and the desired behavior, ensuring that the action can be executed with minimal effort.

The concept of action in the Hook Model builds upon the Fogg Behavior Model (FBM) [Fog02], which posits that for a behavior to occur, three elements must converge at the same moment: *motivation*, *ability*, and a *trigger*. These elements will be

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explained in detail in Section 2.3. If any of these components are missing, the behavior will not occur. Thus, even highly motivated individuals will fail to act if the behavior requires too much effort or if a trigger is absent [EH14].

A crucial component of the *Action* phase is the coupling between *trigger* and *action*. The effectiveness of a trigger depends on how seamlessly it leads to the desired behavior. Ideally, the moment an internal or external trigger occurs, the individual is both motivated and able to act, resulting in an immediate and effortless response [EH14].

From a behavioral design perspective, optimizing the *Action* phase means designing contexts and interfaces that require minimal time, cognitive effort, and uncertainty to complete the desired behavior. In well-designed systems, this phase bridges the gap between the trigger and the reward, ensuring that the user's anticipation of gratification leads to repeated engagement [EH14].

2.2.3 Variable Reward

According to Eyal's Hook Model [EH14], the *variable reward* phase sustains engagement by providing unpredictable yet meaningful reinforcement aligned with the user's internal motivations. This stage leverages the psychological principle that intermittent and uncertain rewards generate higher engagement than consistent and predictable ones. When individuals cannot anticipate the exact nature or timing of a reward, their attention and motivation remain elevated, fostering continued interaction.

Eyal identifies three primary categories of variable rewards: the *reward of the tribe*, the *reward of the hunt*, and the *reward of the self*.

Reward of the Tribe. The *reward of the tribe* encompasses social validation and connection. It reflects the satisfaction individuals gain from recognition, approval, or acknowledgment by others. Social rewards are inherently variable because they depend on external responses, such as praise, likes, or comments, that cannot be fully predicted. These rewards reinforce behaviors by fulfilling the human need for belonging and social comparison, a mechanism also supported by Bandura's social learning theory [Ban77], which emphasizes vicarious reinforcement through the observation of others [EH14].

Reward of the Hunt. The *reward of the hunt* refers to the pursuit of material or informational gains. In digital environments, this can take the form of discovering new content, insights, or data, similar to the unpredictability of finding valuable information in a feed or dataset. The element of surprise and discovery activates dopaminergic responses, reinforcing the behavior that led to the discovery [EH14].

Reward of the Self. The *reward of the self* arises from intrinsic motivation, mastery, and personal growth. These rewards occur when individuals experience competence, completion, or progress toward a goal. Unlike social or material rewards, they are internally generated and linked to self-efficacy and autonomy. Feedback mechanisms

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that highlight achievement or positive progress enhance intrinsic satisfaction without requiring external reinforcement [EH14].

Finite and Infinite Variability. Eyal distinguishes between *finite* and *infinite* variability. Finite variability occurs when the novelty of rewards diminishes over time, leading to decreased engagement. Infinite variability, on the contrary, maintains long-term interest by offering a continually evolving range of experiences or insights. Sustainable systems of engagement, therefore, require designing rewards that retain unpredictability and personal relevance rather than relying solely on repetition [EH14].

Autonomy and Motivation. Preserving user autonomy is essential to maintain engagement. Research shows that individuals are more likely to comply with requests and maintain intrinsic motivation when they perceive that their actions are voluntary [Car13]. This finding is consistent with self-determination theory [RD00], which emphasizes the role of autonomy, competence, and relatedness in intrinsic motivation. Therefore, variable rewards should complement, not undermine, the user's sense of control and purpose.

In summary, the *variable reward* phase is crucial for maintaining engagement in habit-forming systems. By balancing unpredictability with personal relevance and autonomy, designers can reinforce behavior without relying on coercive or extrinsic mechanisms.

2.2.4 Investment

The final phase of the Hook Model is the *Investment Phase*. At this stage, users are asked to contribute something of value to the product - be it time, data, or effort. This investment increases the likelihood of future engagement and subsequent passes through the hook cycle. In contrast to the *Action Phase*, where friction is minimized to encourage immediate behavior, the Investment Phase intentionally introduces effort after users have received a variable reward. The timing is crucial: When asking for investment after a reward, the product leverages fundamental psychological mechanisms that strengthen commitment and future use [EH14].

Psychological Mechanisms. Three tendencies are particularly relevant for user investment. First, the *IKEA effect* describes how people tend to value products more highly when they have invested personal effort in creating or customizing them [NM12]. Second, people have a strong desire to remain *consistent with their past behavior*. Once users have invested time or resources, they are motivated to continue doing so to align with their prior actions [Cia09]. Finally, *cognitive dissonance theory* suggests that people adjust their preferences to reduce the discomfort of inconsistency. If they have invested significant time or money, they rationalize that the product must be valuable [Fes57].

Together, these mechanisms explain why greater investment leads to greater perceived value. As Jesse Schell noted, rationalization drives users to believe that the time or money they spend indicates inherent worth: "Anything you spend time on, you start

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to believe, 'this must be worthwhile. Why? Because I've spent time on it!'" [Sch10]. This psychological loop reinforces user attachment and increases the probability of future engagement [EH14].

Forms of Investment. Investment can take several forms, including:

- **Data:** entering information, preferences, or progress that personalizes future experiences;
- **Skill:** developing competence within the app, which increases both user ability and perceived mastery (aligned with the Fogg Behavior Model [Fog09]);
- **Content:** creating or curating material that provides value for future interactions.

Each investment not only stores value for the user but also *loads the next trigger*, creating a feedback loop that reengages the user in subsequent cycles.

Design Implications. Designers should gradually increase the required investment, from simple, low-effort actions during onboarding to more substantial contributions as users become more engaged. This progression ensures early success while fostering long-term commitment. Although the Investment Phase increases friction, this is beneficial when applied at the right moment, as it deepens user involvement and reinforces habitual use through self-generated motivation [EH14].

2.3 The Fogg Behavior Model

For [Fog09], persuasive technology is about learning to automate behavior change by encoding experiences that will lead to desired behaviors into the underlying technology. Automate, in this context, means using technology to change behavior without human intervention. Fogg's model for automating behavior change is based on three principal factors: motivation, ability, and prompts (formerly called triggers until 2017 [Fog]). The model states that a behavior will only occur if a person is motivated to do it, has the ability to do it, and is triggered to do it. If any of those factors are missing or below a certain threshold, the behavior will not occur.

2.3.1 Motivation in the Fogg Behavior Model

Motivation is what drives people to perform a target behavior. The goal in designing for motivation is to move a user to a higher position in the FBM landscape so they cross the behavior activation threshold. To make the term motivation clear in the context of the FBM, Fogg created a framework consisting of three core motivators, each with two sides. These core motivators account for what motivates human behavior in the context of persuasive design. Each motivator operates on a dimension with opposing forces, creating a spectrum of motivational influence. The FBM does not rank the power

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of these core motivators, as their relative effectiveness depends on the specific context and target behavior. Designers and researchers should consider each core motivator and apply it to their work as appropriate for their specific use case [Fog09].

The three core motivators of behavior are:

- **Pleasure and Pain** represent a primitive motivational dimension characterized by immediate or nearly immediate responses. People respond to what is happening in the moment with little thinking or anticipating. This motivator functions adaptively in activities related to self-preservation and propagation, such as hunger and sex. While powerful, especially pleasure, this motivator type may not always be the ideal approach in persuasive design. [Fog09]
- **Hope and Fear** represent a motivational dimension characterized by anticipation of outcomes. Hope is the anticipation of something good happening, while fear is the anticipation of something bad, often the anticipation of loss. This dimension can at times be more powerful than pleasure and pain, as evidenced when people accept pain (such as a flu shot) to overcome fear (anticipation of getting the flu). Hope is considered one of the most ethical and empowering motivators in the FBM [Fog09].
- **Social Acceptance and Rejection** represents a social dimension that controls much of human social behavior, from the clothes people wear to the language they use. People are motivated to do things that win them social acceptance and perhaps even more dramatically, to avoid being socially rejected. This motivator is likely hardwired into humans and other creatures that historically depended on living in groups to survive. Social technologies like Facebook gain their power to motivate and influence users primarily through this motivator, as people post profile pictures and share content driven significantly by their desire to be socially accepted [Fog09].

2.3.2 Ability in the Fogg Behavior Model

The second major factor in the Fogg Behavior Model is ability, which refers to a person's capacity to perform a target behavior. According to Fogg, optimizing this factor can move users across the behavior activation threshold and trigger the desired action. In real-world design, increasing ability is not about teaching people to do new things or training them to improve, as people are generally resistant to both because they require effort. Instead, to increase a user's ability, designers of persuasive experiences must make the behavior easier to perform. Fogg emphasizes that persuasive design relies heavily on the power of simplicity. In studying simplicity, Fogg found that simplicity is a function of a person's scarcest resource at the moment a behavior is triggered. In general, "*persuasive design succeeds faster when we focus on making the behavior simpler instead of trying to pile on motivation*" [Fog09, p. 6], because people often resist attempts at motivation, but humans naturally love simplicity [Fog09].

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Fogg developed a framework that defines simplicity through six elements that work together like links in a chain. If any single link breaks, then simplicity is lost. These six elements of simplicity are:

- **Time** refers to whether a target behavior requires time and whether that time is available. If a behavior takes more time than a person has available, it is not simple for them.
- **Money** relates to financial resources required for a behavior. For people with limited financial resources, a target behavior that costs money is not simple. For wealthy people, this factor rarely prevents them from engaging in a behavior.
- **Physical Effort** considers the physical exertion required to perform a behavior. Behaviors that require significant physical effort may not be simple, especially for those with physical limitations or when energy is limited.
- **Brain Cycles** refers to the cognitive effort required to perform a behavior. If performing a target behavior causes people to think hard, that might not be simple, especially if their minds are already consumed with other issues.
- **Social Deviance** addresses whether a behavior requires going against social norms or breaking the rules of society. If a target behavior involves social deviance, it is no longer simple, even if it might otherwise require little effort.
- **Non-Routine** considers whether a behavior is part of a person's routine. People tend to find behaviors simple if they are routine activities they do over and over again. When people encounter a behavior that is not routine, they may find it difficult.

2.3.3 Prompts in the Fogg Behavior Model

Prompts are the third essential component of the Fogg Behavior Model. According to Fogg, prompts serve as cues that tell people to "do it now". Without a prompt, the target behavior will not happen, regardless of motivation and ability levels [Fog09; Fog].

Fogg describes three different types of prompts, each matched to different combinations of user motivation and ability:

- **Sparks** are prompts that motivate the user to do a behavior when ability is present, but motivation is low. One has to be careful with sparks, as they may start to annoy the user, leading them to engage in a behavior they did not intend to [Fog09].
- **Facilitators** are prompts that make it easier for the user to perform a behavior when motivation is high, but ability is low. Facilitators reduce friction by making the desired action as simple as possible, often through prominent buttons or streamlined interfaces.

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- **Signals** are simple cues or reminders for users who already have both sufficient motivation and ability. Signals remind the user to perform a behavior at the appropriate moment.

Designing Effective Prompts Through Behavior Chains A key principle in prompt design is that effective prompts should request the simplest possible behavior rather than asking the user to perform a complex action. Fogg emphasizes that designers should map out behavior chains, the sequential user flows they want to achieve, and then systematically prompt each step rather than overwhelming users with complicated initial requests.

A concrete example of this principle comes from Facebook's email notification strategy. Rather than sending users generic reminders to visit the platform, Facebook's emails contained a prominent green "Sign In" button (functioning as a facilitator) that directed users not to the homepage, but specifically to a page to "Find people you know on Facebook." This design created a deliberate behavior chain: the simple initial behavior (clicking "Sign In") naturally led to the next behavior (finding friends), which in turn led to increased overall engagement with the platform. The prompt targeted the easiest first step, signing in, while the page destination facilitated the more valuable engagement behavior of building social connections [Fog].

2.4 Gamification

Gamification is often described as the process of enriching services with motivational affordances to create engaging, game-like experiences and influence user behavior. In the last 10 years, gamification has gained significant attention as a strategy to improve user engagement and promote positive behavior patterns in digital services. Studies have highlighted its potential to increase user activity, foster social interaction, and improve the quality and productivity of actions [HKS14].

In their survey on gamification, [HKS14] lists the 10 elements that are used most frequently in gamification. They include: points, leaderboards, achievements/badges, goals, and progress.

Points and leaderboards are two of the most widely used gamification elements [LBC17] and are found in other self-tracking apps such as Habitica[1] and Habit Garden[2]. A leaderboard offers many possible goals to pursue [LBC17] as the user can aim at different areas of the board and gives the user some sense of competition (see 2.6).

In [LBC17], the authors explain how a leaderboard in general serves as setting a challenging goal and show that the result of a leaderboard is comparable to the outcome as if a difficult or impossible goal were set. The study the authors conducted was about a

[1] https://habitica.com/
[2] https://habitsgarden.com/

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brainstorming task, which has been identified as an ideal context for goal-setting research, but the results are therefore not directly transferable to the app context.

2.4.1 Goals

Without a specific goal, users are essentially operating under a "do your best" goal, which has been shown to be less effective than setting a particular goal [LBC17]. Goal setting works best for simple tasks with well-defined parameters [LB75; LBC17]. [Loc82] compared different levels of set goals and found that even impossible goals do not decrease performance compared to challenging goals, but do improve performance slightly. This suggests that when in doubt about the appropriate level of challenge for a goal, it is better to aim for a more challenging goal.

In [MH22], the criteria that effective goals should meet are specific, measurable, achievable, relevant, time-bound, and challenging. Learning goals should be preferred over performance goals, which means that when formulating a goal, it is better to use a formulation like "explore how to do X" instead of "do X effectively" [MH22].

An app that lets you set and change goals at any time is Goodreads, a social cataloging website that lets users track the books they have read. On Goodreads, users can set a goal for how many books they want to read in a year.

Commitment and Consistency The second postulate in [OH09] states that "*people like their views about the world to be organized and consistent*" [OH09, p. 202]. This is also known as the principle of commitment and consistency [Cia09], which states that once a person publicly commits to something, they are more likely to follow through. This can be leveraged in persuasive systems by asking users to commit to their goals publicly. Latham and Locke [LL91] describe commitment as the moderator of the relationship between goal difficulty and performance. Those with high commitment will show a linear relationship between goal difficulty and performance, whereas those with low commitment will show no such relationship.

2.5 Behavior Change Theory

According to the theory of cognitive consistency, people strive to align their behavior, attitudes, and beliefs. A behavior change can lead to changes in attitude and beliefs over time [OH09].

Intention is the strongest predictor of behavior. To change behavior, the intention must first be changed [OH09].

2.5.1 Breaking Bad Habits

Bad habits are difficult to break because they have often been formed over a long period and become deeply ingrained in behavior. Oinas-Kukkonen and Harjumaa suggest that

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"*persuasive systems should aim at reinforcing proper attitudes and making them easier to stick with even in challenging, spontaneous situations*" [OH09, p. 490]. This implies that systems should not only help users avoid unwanted behaviors but also support them in maintaining desired attitudes and behaviors when faced with temptation or challenging circumstances.

2.6 Persuasive System Design Principle - PSD

The PSD model is a framework for designing persuasive systems. It comprises 28 design principles for designing persuasive systems. For this thesis, only a subset of those principles is relevant.

Reduction A system that reduces complex behavior into simple tasks helps users perform the target behavior. Reduction works by both breaking complex activities into smaller steps and by providing only information relevant to the task [KDO08]. By reducing the effort needed to complete a behavior, systems increase the perceived benefit/cost ratio, which in turn increases a person's motivation to engage in the behavior more frequently.

Common examples include autocomplete functionality, default settings, replacing text input with tapping buttons, and biometric authentication (such as Face ID). A well-known implementation is Amazon.com's one-click shopping, which removes multiple steps from the purchasing process.

In terms of the Fogg Behavior Model, reduction strategies primarily increase ability by making the target behavior simpler to perform, requiring less time and fewer cognitive resources. However, by improving the benefit/cost ratio, reduction strategies also indirectly increase motivation, as users perceive the behavior as more worthwhile relative to the effort required [Fog03].

Tunneling Tunneling is a persuasive method that uses design elements to guide users through a process or experience, helping them reach their primary task [OH09]. This involves breaking complex processes into discrete, sequential steps that guide users toward a desired outcome. When processes that require multiple inputs or decisions are presented all at once, users can become overwhelmed. By creating a guided flow where users progress through one step at a time, with appropriate information presentation and visualization at each stage, systems can significantly improve completion rates and provide opportunities to persuade along the way [OH09].

Progress indicators are a key component of effective tunneling, as they provide transparency about where users are in the process and create a sense of accomplishment at each stage. These indicators leverage the endowed progress effect [ND06], which states that people are more likely to complete a task if they feel that they have already made progress toward the goal. This can be implemented by showing users a progress bar that

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is already partially filled at the start. In this way, users feel that they have already made progress and are more motivated to complete the process.

A typical example of tunneling is registration on websites. To access many sites' services or content, users must complete a registration process that guides them through multiple steps.

Tunneling must be implemented carefully to avoid negative outcomes. If used too aggressively without giving users a way out, users might leave the tunnel in an incomplete or inconsistent state or provide false information to expedite their exit. In the worst-case scenarios, tunneling technologies may border on coercion, depending on how they are designed. To avoid coercion, designers of tunneling technology must make it clear to users how they can exit the tunnel at any time without damaging their system [Fog03]. Giving the user the freedom to exit also aligns with the principle of autonomy and motivation (see 2.2.3).

Self-Monitoring Self-monitoring leverages technology to automate the process of tracking personal behaviors, performance metrics, or status indicators that would otherwise require manual observation and recording [Fog03; OH09]. By removing the burden of continuously measuring and documenting their own activities, these systems enable users to maintain awareness of their progress toward predetermined objectives. The persuasive power of self-monitoring stems from two complementary mechanisms. First, when technology measures and visualizes behavioral patterns, users can easily perceive discrepancies between their current state and their desired goals, which motivates them to adjust their behavior accordingly. Second, self-monitoring systems tap into the intrinsic human desire for self-knowledge and personal insight, making the act of tracking itself potentially rewarding independent of whether it leads to immediate behavior change. The effectiveness of self-monitoring as a persuasive strategy depends on presenting tracked data in ways that are accessible, timely, and meaningful to users, transforming raw behavioral information into actionable awareness [Fog03].

Suggestion Fogg defines suggestion technology as an interactive computing product that proposes a specific behavior at the most opportune moment. The technology works by building on people's existing motivations and cueing relevant behaviors at precisely the right time, essentially communicating "*now would be a good time to do X*" [Fog03, p. 69]. For a suggestion to be effective, it must be both compelling and timely enough that users actually implement the recommended action. Timing is critical because people are more receptive to suggestions when they are in a good mood, when they can take immediate action, when their current worldview no longer makes sense, or when they feel indebted [Fog03].

Social Comparison System users will be more motivated to perform the target behavior if they can compare their performance with others' [OH09]. Social comparison theory holds that people seek information about others' attitudes and behaviors to form their

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own attitudes and behaviors, using others as a comparison point or reality check [Fog03]. Computing technology can leverage this by providing users with information about how their performance compares with others, especially those similar to them [Fog03]. In essence, social comparison enables people to benchmark their performance, reactions, attitudes, or behaviors against those of others [Fog03]. The motivational effect of social comparison is strengthened when users can compare themselves to similar others in terms of age, ability, ethnicity, or other key attributes [Fes54].

Normative Influence Normative influence leverages peer pressure by exploiting people's tendency to change attitudes and behaviors to match the expectations of their in-group, such as classmates, family, work groups, or close friends [Fog03]. Computing technology can leverage this by tracking a person's behavior and sharing the results with their in-group, making behavior visible to peers who matter most [Fog03]. A notable example is Alcohol 101, an interactive educational program that provides college students with actual statistics about peer drinking habits, which many students find to be lower than they expected, positively impacting their attitudes toward drinking [Fog03]. However, connected technologies can also help people resist conformity pressure by showing that at least one other person in their in-group has resisted the same pressure, making it easier for others to dissent [Fog03]. This dual capability means that technology can either reinforce group norms or help individuals make independent choices by expanding their awareness beyond their immediate circle of close friends.

Social Facilitation The principle of social facilitation describes the phenomenon in which people perform better when others are present, participating, or observing. According to this principle, people are more likely to perform a well-learned target behavior if they know they are being observed via computing technology, or if they can discern via technology that others are performing the behavior along with them. Connected products can leverage social facilitation by creating opportunities for virtual presence, enabling others to be present even when performing activities alone. When a person is performing a well-learned activity, such as running on a treadmill, they will likely perform better if a connected product shows that other people are virtually present and performing the same activity. Notably, the representation of others does not need to be realistic to generate social facilitation effects. Simple avatars, abstract shapes, or even bar graphs can effectively represent the presence and performance of others in digital systems [Fog03].

Competition Competition is one of the most powerful group-level intrinsic motivators identified in research on human motivation. Malone and Lepper [ML05] outline seven types of intrinsic motivators, three of which involve interaction among people: competition, cooperation, and recognition. When a competition is established, people become energized and willing to invest significant time and effort toward achieving their goals. Notably, the motivational power of competition is not dependent on extrinsic rewards, such as prizes for winners. The competitive dynamic itself serves as a sufficient motivator for most

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people in many situations. While not everyone is competitive by nature, competition remains an effective motivational strategy for a broad range of users and contexts [Fog03].

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CHAPTER 3

The *Habits Network* Application

3.1 Introduction

Habits Network is a text-based journaling web application designed to help users track activities and social interactions in their daily lives. The application is available as a progressive web app on iOS and Android devices and, since October 2025, also through the iOS App Store. This chapter provides an overview of the design philosophy and goals, the use case, and technical implementation, and gives a detailed overview of all features.

3.1.1 Design Philosophy and Goals

The primary goal of *Habits Network* is to provide users with a simple and sustainable approach to life tracking. The application deliberately focuses on day-level granularity rather than tracking the quantity or intensity of activities. Users track whether an activity occurred on a given day rather than measuring how much or how intensely they performed it. This design decision stems from the philosophy that consistent long-term tracking provides more valuable insights than high-precision tracking maintained only for short periods. By prioritizing simplicity over granularity, the application aims to reduce the effort required for tracking, thereby increasing the likelihood of sustained user engagement.

A distinctive feature of *Habits Network* is its integrated tracking of social interactions alongside other life activities. Users can track not only what they did but also with whom they did it, enabling analysis of how social connections influence habits and behaviors. This dual focus on activities and social context distinguishes *Habits Network* from existing life-tracking applications.

The intended output for users is a comprehensive dashboard that provides insights into their daily lives. The application aims to increase users' awareness of their behavioral patterns, enabling them to identify trends, recognize correlations, and make informed

decisions about their habits. The visualization of personal data serves as a foundation for self-reflection and behavior change.

3.1.2 Use Case

The typical user workflow begins with daily journaling, in which users create text entries that mention activities and people they interact with. Users define their own tracking categories by creating tags for activities they care about, such as exercise, reading, or any other aspect of their lives they wish to monitor. Similarly, users create profiles for people in their social network to track social interactions. Over time, the application aggregates this data and presents it through various visualizations and statistics on individual dashboards for each tag and person.

Users gain value from the application through multiple mechanisms. The visualizations reveal patterns that might otherwise go unnoticed, such as which activities cluster together, how frequently certain behaviors occur, or how social connections correlate with specific activities. Setting goals for activities provides structure and motivation for behavior change. The social features allow users to optionally share their activity data with friends, creating opportunities for comparison, mutual support, and accountability. Users can track both positive habits they want to cultivate and negative behaviors they aim to reduce, with the application supporting both goal types through its streak and goal-setting features.

3.1.3 Technical Implementation

The application was developed using the Python Flask web framework for the back-end, with HTMX and AlpineJS providing dynamic front-end interactions. User data are stored in a MariaDB database. The progressive web app architecture enables installation on mobile devices while maintaining a single codebase across platforms.

As of November 2025, the application has maintained approximately 10 weekly active users since March 2025, providing a user base for the empirical research conducted in this thesis.

3.1.4 Overview of Features and Persuasive Design Principles

Table 3.1 provides an overview of the application's key features, the persuasive design principles and behavior change techniques implemented through each feature, and when each feature was introduced to the application.

3.2 Journaling

When a user opens the app, they are taken to the *Journal* Page (Figure 3.2b). This view shows the last 30 days with the user entries.

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Table 3.1: Overview of application features with associated persuasive design principles and implementation dates

Feature	**Persuasive Design Principles**	**Since**
Tag dashboards with charts	Self-monitoring, feedback	2024-04-11
Activity heatmaps	Self-monitoring	2024-04-11
People tracking and profiles	Self-monitoring	2024-04-11
People bar chart (see 3.13b)	Self-monitoring	2024-04-11
Network graph	Self-monitoring, visualization	2024-04-30
Keyword suggestions	Reduction (see 2.6)	2024-05-13
Sharing tags with friends	Social comparison (see 2.6), normative influence (see 2.6)	2024-06-14
Friends feed	Social facilitation (see 2.6)	2024-07-19
Parent-child tag hierarchy	Reduction, self-monitoring	2024-09-15
Streaks (positive/negative)	Gamification (see 2.4)	2024-11-19
Journaling tag (predefined)	Self-monitoring, streaks	2025-04-19
Tag comparison and leaderboard	Social comparison (see 2.6), competition (see 2.6), gamification (see 2.4)	2025-04-29
Random statistic	Variable reward from Hook Model [EH14]	2025-05-07
Goals with progress tracking	Goal setting (see 2.4.1), commitment, tunneling and endowed progress effect (see 2.6), suggestion (see 2.6)	2025-05-25
Achievements	Gamification (see 2.4), feedback	2025-06-24
Local network graph	Self-monitoring, visualization	2025-07-16

An entry is added in free text form, as can be seen in Figure 3.2a. The design was inspired by the Apple iMessage chat box (Figure 3.1).

In an entry, the user can mention keywords, which can be tags or people. Keywords are mentioned in all caps. When a new entry is added, keyword suggestions for the keywords the user has previously created are shown. When the user taps the suggestion, the current word is replaced with it. This makes it easier for the user to create a new entry and increases the benefit/cost ratio of journaling, the *reduction principle* (see 2.6).

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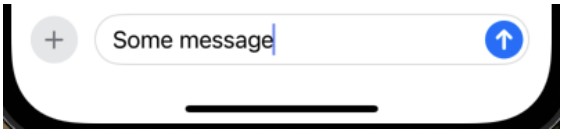
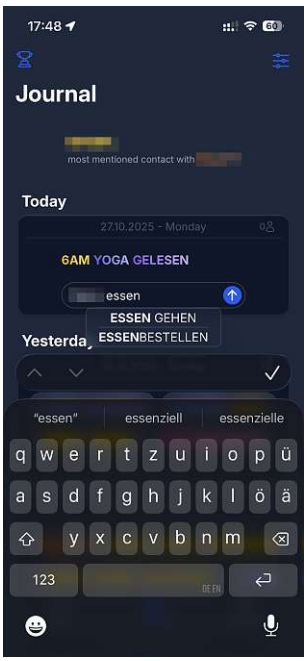
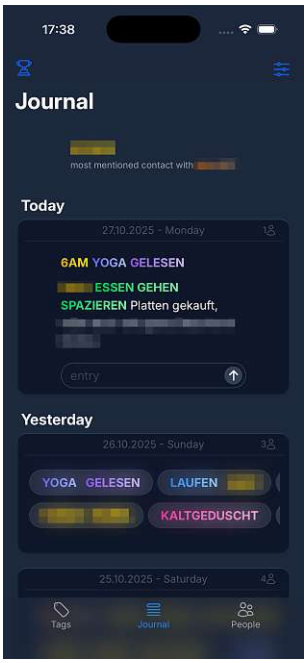


Figure 3.1: The input box of Apple's iMessage was the inspiration for the input field of an entry in *Habits Network*. The goal was to give the user the feeling of chatting with somebody.



(a) Journal input form with keyword suggestions based on the last written characters.



(b) The journal view with the current day open to add a new entry and a preview of the journal entries of the past days

Figure 3.2

3.3 Tags

Tags are keywords that can be mentioned in a journal entry, like *READING*, *EXERCISE*, or *EAT OUT*. They are meant to indicate that a specific activity was performed on that day. An overview of all the tags of a user can be seen in the *tags* tab of the application (Figure 3.3a). From this overview, the user can see the day heatmap of their tags and check for any streaks. More about streaks and heatmaps in their dedicated subsections (3.3.2 Heatmap, 3.3 Streaks).

When a user clicks a colored keyword, they are taken to the keyword's Dashboard (Figure 3.3b). The tag dashboard contains metrics and six charts showing the tag's usage. Tags have certain properties that the user can set through the tags' settings (Figure 3.4b).

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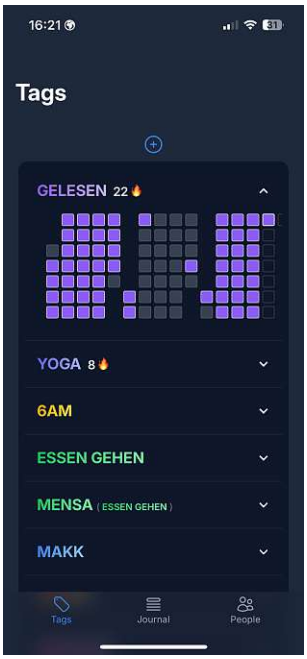

(a) The view of all tags with a button to add a new tag and one tag expanded to see the active days of the year.

(b) The view of a tag. Statistics in the top area, a heatmap in the bottom, and a progress bar in between.

Figure 3.3

Influence A tag can be set to have a positive, neutral, or negative influence on the user. By default, a tag has a neutral influence, but the user can change it to positive or negative. If the tag has a non-neutral influence, a streak can be started.

Streaks A streak is a sequence of days in which a user has mentioned a tag in a journal entry if the tag has a positive influence, or has not mentioned the tag if it has a negative influence. The streak is broken if the user mentions the tag and it has a negative influence, or if the user does not mention the tag and it has a positive influence.

When the user has a streak, the tag view displays a progress bar showing how long the current streak is relative to the user's longest streak. In Figure 3.3a, the user also gets an indicator for a streak: the flame emoji for positive streaks and a lock emoji for negative streaks. This feature, and especially the flame icon, were inspired by Snapchat, which uses the flame to indicate a "streak" of days in which a user has sent a photo to another user. In this way, the user is rewarded for continued effort in something that they define as good for them.

Counting a Tag towards another Tag In a tag's settings, the user can select a single parent tag for the tag. In this way, the tag is counted toward the parent tag and included in the parent tag's statistics. An example of this is the tags *SWIMMING* and

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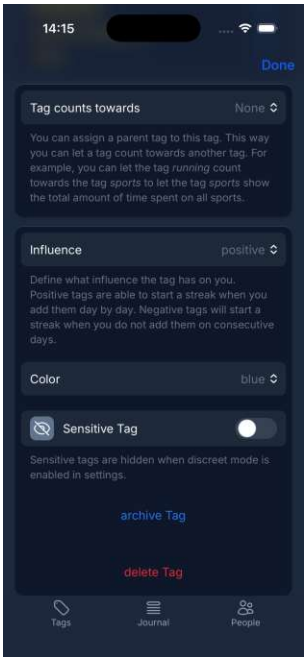
(a) A user can share a tag with others through the settings of the tag



(b) A user can define the properties in the settings of the tag

Figure 3.4

RUNNING, which can be counted towards the parent tag *EXERCISE* to get a statistic on how often the user has exercised in total. Child tags of a tag can also be seen in the parent dashboard (see Figure 3.6a).

Sharing a Tag with a Friend In the Settings of a tag, the user can share the tag with friends who also use the app (see Figure 3.4a). When someone shares a tag with the user, the user can access the dashboard for the tag owned by the other user. When another user can see the tag, the principle of *social comparison* (see 2.6) comes into play. Suppose the other user also tracks the same activity. In that case, the user can compare their activity with that of the other user, which might motivate them to increase or decrease their activity in that area. Many people use the app to track private activities, such as *DRINKING ALCOHOL* or *SMOKING*. When users can see that others are tracking these activities and doing less of them, they might be motivated to do the same. The effect of normative influence (see 2.6) is even stronger when people are similar to each other, as is the case here, since one tends to share such activities only with close friends.

Tags can also be marked as *sensitive*, so they are hidden when the *discreet mode* toggle is activated in the settings. Discreet mode is a privacy feature that becomes relevant when a user shows the app to someone else but wants to hide specific entries.

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3.3.1 Connecting and Comparing Tags

When user A shares a tag with user B, user B can connect it to one of their own tags. Theoretically, a tag from another user could be connected to any tag from the user, but in practice, users mostly connect tags that have the same meaning to each other. For example, user A has a tag *READING* and shares it with user B, then user B can connect user A's *READING* tag to their own *GELESEN* tag. After connecting the two tags, the *READING* tag is rebranded to *GELESEN* in user B's app. The tags' activities can now be compared by tapping on the *Compare* button on the tag dashboard (see Figure 3.5a). In the line chart in the comparison view (Figure 3.5b), only the user A and user B tags are shown in the comparison if the view is accessed from user A's tag, like in Figure 3.5a. If user B has connected more tags from other users to their own tag, those tags can also be added to the comparison. If the comparison view is accessed from user B's own tag (see Figure 3.3b), all connected tags are shown in the line chart by default. The leaderboard leverages the principles of social comparison 2.6 and competition 2.6, and it is a gamification element (see 2.4). By comparing their activity with others, users can be motivated to increase their activity in the respective area.

The rebranding of the shared tag to the user's own tag is done to decrease cognitive load for the user. The name and the color of the tag are changed to match the user's own tag. This way, the number of different tags user B is exposed to is reduced, and the user can focus on the activity of the tag instead of being distracted by other names and colors for the same activity. In this example, user A shares their *READING* tag with user B, who has the *GELESEN* tag. For user B to be able to connect their tag to user A's tag, user B does not have to share their *GELESEN* tag with user A. Also, when the two tags are connected, user A does not get access to user B's *GELESEN* tag.

3.3.2 Charts

This subsection includes the charts in a tag's dashboard.

Activity Heatmap The heatmap is the first visualization in the dashboard of a tag. Its purpose is to give the user a quick overview of the tag's activity. In Figures 3.3a and 3.3b, the heatmap appears in the tag overview and the tag view. Every day of the year is represented as a square in the heatmap. In contrast to a heatmap, which distinguishes the intensity of an activity on a given day, as shown in a Profile on GitHub, the heatmap in the app is binary. In a heatmap where activity intensity is encoded, hovering over an entry displays the activity's intensity. This is not necessary in the app, since the activity's intensity is not relevant. When a user clicks a square in the heatmap, they are taken to the journal view for that day.

Number of days with other people This vertical bar chart (see Figure 3.8a) shows how often the tag co-occurred with people in journal entries.

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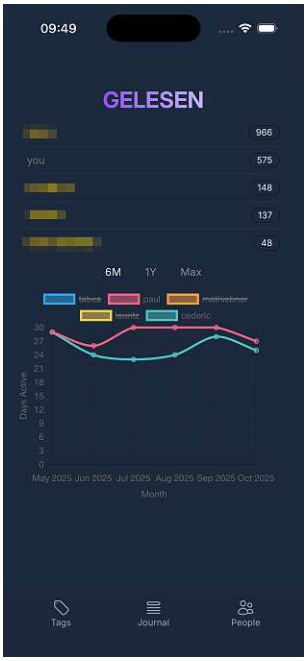
(a) The tag view of a shared tag from another user after the tags have been connected.



(b) The activity of connected tags can be compared with each other. More people can be added to the comparison if they are also connected to the user's tag.

Figure 3.5

Tag - Contact Bubble Chart This chart (see Figure 3.6b) aims to visualize the co-occurrences of the tag and people as yellow bubbles. On the x-axis, one can see the number of days a contact was mentioned together with the tag, and on the y-axis, the conditional probability that the tag was mentioned given that the person was mentioned. So, if the y-axis was 1, then every time the person was mentioned, the tag was also mentioned. Bubbles in the top right therefore indicate that a person occurred relatively often with the tag and that, in the past, the chance was relatively high (relative to the other people the user tracks) that, when the person was mentioned, the tag was also mentioned.

Line Chart The line chart shows the trend in the number of days per month for the keyword. Users can switch between six months, one year, and the maximum time span. The line chart can be seen in Figure 3.6b.

Pie Chart The doughnut chart (see Figure 3.6a) shows the percentage distribution of the days the user used the tag. It is calculated over the entire time the user used the app.

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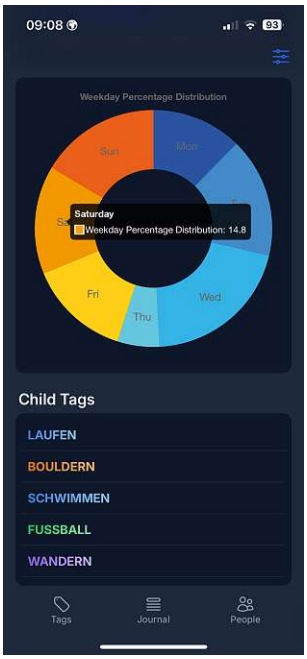

(a) Pie chart of the weekday distribution and the child tags of the tag "sport".

(b) A tag's dashboard includes a bubble chart of a tag that shows what people occur together with the tag, and a line chart of the frequency of the tag per month.

Figure 3.6

Cosine Similarity The cosine similarity between two keywords (see Figure 5.23b), in this case a tag and a person, is the cosine of the angle between their boolean vector representations. The keyword can be described as a boolean vector of length equal to the number of entries of the user, where an entry of the vector is 1 when the keyword occurs in the entry and 0 if not. The similarity is then calculated with the cosine between those vectors. In other words, it is the number of entries where the keywords occur together, divided by the square root of the product of the number of entries of the two keywords. Let A and B be the set of entries where the keywords occur, then the cosine similarity $\cos(A, B)$ is defined as

$$\cos(A, B) = \frac{|A \cap B|}{\sqrt{|A| \cdot |B|}}$$

This measure was introduced to add another measure of similarity in addition to the absolute number, which can be dominated by people who are just very present in one's life.

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Journaling Tag

The predefined *JOURNALING* tag (see Figure 3.7a) is special: it is automatically created when the user registers for an account and cannot be deleted. It does not have to be mentioned; it is only there to show the user which days they created an entry and which days they did not, and how this behavior changes over time. The tag is defined as having a positive influence to get a streak. Besides sharing with other people, there are no options to modify the tag.

(a) The journaling tag of a user

Figure 3.7

3.4 People

People are keywords that, like tags, can be mentioned in a journal entry. A Person is created with their handle (how the user will mention the person in an entry), their full name, and their gender.

A person's dashboard is very similar to a tag's dashboard, with some differences.

3.4.1 Charts

The charts in both the dashboard for a tag and for a profile are: the heatmap (see 3.3.2), the number of days with other people (see 3.3.2), and the line chart (see 3.3.2).

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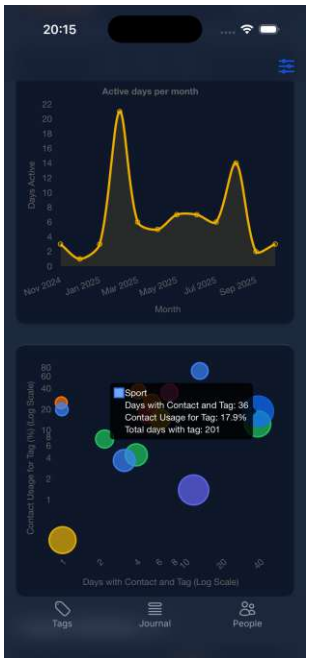
The bubble chart (see 3.3.2) is also available on a person's dashboard, but the bubbles represent tags rather than people. The cosine similarity 3.3.2 and the number of days with another keyword 3.3.2 exist in a person's dashboard for tags and for people (see Figure 3.8a). The weekday percentage pie chart 3.3.2 is not present in a person's dashboard.

(a) In the profile of a person, the co-occurrences are shown for both other people and tags. The percent number is the cosine similarity, and the number on the left is the absolute number of co-occurrences.



(b) Line chart and bubble chart with open tooltip in the dashboard of a person

Figure 3.8

Local Network The local network graph (see Figure 3.9a) of a person aims to give a sense of how connected the person is in the user's network, which tags mostly occur with the person, and how often, all in one chart. The local network or 1.5 network is the subgraph defined by the constraint that all nodes in the subgraph must have a direct edge to the particular person's node. The graph does not include the person itself because, by that definition, their node would have an edge to every node in the network, which would make the graph harder to read.

The nodes in the graph are sized by the number of co-occurrences with the person. The opacity of a person's node decreases by the number of days that have passed since the last mention of the two people together in an entry. The opacity and thickness of the edges are also a function of the number of co-occurrences of the keywords. The graph

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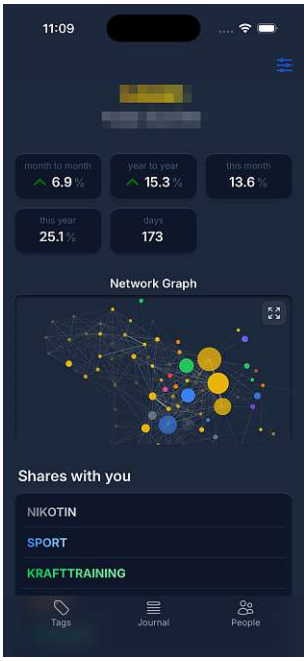
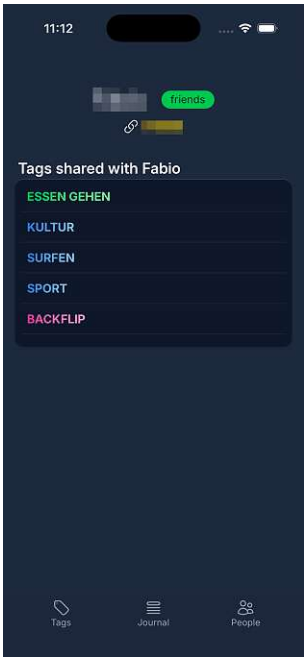
can be zoomed in and panned. Nodes can be dragged around, and tapping or clicking a node in the graph highlights the surrounding nodes.



(a) View of a contact profile that is associated with a real user of the app that shares tags with the user.



(b) The user profile of a user who accepted the friend request and is associated with a contact in the app.

Figure 3.9

3.4.2 Feed

From the people view (Figure 3.11a), the user can see a preview of the first three entries in the friends feed and, when clicked, is taken to the full friends feed. The friends feed shows, in chronological order, which keywords friends used in their entries and when they used them. Only tags that are shared with the user are shown in the feed (see Figure 3.10a).

In the feed, the principle of *social facilitation* (see 2.6) is incorporated into the app, as a user can observe others engaging in behaviors they are tracking, such as exercising or reading. Through the feed, the user is also reminded that others can observe their behavior, which might motivate them to engage in an activity they track more often. On a meta-level, the feed also serves to remind the user that others are using the app, which might motivate them to use it more often.

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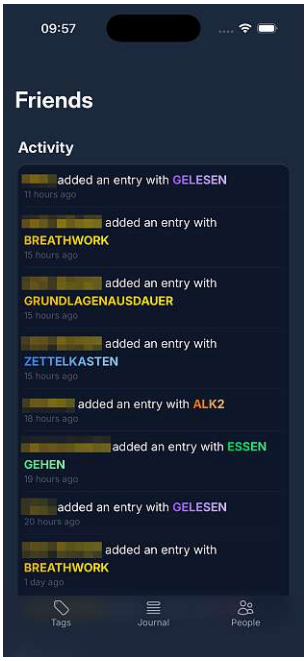
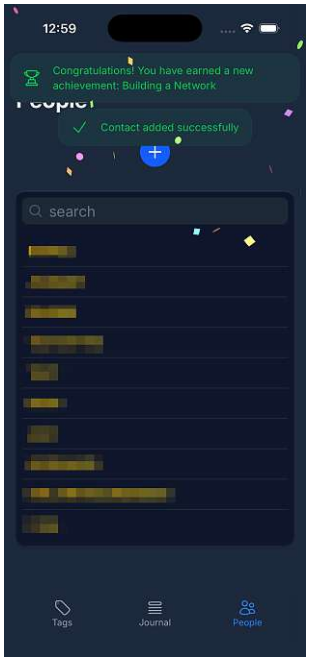


(a) The feed view shows what keywords the user's friends have mentioned in their entries.



(b) When the user performs an action that unlocks an achievement, a toast notification with confetti is shown.

3.4.3 People Statistics

The second tile on the people view (Figure 3.11a) takes the user to the people statistics view. This view shows three charts about the people the user has mentioned in their entries. The first chart (Figure 3.11b) shows how many people the user has mentioned each month and how many were new. The gray line shows how many people were mentioned for the first time in that month, while the yellow line shows the total number of people mentioned in that month.

In the line chart, the user can change the timespan to six months, one year, or the maximum time span of the user's entries. The third graph is a horizontal bar chart that shows the number of mentions per person (Figure 3.13b). The user can change the month displayed in the bar chart using a date picker above the chart. One can also select a full year or the maximum time span of the user's entries.

3.4.4 Network Graph

The network visualization is an interactive, force-directed graph built with the D3.js library. People are represented as nodes, and if the journal contains an entry that mentions two people, there is an edge between their nodes. The graph is interactive, so the user can drag the nodes around and zoom in and out. The user can also click a node to see the person's name and information about the number of people the person knows

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(a) The people view with a preview of the friends feed and a button to the people statistics and the network graph.

(b) Chart of people the user has mentioned per month in yellow and the new people mentioned in gray in the statistics view.

Figure 3.11

in the network, the total number of days the person was mentioned, and when the person was last seen. The graph is shown in Figure 3.12.

The node's size encodes the number of entries that the user has made with a person. The more entries the user has made with a person, the larger the node.

The time elapsed since the last entry with a person is encoded in the node's opacity. The longer it has been since the last entry with a person, the more transparent the node is.

3.5 Goals

The goals were introduced on June 1st, 2025. Tag goals can be set to motivate the user to mention a tag more or less often until a specific date. A goal is defined in a three-step process (see Figure 3.14a). First, the user selects a tag for which they want to set a goal. This step is skipped if the user creates a goal from the tag's goals overview (see Figure 3.13a). The second step is to define the properties of the goal (see Figure 3.14a). In this step, the user selects an end date for the goal, whether they want to mention the tag more or less often, and above or below how many days they want to mention the tag until the end date. In the final step, the user gives the goal a name. These steps are shown in a modal dialog. There is a progress bar at the top of the modal that shows the

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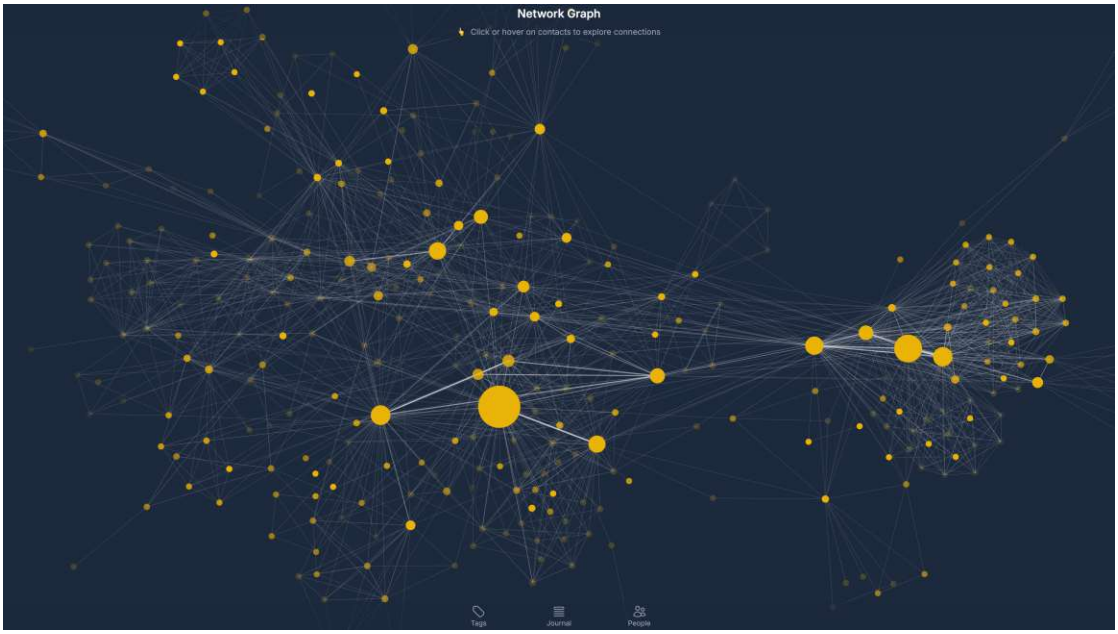

Figure 3.12: Interactive Network Graph of people the user has mentioned in their entries.

user how far they are in the goal creation process. This progress bar uses the endowed progress effect (see 2.6) to motivate the user to complete the goal-creation process. The progress bar shows that the user is already 20% in the process when they start creating a goal. After selecting the tag, the progress bar shows 50%, and after defining the goal's properties, it shows 90%.

If the tag the user creates the goal for is set to have a positive or negative influence, the goal's setting is prefilled to match that influence. If the tag has a positive influence, the goal is set to mention it more often; if the tag has a negative influence, the goal is set to mention it less often.

There is also a suggestion for how many days the user should set the goal to (see Figure 3.14a) to make the goal-creation process easier for the user and to make the application more persuasive in general (see 2.6). As was mentioned in 2.6, the success of suggestions as persuasive technology tools in general largely depends on the context in which the user receives them. Since the user is already in the context of setting a goal, the suggestion fits the user's context. The suggestion is based on the user's past behavior, and its quality depends on the data the user has entered. The suggestion is the average of the last 90 days before the current date, or, if that was less than 90 days ago, the average since the tag or one of its children was first used, plus or minus 1, depending on whether the user wants to mention the tag more or less often. The reason the suggestion is plus or minus 1 is to make the goal slightly more challenging than the user's average behavior, motivating the user to change their behavior (see 2.4.1). In 2.4.1, it was described that goal setting works best for simple tasks with well-defined parameters Since

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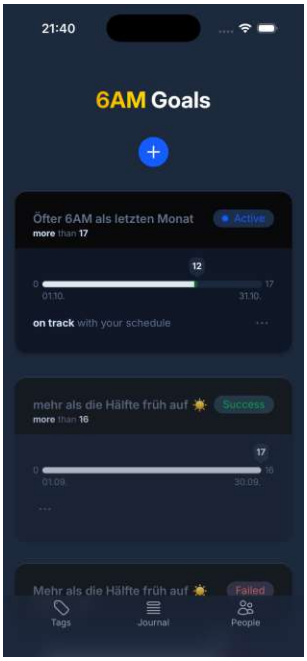

(a) All goals of a tag can be accessed from a tag's dashboard

(b) A line plot of the number of mentions of female and male contacts and a horizontal bar chart of the number of mentions of a contact.

Figure 3.13

goals are defined by the number of days the user wants to perform an activity within a specific timeframe, all goals can be framed as simple tasks with well-defined parameters, regardless of the underlying activity's complexity.

The goals overview (see Figure 3.14b) shows all the user's current goals. For each goal, an indicator shows whether the user is on track, ahead, or behind it. The white progress bar shows linear progress from the start date to the goal's end date. If the user wants to mention a tag more often and has done so more than necessary to stay on track, the amount ahead is shown in green in the progress bar. If the user is behind their goal, the amount behind is shown in red in the progress bar. The same applies if the user wants to mention a tag less often: if it's mentioned less often than necessary to stay on track, it's shown in green; if it's mentioned more often, it's shown in red.

If the user reaches their goal, the goal status changes to *success*; if they do not reach it by the end date or mention the tag more often than necessary, the status changes to *failed*. Users can delete goals at any time.

Setting goals in general is a form of commitment 2.4.1, and for this commitment to be most effective, the person who committed should know that others are aware of and can observe their progress toward the goal. This was the plan when implementing the goals

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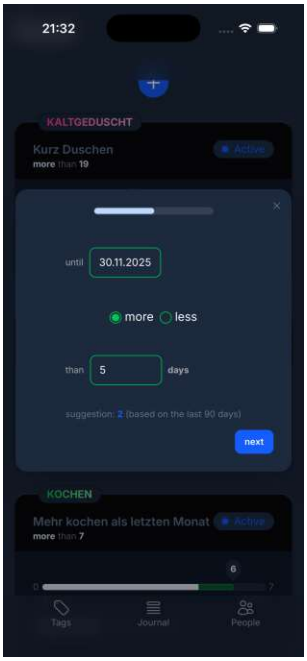
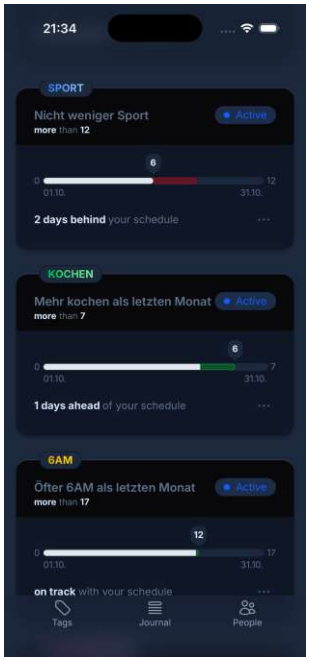


(a) The second step of the goal creation process. Here, the end date, more or less, and the number of days are selected.



(b) Current goals are displayed with an indicator that shows whether the user is on track, ahead, or behind their goals

Figure 3.14

feature, but due to time constraints, sharing a goal with friends was not included in this thesis.

3.6 Achievements

Achievements were introduced on June 24th, 2025, to reward users for their behavior in the app and to enhance gamification (see 2.4). The list of achievements covers the app's features, from basic to advanced, and is meant to guide users through the app and show them what is possible. When the user does something in the app that unlocks an achievement, a toast notification appears at the top of the screen to inform them, and confetti falls from the top (see Figure 3.10b). The achievements are divided into three different categories: bronze, silver, and gold. The 20 bronze achievements are meant to be easy to unlock and to cover the app's basic features. The four silver achievements are meant to be more challenging and to be earned after using the app for a while, like adding 10 tags or 100 people. The two gold achievements are adding 1000 people and 1000 entries; the latter is already unlocked by six users as of August 2025. No user has yet unlocked the achievement for adding 1000 people. The achievements feature is also meant to replace a user onboarding tour that shows users the app's features when they first register.

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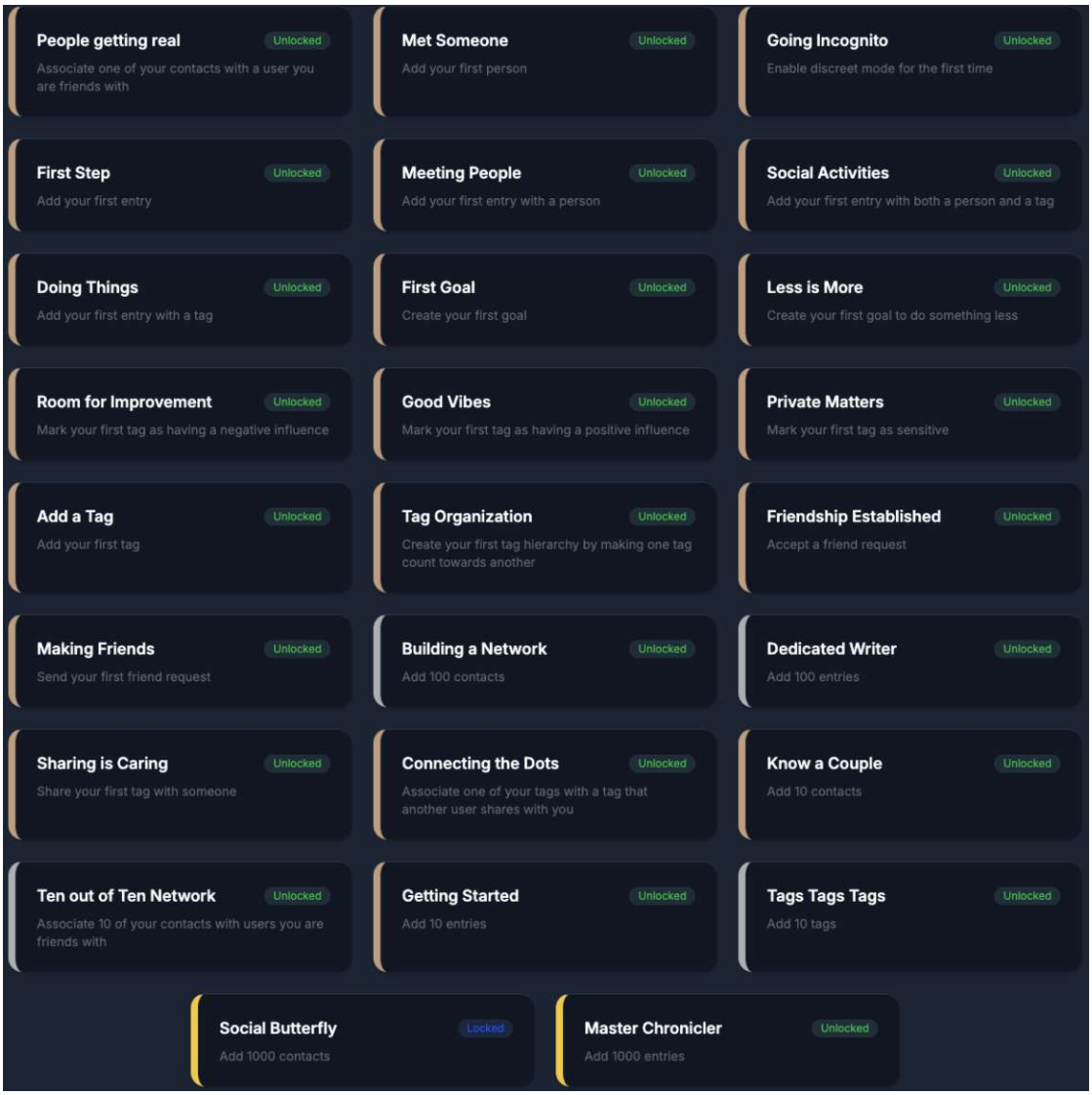

Figure 3.15: List of achievements available in the app

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3.7 Random Statistic

The random statistic was introduced on May 7th, 2025, to give users a variable reward when creating a new journal entry. The random statistic is at the top of the journal view, above the newest day (see Figure 3.2b). When the user creates a new entry and 12h have passed since the last entry, the random statistic changes to a new one. The random statistic is chosen from a pool of functions: the number of entries with a keyword, the number of days since mentioning a keyword, the longest streak of a tag with a positive influence, the most extended period between mentioning a tag with a negative influence, the most used tag with a person, and the most used person with a tag. Those functions are chosen randomly and with a random tag and contact as input. The functions that indicate how many days have passed since a keyword was mentioned were chosen because they are not available elsewhere in the app and provide the user with new information about their behavior. The other statistics are also available in a keyword's dashboard and should encourage users to explore it more often. The keywords in the random statistic are clickable and take the user to the keyword's dashboard. The functions or statistics that received the most engagement are described in the User Behavior Analysis chapter 5.1. The random statistic is meant to give the user a variable reward when creating a new entry in the app. When implementing this feature, I chose a 12-hour cooldown to avoid overwhelming users with changing statistics when they create multiple entries in a short time span. The 12-hour interval was chosen so that, in the ideal case, the user gets a new statistic every day when they create an entry in the morning and in the evening. This feature was implemented because of the importance of *variable reward* in the hook model [EH14]. What was not clear when implementing this feature was which action should lead to the variable reward. Now the action is creating a new journal entry, but it could also be opening the app itself. I decided to go with entering an entry because it is the one thing that the user has to do to get value from the app. This decision will also be discussed in the discussion chapter 6.

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CHAPTER 4

Methods

The empirical evaluation of the *Habits Network* application used a mixed-methods approach. The goal was to identify what motivates users to engage with the app, which features they use, and which persuasive design elements are effective in bringing users back to the app. This was investigated through semi-structured interviews with ten users (ranging from five to 17 months of usage) who were asked directly about their motivations and usage patterns, while their actual usage behavior was analyzed using app analytics data collected during their individual usage periods. Four expert interviews with professionals in visualization, UI/UX design, and behavior change provided insights into what could be improved or done differently. Additionally, the System Usability Scale (SUS) was administered to all ten user interview participants before the interviews to make the app's usability comparable to other applications.

4.1 User Behavior Analysis

User behavior data was collected using PostHog, a product analytics platform that tracks user interactions within web applications. Event tracking was implemented incrementally throughout the app's development process. When developing new features, event tracking was typically added either during initial implementation or in subsequent updates. Backend event tracking was the preferred method, with frontend tracking used only when backend tracking was not feasible due to technical constraints.

Two approaches were used to measure feature usage. First, explicit events were defined and triggered when users performed specific actions, such as viewing the achievements page or creating a goal. Second, for features without explicit event tracking, usage was measured by analyzing URL access patterns via PostHog's pageview events. This URL-based approach was used for person dashboard views and feed views, where the presence of specific URL patterns indicated feature usage.

For frontend-tracked events, a significant limitation exists: browser ad blockers can prevent PostHog's frontend JavaScript library from running, leading to unreported events. This limitation affects person dashboard views and feed page views, where the reported numbers represent lower bounds rather than complete counts. Backend-tracked events are not subject to this limitation as they execute server-side before content is delivered to the browser.

Following the completion of user interviews, all event data for the ten study participants was extracted from PostHog for analysis. A Python script was developed to query PostHog's API using HogQL, PostHog's SQL-like query language. The script retrieved all relevant events for the study participants, filtered by their user identifiers and a cutoff date of August 17th, 2025. The retrieved data was processed and visualized using Python's matplotlib library.

4.2 Expert Interviews

The expert interviews were conducted with four professionals specializing in visualization, UI/UX design, and behavior change. They were recruited through the thesis supervisor's professional network.

All expert interviews were held in person, either in the experts' offices or at a meeting room at their university. The interviews ranged from 65 to 95 minutes, depending on the depth of discussion and participants' engagement.

Before interviews, all participants gave their informed consent, agreeing to the recording and use of their data for research purposes. The consent form (see A.4) outlined the study's purpose, confidentiality measures, and participants' rights, ensuring ethical compliance throughout the research process. Before the interviews, participants received a video link that explained the app and its features in detail to ensure they had a clear understanding of the app. The experts were asked beforehand what their area of expertise is to tailor the interview questions accordingly. The experts' expertise lay in very different fields; therefore, the questions asked did not cover all the research questions but focused on the interviewees' areas of expertise. During the interviews, participants were encouraged to share their professional insights on the app's design, usability, and behavior change strategies. The interviews were semi-structured, allowing for a balance between guided questions and open-ended responses to capture rich qualitative data. The interviews were audio-recorded with Apple's Voice Memos app and later transcribed, as with the user interviews (see 4.4). The interview data were analyzed using qualitative content analysis following the method proposed by Mayring [May00]

4.3 System Usability Score

The System Usability Scale (SUS) was administered to 10 participants who had been using the application for 5 to 17 months. After agreeing to participate in an interview,

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participants were asked to complete the SUS questionnaire beforehand. The questionnaire was distributed individually via WhatsApp or Signal, allowing participants to respond directly in the chat by either returning the completed form or sending their ratings as numerical responses.

4.4 User Interviews

The user interviews were conducted with the same ten participants who had been using the application for five to 17 months. Participants were recruited through direct outreach via messaging platforms such as WhatsApp and Signal. All participants provided informed consent before the interviews, agreeing to the recording and use of their data for research purposes. The consent form (see A.4) outlined the study's purpose, confidentiality measures, and participants' rights, ensuring ethical compliance throughout the research process. The interviews were scheduled at times convenient for the participants, ensuring minimal disruption to their daily routines.

The interviews were all held online via Zoom. This approach was chosen to accommodate participants' varying locations and schedules, allowing for greater flexibility and comfort. The interviews ranged from 40 to 80 minutes, depending on the depth of discussion and participant engagement. During the interviews, participants were encouraged to share their experiences with the application, including their motivations for using it, perceived benefits, challenges encountered, and suggestions for improvement. The interviews were semi-structured, allowing for a balance between guided questions and open-ended responses to capture rich qualitative data. The interviews were audio- and video-recorded using Zoom's recording feature, with participants' consent, to ensure accurate transcription and analysis of the collected data.

All interviews were transcribed using CapCut, which produced an initial automatic transcription that was then manually corrected while watching the interview videos. The interview data were analyzed using qualitative content analysis following the method proposed by Mayring [May00].

The interviewees were between 19 and 29 years old. Eight participants were students, and two were working professionals. The interview guide (see A.2) was developed to explore key themes related to user experience, behavior change, and application features.

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CHAPTER 5

Findings

5.1 User Behavior

User behavior data was collected using PostHog, a product analytics platform that tracks user interactions within the application. Event tracking was implemented incrementally during development. Not all features had tracking enabled when they were initially released to users. Some tracking implementations were added several months after the corresponding features had already been in use. Therefore, the charts presented in this section display data from the point at which tracking began for each feature, rather than from the feature's initial release date. This means that early usage patterns for some features are not captured in the data.

The ten participants who were interviewed varied significantly in their usage patterns and time spent with the app. Table 5.1 presents an overview of the participants' engagement levels, showing the number of journal entries created, the date they started using the app, and their average entries per day until their interview date. These statistics demonstrate diverse levels of engagement, with 0.81 to 3.83 entries per day and total entry counts ranging from 242 to 1960.

5.1.1 Usage Time

Figure 5.1 displays the distribution of times when individual users created entries. Data collection began when each user created their account and concluded on August 17th, 2025. The visualization reveals diverse usage patterns across users, showing variation in both entry frequency and temporal patterns. This variation reflects differences in app engagement levels as well as differences in account creation dates and adoption periods. Individual usage patterns do not exhibit a uniform temporal distribution. Some users show increased activity in the evening or late evening, while others show higher activity in the morning, with an additional spike in the evening.

Participant ID	Journal Entries	User Since	Entries/Day
1	476	2024-04-10	0.81
2	1348	2024-08-14	2.90
3	1960	2024-04-04	3.28
4	242	2025-03-20	0.98
5	618	2024-10-23	1.56
6	1222	2025-01-06	3.83
7	1607	2024-07-02	3.17
8	1168	2024-11-07	3.07
9	1113	2024-04-01	1.85
10	597	2024-03-24	0.98

Table 5.1: Overview of participant engagement levels showing total journal entries, account creation date, and average daily entry rate up to the interview date.

Figure 5.2 presents the entry frequency by hour of day aggregated across all users, with each user weighted equally to ensure users with lower activity levels are represented proportionally. The data reveals several distinct temporal patterns. Activity begins to increase at 6 AM, reaching a local maximum at 11 AM with 6.1% of entries. Following a sharp decline, entries increase again to 6.2% at 2 PM, then decline sharply at 3 PM. Entry frequency subsequently rises throughout the evening, peaking at 9 PM with 7.4%. After this peak, activity gradually decreases until midnight, where frequency drops from 5.4% to 1.4% at 1 AM. Minimal activity occurs between 2 AM and 5 AM, with entry creation resuming its upward trend at 6 AM.

5.1.2 Usage of New Features

Achievements Figure 5.3 presents a timeline visualization of achievement-related user behavior. Blue dots indicate instances when users viewed the achievements page, while green dots represent when users unlocked new achievements. The achievement system is designed to guide users in exploring different parts of the app and to reward them for using it in various ways.

Achievements can be considered successful if they motivate users to discover and use new features. While it is not directly observable whether a user engaged with a feature specifically because of viewing an achievement, behavioral patterns in the data provide indicators of this causal relationship. When a user views the achievements page and subsequently unlocks an achievement, this suggests the user may have unlocked it after discovering it in the achievement list. When multiple occurrences of viewing the achievements page followed by unlocking achievements appear in a short time frame, this pattern strongly indicates that the user was systematically going through the achievement list and unlocking achievements one after the other. The chart shows that the achievement system successfully guided users to discover and try new features.

Upon the initial release of the achievement feature, the data shows a noticeable spike in

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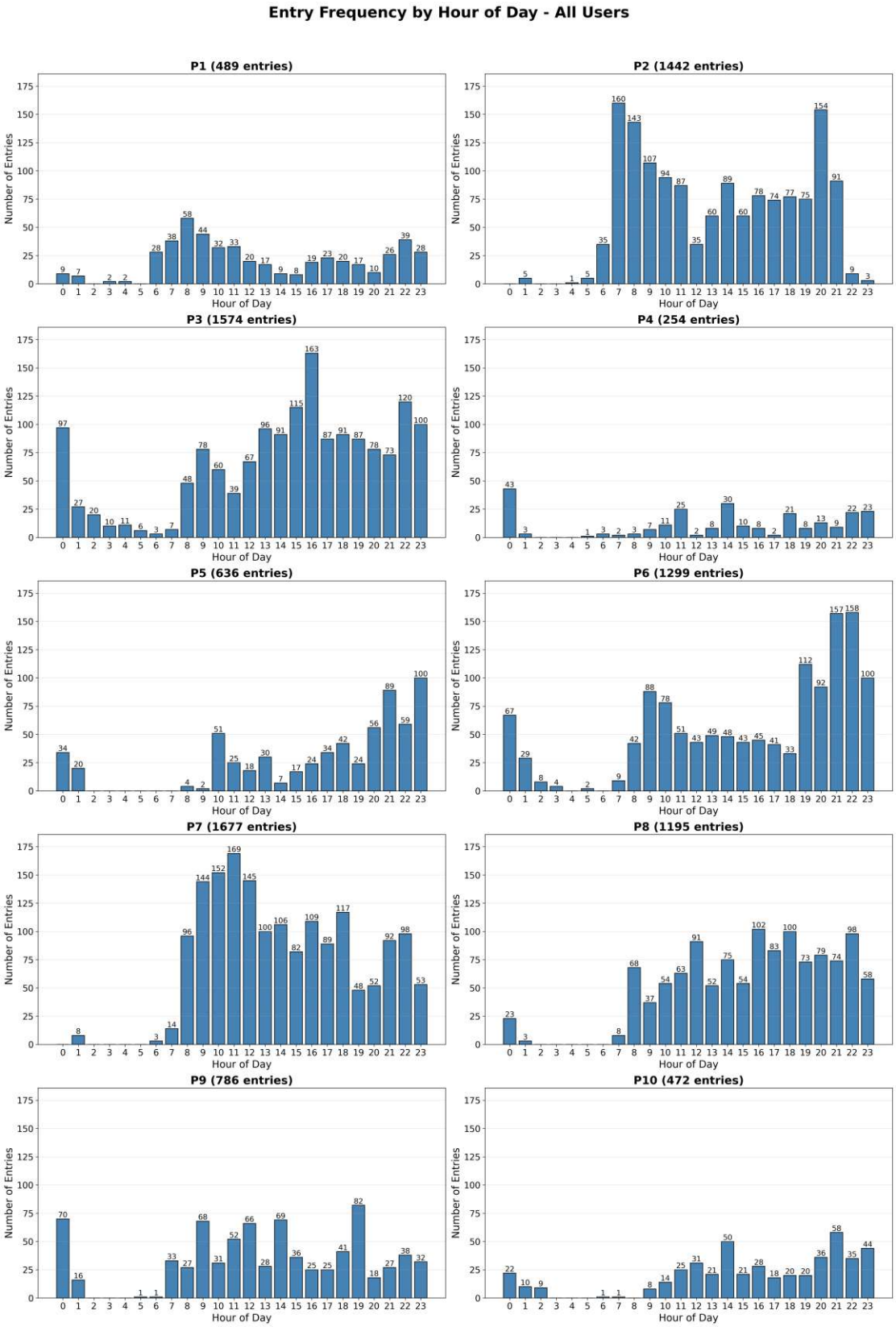

Figure 5.1: Entry Frequency by Hour of Day per User. The start is the date the user created their account, which is different for each user, and the end is August 1st, 2025.

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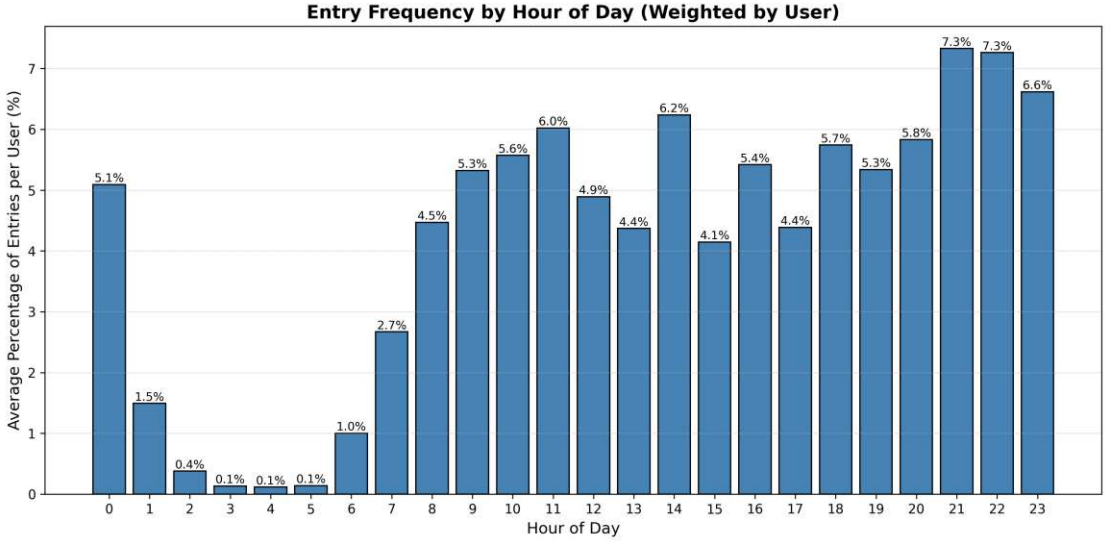

Figure 5.2: Entry Frequency by Hour of Day for all users combined.

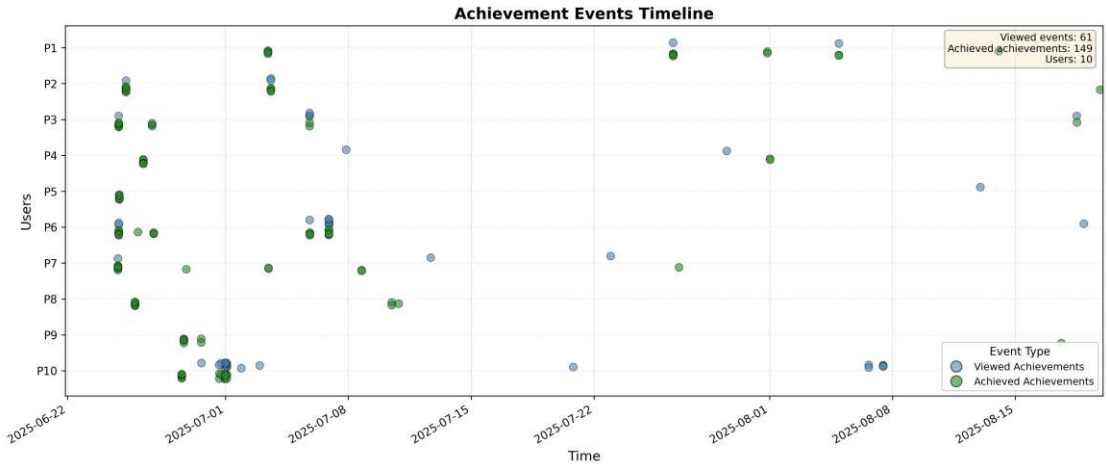

Figure 5.3: Timeline of the users viewing the achievements

users viewing the achievements page. Several users unlocked one or two achievements immediately following the feature release.

On July 1st, 2025, User 10 exhibited a pattern of viewing the achievements page, then unlocking multiple achievements within a short period. A few days later, User 2 displayed similar behavior, unlocking multiple achievements after viewing the achievement page twice. Users 3 and 6 demonstrated comparable patterns simultaneously, unlocking multiple achievements after viewing the achievement page several times. While additional users showed evidence of systematically working through the achievement list within short time frames, this behavior was most prominent for these four users.

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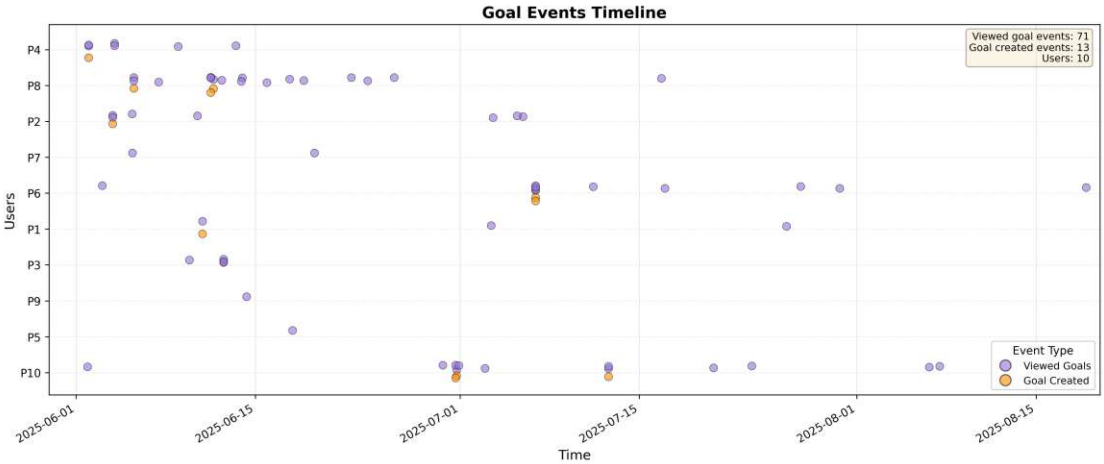

Figure 5.4: Timeline of the users viewing and creating goals. A couple of users used the goal feature right after it was released, but stopped using it shortly thereafter.

Goals To analyze the adoption of the goals feature, two events were tracked: when users viewed their goals (either in the overview displaying all goals or in tag-specific goal views) and when users created new goals. Figure 5.4 presents these events by user in corresponding colors.

When the goals feature was released in early June, adoption was not immediate across all users. Users gradually discovered the new feature, with viewing events appearing at different times for different users. Several users, including Users 4, 8, 2, and 1, created goals immediately upon discovering the feature. Some users created multiple goals in the weeks following the release. However, goal creation activity declined significantly after mid-July, with almost no new goals created beyond this point. Similarly, the frequency with which users viewed their goals went almost to zero until the interviews, indicating that both goal creation and goal monitoring had largely ceased.

Random Statistic The random statistic feature was introduced in mid-May to increase user engagement with charts and tag dashboards. Success was measured by tracking when users clicked on keywords displayed in the random statistic, which would navigate them to view and analyze their data for that keyword.

Figure 5.5 shows that initially, many users clicked on random statistic keywords. However, this behavior decreased significantly from early June onward, suggesting that the novelty of the feature primarily drove initial engagement. User 27 demonstrated exceptionally high engagement, clicking on keywords more frequently than any other user, but this activity also ceased after August. User 14 represents an exception to the general pattern, continuing to click on random statistics to navigate to keywords periodically. For most users, the random statistic feature did not serve as a meaningful way to access keyword-specific data analysis.

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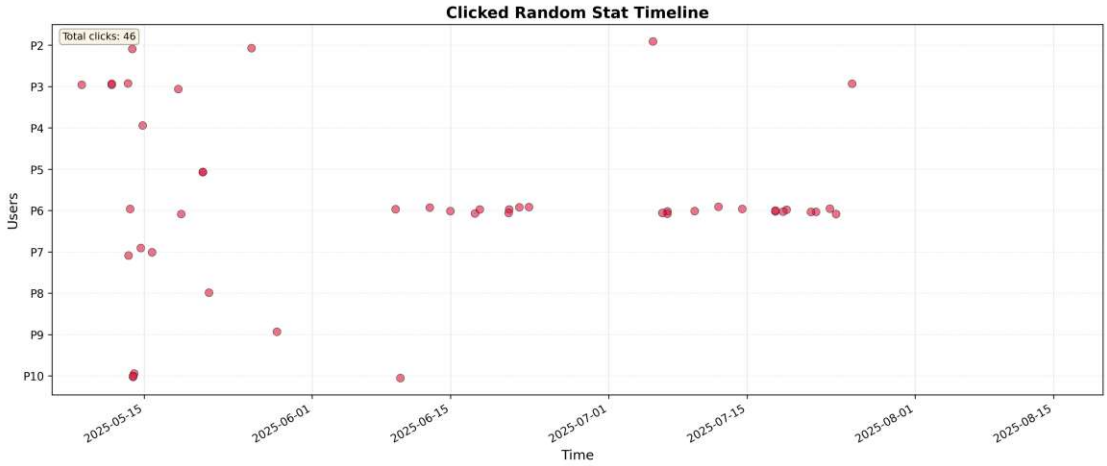

Figure 5.5: Timeline of users clicking on a keyword in the random statistic

Figure 5.6 displays which categories of random statistics users clicked on. The most frequently clicked category was "*most used contact with tag,*" dominated by User 27's engagement with this statistic type. The second-most-clicked category was "*most used tag with contact*", with 10 total clicks across all users. The "*number of days with tag*" category showed the highest diversity of users who clicked, with engagement distributed across multiple users. This pattern is consistent with the expectation that all users track tags, while not all users track people, explaining why categories involving people (contacts) showed fewer unique users. The category with the lowest engagement was "*longest streak*", which received only three total clicks during the observation period ending August 17th, 2025.

5.1.3 What Charts Users Look At

Network Chart The network graph event was tracked when users opened the network visualization, displaying their tracked people as nodes with connections between them. Figure 5.7 reveals that users who track people demonstrate high initial engagement with the network graph when they first begin using the app. This viewing frequency typically decreases after approximately 1 to 1.5 months, then stabilizes at a consistent rate of several views per month. User 10's usage pattern shows high network graph view counts even though the user reported having stopped tracking people. The user might have been curious to see what the network graph looked like with the previously tracked people. User 5 did not track people, so it shows no network graph activity.

All Tags The tag overview page lets users quickly view heat maps for all their tracked tags. Figure 5.8 shows heavy engagement on a consistent level by User 2 throughout the observation period. Users who created their accounts during the observation period, including Users 4, 6, 7, and 8, demonstrated high engagement with this view during their

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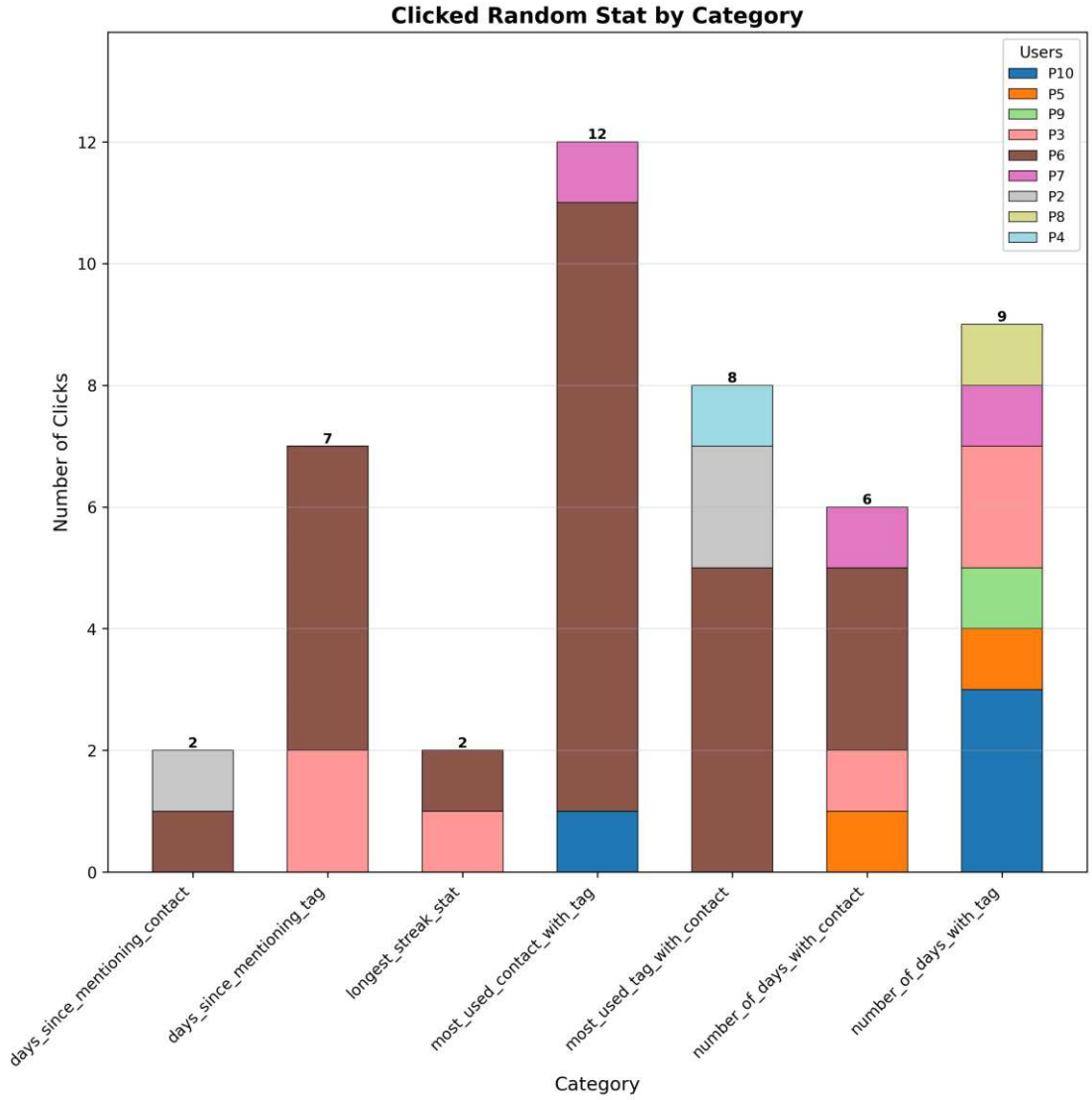

Figure 5.6: Stacked bar chart showing which categories of random statistics users clicked on.

first month of usage, exhibiting a pattern similar to the network graph viewing behavior. After this initial period, engagement stabilized at a consistent level for these users. User 5 shows significantly lower engagement with the tag overview page than other users. This difference is likely attributable to User 5 tracking considerably fewer tags than other users. When examining individual tag dashboard views, User 5 demonstrates frequent usage, suggesting this user finds it more convenient to navigate directly to specific tag pages rather than using the overview page.

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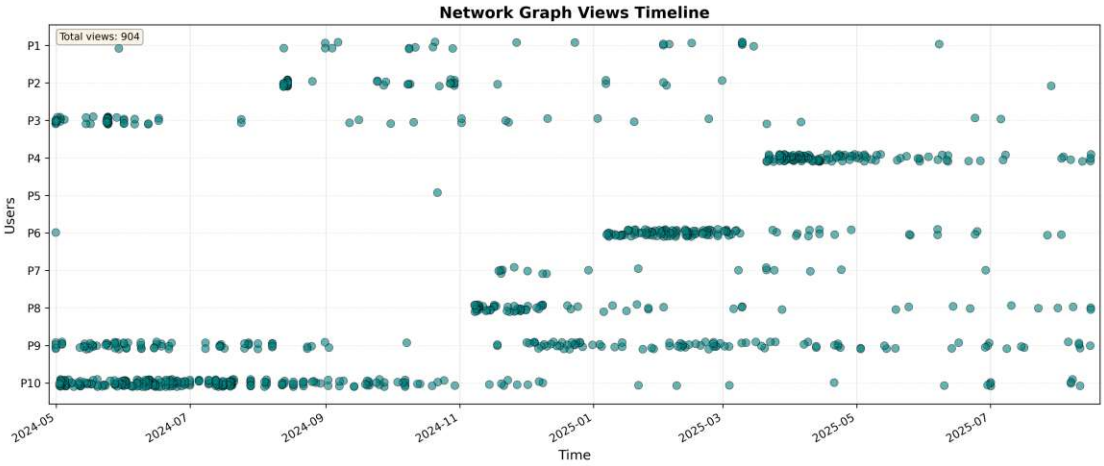

Figure 5.7: Timeline of users viewing the network graph.

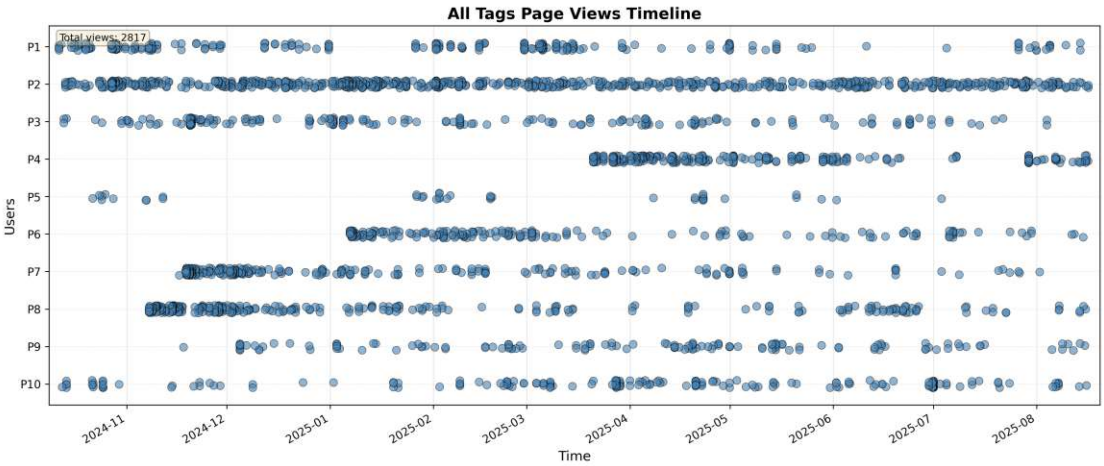

Figure 5.8: Timeline of users viewing the tag overview page.

Tag Dashboard Figure 5.9 shows viewing behavior for individual tag dashboard pages that is very similar to the tag overview page pattern. Users visit tag dashboards regularly, though engagement can dip during occasional periods. New users, including Users 6, 7, and 8, showed increased tag dashboard views after account registration, though this pattern is less pronounced than the initial engagement spike observed on the tag overview page. Consistent usage across users indicates that viewing individual tag dashboards is a core activity in the app for analyzing specific tracked behaviors.

Person Dashboard The person dashboard data presents a different measurement scenario because view tracking was not implemented in the Flask backend, so the frontend analytics tool is used instead. Since browser ad blockers can block frontend analytics, the data provides a lower bound for actual views rather than complete coverage. This

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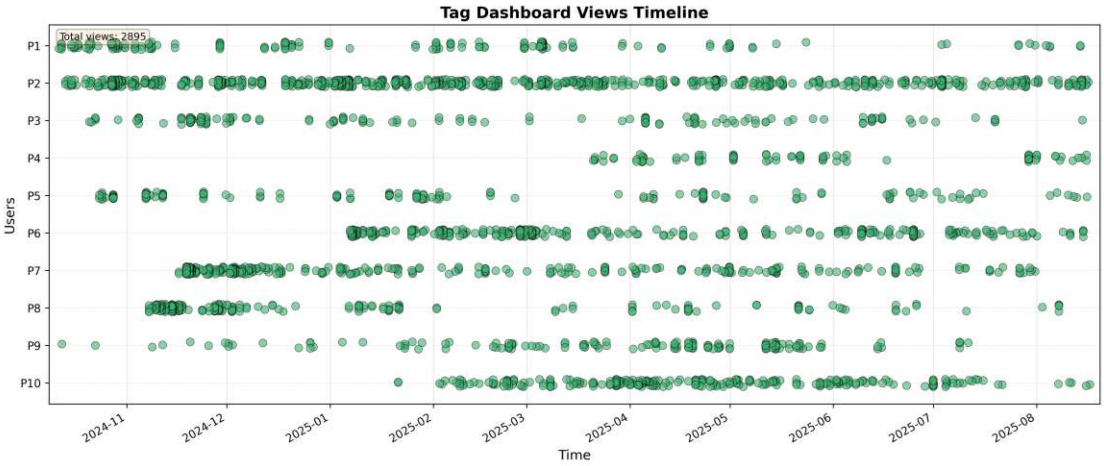

Figure 5.9: Timeline of users viewing individual tag dashboards.

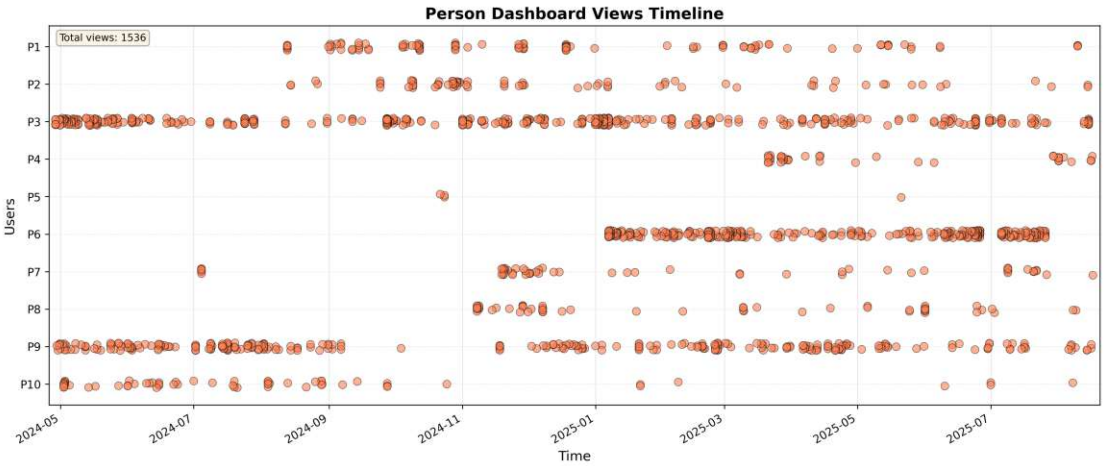

Figure 5.10: Timeline of users viewing person dashboards.

limitation is evident in User 6's behavior, which shows a sharp decline in mid-July despite this user continuing to track people, suggesting the adoption of an ad blocker for subsequent app usage. Users who track people regularly view their people's dashboards. User 2, who did not track people until September 2025, still viewed person dashboards, likely to access tags that were shared with them by their friends. User 10 stopped tracking people during the observation period and correspondingly ceased visiting person dashboards. User 5 does not track people at all, resulting in almost no engagement with person dashboards.

People Statistics The people statistics page shows varied relevance across users, even among those who track people. Figure 5.11 reveals that User 1 rarely viewed the statistics page despite tracking people. User 8 also stopped visiting the people statistics page after

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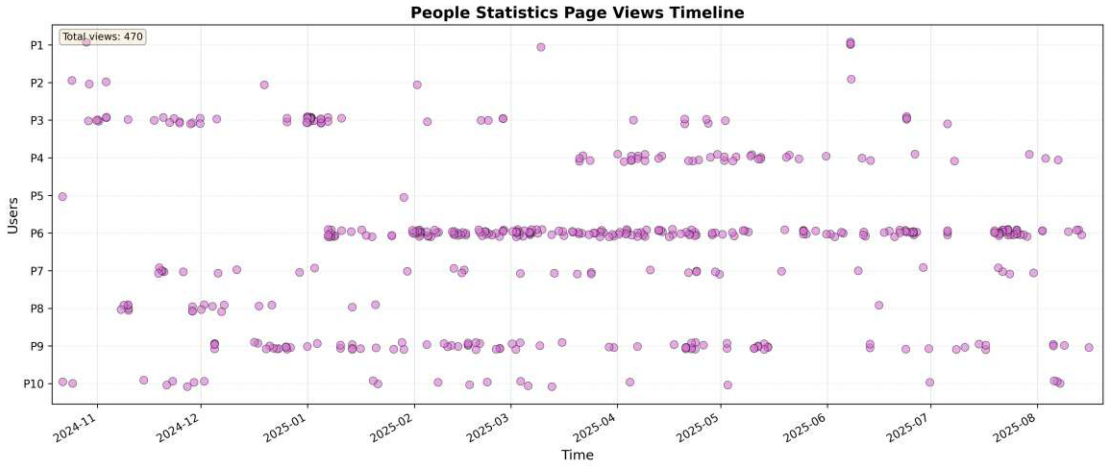

Figure 5.11: Timeline of users viewing the people statistics page.

an initial period of engagement. In contrast, some users, including User 9 and User 6, visit the page very frequently, demonstrating a strong interest in analyzing their people data. Users who do not track people correspondingly do not visit the page, as this feature requires tracked people data to provide meaningful insights.

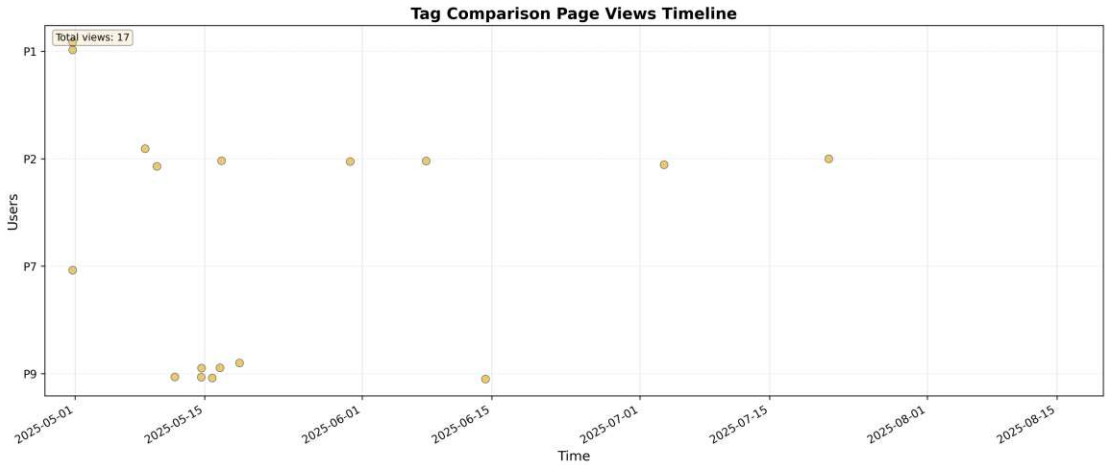

Figure 5.12: Timeline of users viewing tag comparison pages.

Tag Comparison User interviews revealed that some users did not know that they could compare tags or how to access the comparison functionality. Figure 5.12 shows that until the interviews started, only two users had viewed the comparison page more than five times.

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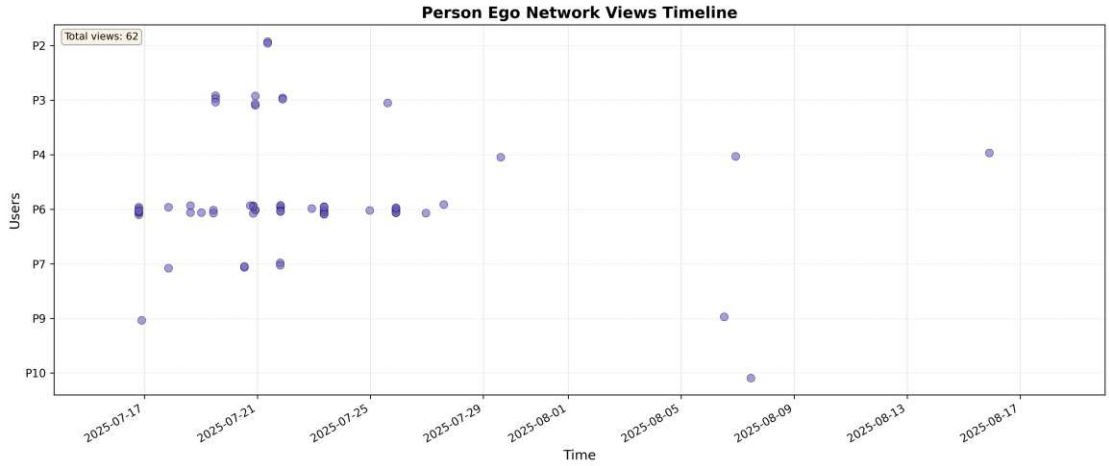

Figure 5.13: Timeline of users viewing a person's ego network.

Ego Network Graph The person's ego network feature is relevant only for users who track people. Figure 5.13 shows that some users, such as User 6, initially showed interest in the graph but stopped viewing it after approximately one month. Almost no users viewed this graph regularly throughout the observation period. After approximately 1.5 months, viewing activity declined substantially, with very few users accessing the feature. This pattern indicates insufficient perceived value for the ego network visualization, despite the feature being easily accessible on pageload without scrolling via a single button click from a person's profile page.

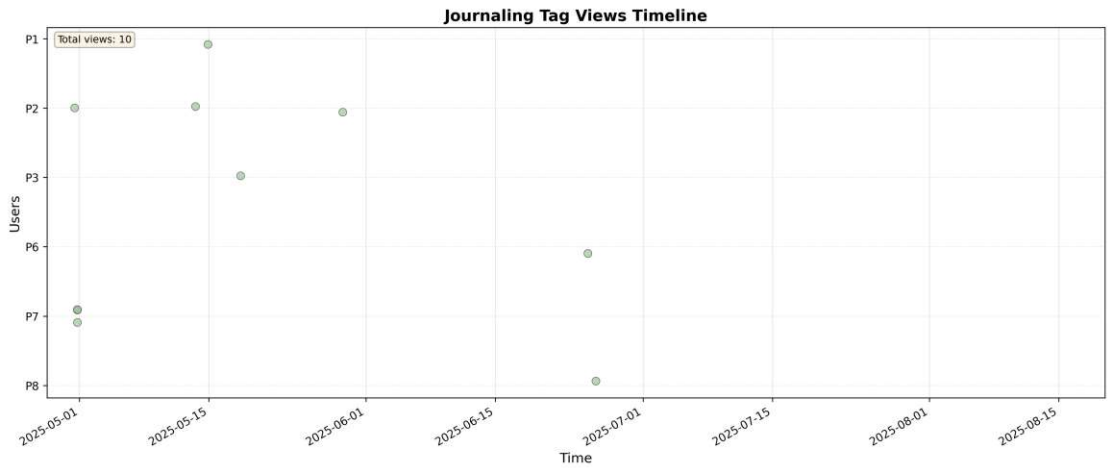

Figure 5.14: Timeline of users viewing the journaling tag.

Journaling Tag The journaling tag showed almost no repeated engagement throughout the observation period. Figure 5.14 reveals that many users viewed the tag only once and never returned to it. This pattern is likely attributable to the journaling tag's position

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in the tag overview, which becomes less prominent as users create and use more tags. As users expand their tag collections, the probability of encountering the journaling tag diminishes over time. The lack of a dedicated location for the journaling tag, beyond its position within the tag overview, contributes to its declining discoverability and subsequent lack of engagement.

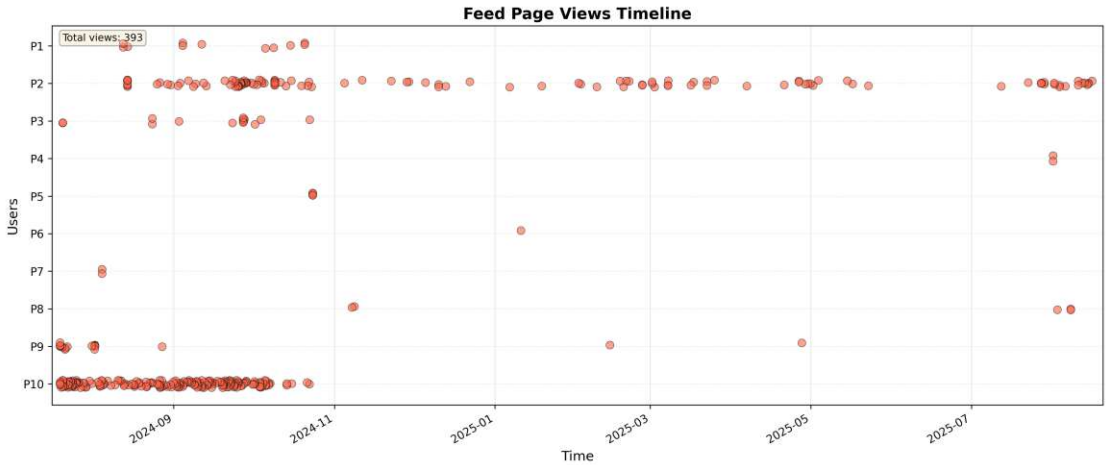

Figure 5.15: Timeline of users viewing the feed.

Feed The feed page presents the same measurement limitation as the person dashboard, with no backend event tracking implemented. The data in Figure 5.15 represents a lower bound derived from the frontend analytics tool, which users can block. An additional complexity affects data interpretation: before October 6th, 2025, the view also included the option to search for other users, meaning views of the page before that date might indicate users' interest in searching for other users rather than viewing their feed. The chart shows continuous engagement from P2, but other users visited the page only occasionally. Engagement increased slightly following the user interviews, though not substantially.

5.2 Expert Interviews

Four experts from Austrian universities with backgrounds in behavior change, persuasive technology, and human-computer interaction were interviewed to provide feedback on the application's design and features. Table 5.2 provides an overview of each expert's area of expertise.

The analysis of the expert interviews revealed several key insights regarding the app's design, usability, and potential for behavior change. The topics covered can be divided into the following themes:

- positive effects of the social aspect of the app

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Table 5.2: Overview of expert interviews

Expert	Area of Expertise	duration (minutes)	Key Points of Critique and Recommendation
Exp1	Data visualization, sports technologies, well-being technologies, user perception studies, UI/UX Design	95	Missing reference points for interpretation; visualizations too low-level, need higher abstraction
Exp2	Human-Computer Interaction, attention management, productivity systems, digital self-control tools, intelligent user interfaces	70	Insufficient praise for habit completion; random statistic should be prominent, and first thing users see
Exp3	Educational technology, usability evaluation, cognitive science, cognitive biases, Cognitive Flow Theory, digital game-based learning	78	Technical terminology (cosine similarity) may deter users; ego network visually appealing but has low practical utility due to cognitive overload; bubble chart too complex
Exp4[a]	Visual analytics, visualization, similarity search, 3D object retrieval, image processing, archaeological applications of visualization	70	Use sparsification in the network graphs; add labels to the graph; render graph with canvas for better performance; remove singletons from the graph

[a]Exp4 provided valuable feedback and insights on the technical implementation of network visualizations. While highly appreciated, this input was not directly applicable to answering the research questions and is therefore not discussed in the subsequent analysis.

- Too complicated/ too much/ too many features
- Not enough praise
- Missing information or context

5.2.1 Positive Effects of the Social Aspect of the App

Two of the interviewed experts noted that the format of tracking social interactions alongside activities is unique and something they have not seen before. They assume that it has positive implications in different ways.

Help Put Events in Context When trying to remember past events, people often struggle to recall specific details. Social interactions can serve as powerful memory cues, helping users remember more clearly and contextualize their experiences.

Exp1 elaborated on this benefit by explaining that users often struggle with raw data alone:

> *What is kind of good about it is that you get an extra contextual layer to it because what we often see in these kinds of applications is that if users just*

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> *get raw data, so let's say they just get raw step data, for example. Users then often do not really know what to do with that because they cannot then really revert back to what happened exactly on that particular day that made me move so much more. [...] We also see that users find it way easier to also remember things if they can contextualize it to something else or if there's something that gives them the extra cue. I get the feeling that probably if you can show that you do certain activities together with someone else and maybe on certain points during the day or something or maybe during certain months you may be then be able to remember better kind of what you did that particular month or week.*

This insight highlights how the app's social tracking feature provides contextual anchors that help users make sense of their activity data and improve recall of past experiences.

Helps Keep Social Connection Exp2 particularly emphasized the value of the social connection feature, noting that it represents a departure from typical goal-tracking applications. They expressed enthusiasm for the concept of tracking activities in connection with others, stating that they "*really loved*" this aspect of the app.

Unlike applications such as Duolingo that may include friend features, the app's approach acknowledges the collaborative nature of activities - for example, "*I did it with Alex, I am going for a run with Alex 6 times per month.*" Exp2 highlighted that this feature addresses a broader goal of maintaining social connections in an era where people are "*too much addicted, too much concentrating on ourselves and on our goals, forgetting that bonding is one of the biggest source of happiness.*"

Social Comparison and Motivation Exp1 also discussed how the app's social features could influence motivation through social comparison, while cautioning about potential negative effects:

> *What I think this application could work quite well for is people who actually feel more motivated when they can show like I do these things together with certain people. Because it's also kind of known in sports and psychology in general that for some people this works very well, and for others it doesn't, and it also has to do a lot with how similar the other person is. For example, let's say that person does a similar amount of sports to you, then you're more likely to compare yourself to that person because you feel like it's more even in that sense. Then people get more motivation from that. But let's say you see in this application that some people do way more than you, it can actually work the other way, so then it might actually be demotivating.*

This observation highlights the dual nature of social comparison in activity tracking. While seeing activities done with similar others can enhance motivation, significant

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disparities in activity levels could discourage users. This suggests the need for careful consideration of how social comparison features are designed and presented to users.

5.2.2 Too Complicated

Exp2 found the app overwhelming due to the simultaneous presentation of numerous features, including habit and social interaction tracking and goals. The features are scattered across different parts of the interface, making it unclear what users should prioritize to achieve satisfaction. This created uncertainty about where to direct attention when everything is presented at once. The resulting learning curve was compared to the paradox of choice, where presenting 27 varieties of jam leads to decision paralysis - users may postpone engagement with the app until they have time to understand all features, risking abandonment.

To address this complexity, Exp2 suggested understanding user values before presenting features, then grouping functionality based on individual priorities such as maintaining a healthy lifestyle, staying connected with close ones, or competing with others. This could be implemented through an initial settings or value assessment to determine what matters most to each user. Additionally, features could be introduced more gradually rather than all at once.

5.2.3 Missing Information or Context

Missing Reference Point for Interpretation One expert noted that while the app's visualizations provide interesting information about activities with others, they may lack a clear reference point for interpreting progress or change. Without a baseline, users might find it challenging to determine whether they are improving. The expert emphasized that comparison frames are often essential in sports or social tracking applications to help users evaluate their own behavior. Exp1 highlighted comparison charts as a particularly effective visualization for addressing this need, as they can clearly show where a user stands relative to others, making it immediately apparent whether they are performing better or worse than their peers. They also point out that goals can serve as another reference point.

Random Statistic and Positive Reinforcement Both experts identified a lack of positive feedback when users add journal entries. The app implements a variable reward system that displays a random statistic after each entry submission, with a 12-hour cooldown. However, both experts questioned the effectiveness of this mechanism as a reward.

Exp2 found that the app offers limited rewards that help users feel positive about their engagement. While they noted the presence of achievement notifications - such as "*oh you registered you made this first step here you go*" - they expressed confusion about their purpose and accumulation: "*I did not understand where it's accumulated, what it replicates for me.*" More critically, they pointed out that when users perform a positive

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action, such as completing a habit, "*there is no praise. You get nothing. Now here it is actually time to praise the action of the user because the user made the habit.*"

The expert suggests that the random statistic should not be updated on entry creation, but rather upon opening the app. They criticized the attempt to artificially enhance this basic interaction, stating "*you don't need to do artificial yohoo, you added it, congratulations, you know how to use your phone.*" She also agrees that after opening the app, creating an entry will happen automatically. However, they emphasized that this type of simple incremental feedback is currently missing from the app.

In contrast, the expert argued that the random statistic feature deserves prominent display because it reveals patterns users cannot perceive through daily use. They explained: "*this is something that you cannot see day by day this is your reward right put in it everyday and then it helps you to see bigger picture oh I actually working already for 26 days this is something that you don't see on everyday basis that's why it should be bigger because this is something that your app gives that you cannot see.*" They noted that the statistic appears "*so small compared to everything*" and questioned "*why it's so small it's a great achievement.*" Using the example where the app displayed "*26 days you're writing your master thesis*", they argued that this represents "*a great achievement*" and should be prominently featured. They suggested that "*this should be the first thing people see when they open the app and it should be something encouraging,*" proposing that the random statistic should occupy a much larger space, potentially "*the whole screen should be only for this with this with this juhuu and sound.*" The expert further argued that the journal input page, which they characterized as "*just a kind of archive,*" should not be the main page because users "*cannot track of this whether I was good or not.*" Instead, they suggested that analysis of the journal data - "*what the app is doing with this archive*" - should be more prominently displayed.

Exp1 raised a different concern about the random statistic feature, questioning both its nature as a reward and its relevance to user input. They asked, "*what do users exactly see as a reward, because just kind of seeing a random fact is that really a reward though?*" Additionally, they noted that the displayed statistics are not related to the data the user just entered, suggesting a disconnect between the user's action and the supposed reward.

5.2.4 Charts

Heatmap Exp2 identified the heatmap visualization as the most attractive feature of the app from a personal perspective. However, they noted a significant discoverability problem, reporting difficulty locating the visualization even immediately after viewing a tutorial demonstration. This feedback highlights a tension between the visualization's inherent appeal and its accessibility within the app's interface.

Cosine Similarity Exp3 provided feedback on the cosine similarity visualization, noting that its comprehensibility depends on the target audience. While acknowledging that some users with technical or mathematical backgrounds might understand it, they

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expressed concern that approximately half of potential users could be deterred by the technical presentation. They explained: "*Ich befürchte, dass die Hälfte aller Personen, die ich kenne, dadurch abgeschreckt ist und das nachher nicht mehr nutzen wird. Weil die lesen sich das durch und denken sich was, wozu, warum?*"

The expert emphasized that the primary consideration should be making the purpose immediately clear to users: "*wichtig ist immer, dass das Ziel unmittelbar verständlich ist für die Personen, was kann ich mit dieser Information machen, was ist die Fragestellung, die ich damit beantworten kann.*" They argued that users need to understand what they can learn from the visualization rather than the underlying technical mechanism, stating "*es geht darum, was mach ich damit oder was kann ich damit herausfinden.*"

They proposed leading with a one-sentence explanation of what insights users can derive from the chart, while relegating the technical term "*Cosine Similarity*" and mathematical details to supplementary information such as notes or links for users who are interested in the methodology: "*dass man das in vielleicht in einem Satz irgendwie erklärt, was man rauslesen kann und ob das Cosine Similarity ist, das kann man ja vielleicht mit den Anmerkungen noch zusätzlich erwähnen und man kann's verlinken oder so wenn sich Leute vielleicht dafür interessieren.*"

Ego Network Exp3 provided extensive feedback on the ego network visualization, praising its aesthetic quality while raising concerns about its usability. They described it as looking "*wunderschön*" and "*wie ein Kunstwerk,*" particularly appreciating its interactive nature. However, they identified a fundamental tension between the visualization's comprehensive data presentation and users' ability to extract actionable insights.

Their primary concern centered on cognitive overload. With 40 nodes and 100 edges displayed simultaneously, Exp3 questioned whether users could effectively answer specific questions despite the wealth of information presented. They explained: "*du hast 100 Kanten, du hast 40 Knoten und du hast ja eigentlich nur bestimmte oder Reihe von Fragen die du damit rauslesen möchtest.*" They observed that while the visualization contains extensive information that "*könntest dich stundenlang damit beschäftigen und hunderte Fragen beantworten,*" its practical utility remains limited because users must click each element to identify it, and adding labels would make the visualization overwhelming. Exp3 summarized this paradox: "*du bietest Übersicht an und das schaut wunderschön aus und so aber der nutzen den man draus ziehen kann ist dann trotzdem erschreckend gering.*"

To address these usability issues, Exp3 suggested implementing filter mechanisms to help users focus on specific questions. They proposed filters could highlight particularly relevant information, such as nodes that are closely connected or, conversely, nodes that are distant but might benefit from stronger connections. They suggested visual techniques like blurring irrelevant elements while highlighting those pertinent to the user's current question: "*man könnte zum Beispiel mit Ausblurren arbeiten, oder? Also, dass du alles was für die Frage, welche Knoten hängen besonders eng miteinander zusammen, könnte man alles andere ausblurren was rundherum ist.*"

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Bubble Chart Both Exp3 and Exp1 expressed concerns about the bubble chart's complexity and usability for typical users. Exp1 questioned whether users without technical backgrounds would understand the visualization's purpose, stating "*was macht man als Person damit?*" While acknowledging familiarity with such visualizations from statistical analysis, they doubted their relevance for users without backgrounds in informatics or statistical evaluation: "*die Frage ist halt jetzt, wenn man sie nur in einen bestimmten Nutzer oder Nutzerinnen reinversetzt, die vielleicht kein, was weiß ich, Informatikstudium hinter sich hat.*" When asked about the chart's potential, Exp3 responded that they did not see value "*in der derzeitigen Form weil ich muss wirklich sehr sehr sehr viel machen um mir irgendwelche relevanten Antworten ausgeben zu lassen.*"

Exp3 identified significant interaction challenges, particularly on mobile devices. With 40 bubbles displayed simultaneously, they noted that bubbles overlap, making them difficult to select accurately: "*die Bubbles sind übereinander gelagert, ich muss mit meinen Wurstfingern da irgendwie keine Ahnung, irgendwelche treffen.*" Since bubbles lack labels, users must click on each one individually to identify it, but this creates a memory problem: "*Nach spätestens 5 Bubble weiß ich nicht mehr, was ich am Anfang gemacht habt.*" They characterized the overall presentation of 40 people in a single view as "*unübersichtlich.*"

To address these issues, Exp3 suggested reducing the number of simultaneously displayed bubbles to approximately 10, arguing "*ich glaube, dass man mit, ich weiß nicht, Hausnummer, keine Ahnung, maximal 10 Bubbles, dass man dann was Sinnvolles damit machen könnt.*" They proposed implementing user-controlled filtering mechanisms that would allow people to activate or deactivate specific individuals, or offering preset filters such as top 10 or bottom 10 based on interaction frequency. They also suggested showing people with whom the user had fewer interactions as an alternative view.

Exp1 raised an additional concern about the chart's use of a log scale, noting that "*the majority of users are not really going to understand what a log scale really means.*"

Metric Abstraction Levels Exp1 provided a framework for understanding the app's visualizations in terms of metric abstraction levels, distinguishing between low-level, mid-level, and high-level metrics. They explained that simple visualizations, like line graphs, represent low-level metrics that users generally understand well. In contrast, higher-level abstractions are substantially more complex for users to comprehend, even though they typically convey more concrete behavioral insights.

Exp1 characterized the app's visualizations as predominantly low-level, noting that even the cosine similarity feature, while representing a mid-level metric positioned one abstraction layer above basic numerical data, does not reach the highest levels of metric abstraction they have observed in other applications. They observed that, despite collecting interesting data, the app presents information at a relatively raw level, without higher-level abstraction layers. Exp1 suggested that incorporating reference points could provide an accessible path toward adding meaningful abstraction while maintaining user comprehension.

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5.2.5 Notifications as External Triggers

The experts held diverging opinions on the role and effectiveness of notifications in maintaining user engagement with *Habits Network*.

Exp2 expressed a strong preference against notifications from a personal perspective, stating that "*we just have too many*" notifications in general. However, from a developer's perspective, they acknowledged the value of notifications as an engagement mechanism: "*As a potential developer, I would always try to introduce notifications, and then it's already up to the user to turn them off if they find them too much.*"

Exp2 particularly emphasized the potential of notifications to implement variable rewards, noting that "*this is also the opportunity for the random reward right.*" They argued that notifications need not be limited to reminders but could serve multiple functions: "*a notification doesn't always have to be: 'oh don't forget to put some habit', just saying: 'the last 2 weeks you went for running 5 times do you want to continue doing?'*" nudges users toward the habit and encourages them to document their activities. They cited Duolingo's approach of introducing random notifications where "*you never know what it's gonna be. Sometimes it's encouraging, sometimes it's a reminder, sometimes it's telling about that your friend has done the German class and you haven't.*"

In contrast, Exp1 questioned the fundamental effectiveness of notifications as a driver of user engagement. Based on their research experience, they stated: "*I never feel like notifications are the determining factor of whether or not it keeps a user's attention to that. It's always the visualisation.*" They suggested that the effectiveness of notifications may depend on the type of application, noting that "*for physical activity apps, then probably notifications are not super motivating because you already have a set routine, your daily life is already going to influence a lot what your behavior is gonna look like so a notification is not going to change that.*" Instead, Exp1 argued that data visualizations hold greater motivational potential: "*I feel like data visualizations can - if they tell a user clearly, like if you can do this very small thing, then you can, for instance, improve - be much more motivating compared to a notification.*"

5.3 System Usability Scale

In total, 10 participants answered the System Usability Scale (SUS) questionnaire (see A.3). The results yielded a mean SUS score of 77.00 (median: 78.75, SD: 9.47), indicating good usability according to standard SUS interpretation guidelines, where scores above 68 are considered above average [BKM09]. To provide further context, Figure 5.17 displays boxplots for each SUS question, illustrating the distribution of responses and a boxplot of the overall SUS score for each participant. The scores ranged from 57.50 to 90.00, demonstrating that while most users found the system highly usable, there was some variation in perceived usability. The relatively low standard deviation of 9.47 suggests that users had generally consistent experiences with the application.

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Although three participants used only tags in their diary entries and the remaining seven used both tags and people, no clear pattern emerged between these usage styles and the SUS scores.

The time participants had been using the app showed a moderate but not statistically significant ($r = -0.3191$, $p = 0.388$) correlation with their SUS scores (see Figure 5.16 for a bubble chart of the normalized sus scores per participant and days of usage), indicating that shorter usage durations might be associated with higher usability perceptions. This could be due to changes in the app over the observation period that increased perceived usability, but this trend is not strong enough to be conclusive. This also does not suggest that the application's usability was consistently experienced across different user types and levels of engagement, since there is insufficient evidence. For the per-question analysis, Figure 5.18 shows the normalized answers to each SUS question plotted against users' sign-up dates. The red line indicates a linear regression fit to the data points. It seems like there is a slight upward trend in scores for questions 2, 4, 6, and 10, but again, this is not statistically significant. For more meaningful results, a larger sample size would be necessary to draw more definitive conclusions about how different usage patterns impact perceived usability.

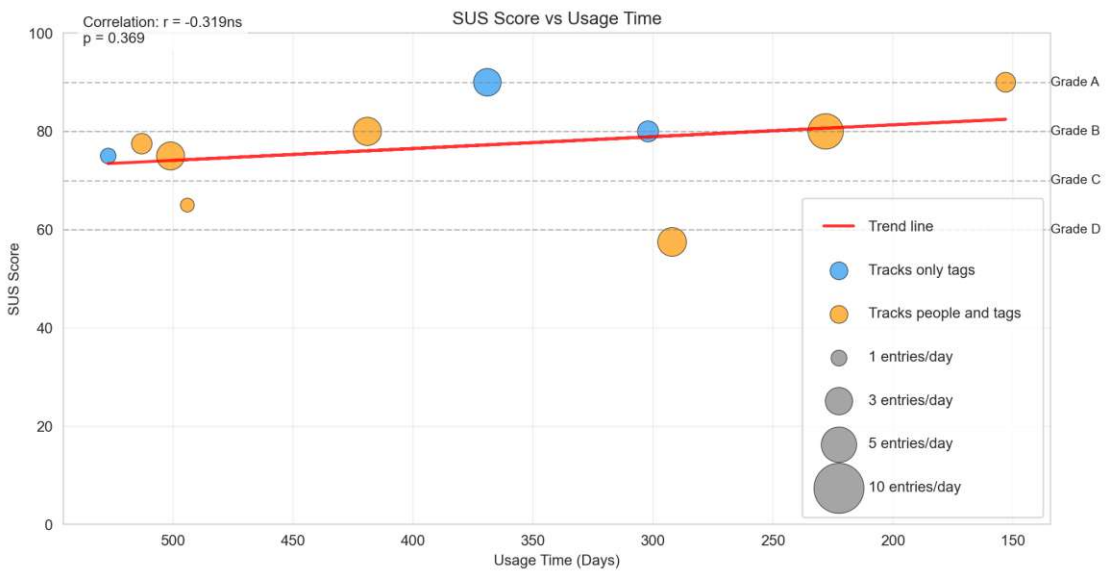

Figure 5.16: Bubble chart with the SUS score as the y axis and the number of days used until the interview on the x axis. The color encodes whether the user tracks only tags or both tags and people. The size of the node encodes the number of entries the user made per day.

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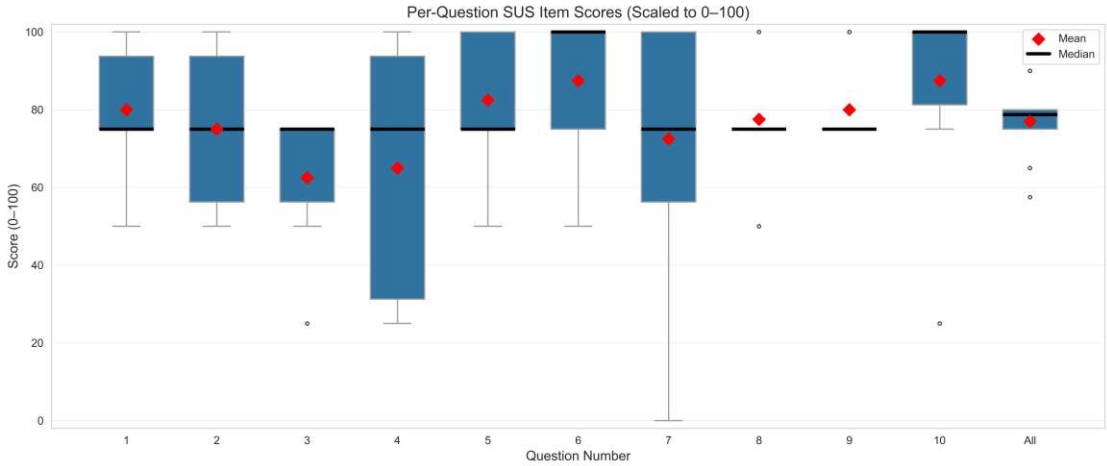

Figure 5.17: Boxplots for the normalized answers to each SUS question and the overall SUS score on the right

5.4 User Interviews

Ten participants were interviewed to gather insights about their experiences with the application. Table 5.3 provides an overview of participant demographics and key usage characteristics. This section is divided into the three research questions. The Topics covered in the interview are presented in the respective subsections.

Table 5.3: Overview of user interview participants

ID	Sex	Age	Occupation	Usage before interview (months)	SUS	Usage
P1	m	26	student	16.2	65.0	People, Tags
P2	m	26	project manager	12.1	90.0	Tags only
P3	m	24	student	16.5	75.0	People, Tags
P4	m	23	student	5.0	90.0	People, Tags
P5	m	25	student	9.9	80.0	People, Tags
P6	m	25	student	7.5	80.0	People, Tags
P7	m	25	student	13.8	80.0	People, Tags
P8	m	25	student	9.6	57.5	People, Tags
P9	f	19	student	16.9	77.5	People, Tags
P10	f	29	medical assistant	17.3	75.0	Tags only

5.4.1 RQ1

What motivates users to use the app?

To understand user motivations, the interview began by asking participants why they use the app. Habit tracking for awareness emerged as a primary motivation for many

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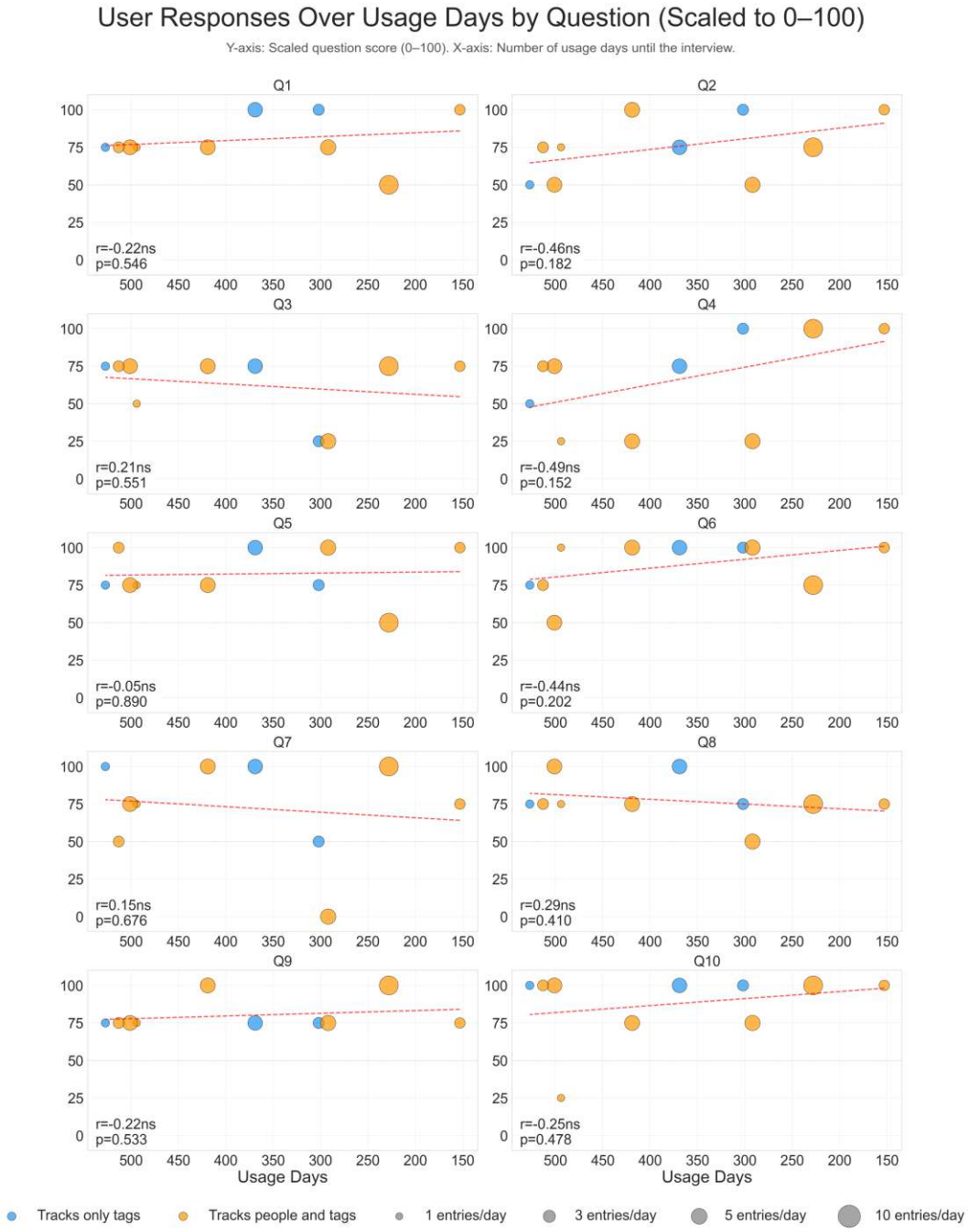

Figure 5.18: Normalized question answers with sign-up date of users

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participants (P1, P2, P7, P8), who wanted to become conscious of both positive and negative behaviors, with P2 noting they successfully quit nicotine by seeing their tracked data. Social tracking motivated several participants (P3, P4, P6) to understand meeting patterns, frequencies, and activities shared with friends. The app's visualizations and statistics attracted multiple participants (P4, P5, P6, P8, P9), with some (P5, P9) specifically valuing how visible progress motivates continued activity even when physical results aren't immediately apparent. Other motivations included using the app as a diary (P8) and for specific tracking needs, such as medication and alcohol consumption (P10).

Tracking People Increases the Motivation to Track Certain Tags

One user reported that they would not stop tracking certain tags because that would mean losing data about the people they track. The user particularly mentions that using a tag with a person gives them the information on how that person influences them.

Adding an Entry Right After an Activity With a Positive Influence

Multiple participants reported experiencing positive emotions when entering data after completing activities they felt good about. P1 explained that they tend to enter positive activities, like a workout, immediately because it is fulfilling to do so, while they often enter negative tags like alcohol later in the evening or the next day: "*ein Training, das für mich positiv assoziiert ist, trage ich unmittelbar danach oft ein, weil es irgendwo auch erfüllend ist, das einzutragen. Hingegen einen negativen Tag wie Alkohol trage ich oft erst am Abend oder am Tag danach eigentlich ein.*" They also explained that when they enter positive activities, it reinforces them and increases the likelihood of doing them again the next day because they experience the positive feeling not only during the activity but also when entering it. Similarly, P9 mentioned that they look forward to creating entries when they have done something they are proud of, such as going for a run, noting that they enter them immediately because they are excited about having completed the activity. P10 described the immediate satisfaction of logging positive activities: "*ich hab mich schon gefreut wenn ich was gemacht hab und dann hab ich's eingetragen so wie Sport weil dann hab ich mir gedacht cool, ich hab das gemacht und jetzt kann ich das eintragen und dann ist dieser blaue Punkt bei mir halt in diesem Raster und ja das gibt so einen kurzen Dopaminschub.*" P6 explained that they often immediately enter specific activities, such as planks, after completing them because of the sense of accomplishment, wanting to enter them right away. However, they noted that more general entries about whom they saw doing what are always entered in the evening.

While 4 participants experienced positive feelings during data entry, others reported positive emotions when viewing their logged activities afterwards. P3 explained that while they do not feel compelled to enter sports activities immediately, they feel good when looking back and seeing that they have been active in recent days. P2 takes a more strategic approach, actively creating entries immediately after positive activities because they read in *Atomic Habits* that this timing can enhance the behavior's positive effect, though they note this practice is not essential to their routine.

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Evening Routine

One participant described integrating the app into their evening routine as part of their daily reflection. They explained that they initially tried to make entries immediately after each activity, but found this approach impractical because they did not always have their phone with them. By switching to an evening routine in which they review their day and make all entries at once, they found they used their phone less throughout the day while maintaining a better overview of their activities.

User Primary Focus Is on Inputting Data Rather Than Analysis

Five participants explicitly reported that they primarily open the app to create entries rather than to analyze their data. P1 explicitly stated that they use the app primarily to add entries, and that they use it more for tracking than for actually analyzing data. P1 further explained that they have to actively think about viewing charts for this to happen, and that the focus lies on entering data. P9 confirmed they would not open the app only to view statistics, stating they must have come in to enter something first. Similarly, P5 reported that they open the app always intending to add a new entry and might then look at statistics, noting they look at tags almost exclusively only when entering them. P4 explained that they look at dashboards after events or when they have time and have just entered something, particularly after gaps in usage or after frequent recent entries. P2 described their routine of entering data in the evening and then spending additional time viewing statistics, specifically mentioning they enter their tags first and then spend two to three minutes analyzing tags they are currently interested in.

Users Need Something to Track

Three participants reported a decrease in motivation to use the app during periods when they had little to track. P10 explained that during busy work periods, they found themselves in a mode where there was nothing meaningful to enter, which led to reduced app usage. P1 described feeling uncomfortable when days went by empty, as it gave the impression that something had been forgotten. Attempting to resolve this by entering a *WORK* tag also felt unsatisfying, as work was not perceived as a positive activity worth tracking and therefore did not provide the same sense of accomplishment. P7 reported not opening the app for 1.5 weeks while they were sick and at home, with nothing to enter. P9 emphasized that they prefer meaningful or cool things to track, noting that when they have only one person or nothing to enter, there is no appeal to opening the app. They compared this lack of motivation to the reluctance to answer messages, explaining: "*manchmal hat man auch keine Lust drauf*" (sometimes you just don't feel like it), even though the task takes little time. They noted that, similar to messages where one does not really get anything out of replying, they feel they do not get anything out of entering data when they lack meaningful content to track.

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Writing Something Other Than Keywords

Participants used different approaches to add text beyond keywords in their entries. One participant actively writes additional context to ensure they can understand what happened when looking back at entries years later, as tags alone would not provide sufficient detail about conversations or activities. Another participant initially wrote more detailed descriptions, but stopped after finding that doing so did not provide enough value. A third participant considered adding more text but decided against it due to the additional effort required, preferring to stick to quick and simple keyword-based entries.

Creative Investment Through Hierarchical Tag Systems

One participant demonstrated substantial investment through the creative use of the hierarchical tag system. P2 created a "Monk Mode" tag that consolidates all negative behaviors they want to avoid, using the streak functionality to track their progress. This sophisticated approach demonstrates deep engagement and personal investment in customizing the system to support behavioral goals. This could also be seen as an indicator of how a persuasive design element (streaks) can motivate users in their real-life goals.

5.4.2 RQ2

Stats

Users' appreciation of the numerical statistics (see 5.19a) at the top of a keyword's dashboards varied significantly. P1 acknowledged the utility of percentage statistics but struggled to extract meaning from them, noting that visual elements speak to them much more than the numbers at the top. P2 similarly does not look at the percentage numbers, explaining that they can see roughly from the graphs how often they have done something, and the exact percentages are not tangible for them. P3 thinks the numbers at the top could be reduced so the heatmap fits better on the screen, suggesting they do not prioritize them. P10 found the numbers at the top too exhausting to look at, stating it feels strenuous to process information like a 21% increase in some keyword. P5 mentioned that they find the month-to-month statistics and the difference to the previous period interesting, particularly when things are going well, and they enjoy looking at those numbers. However, they primarily use the app to enter activities and check these statistics only occasionally.

P8 showed a different preference depending on the dashboard type. For people dashboards, they do not find the change statistics at the top useful; specifically, they say month-to-month and year-to-year changes are not decisive for them, while they do look at the total days statistic. However, for tag dashboards, they do look at these change statistics and find them more interesting, particularly the year-to-year comparisons, noting that this will become more valuable once they have a full year of data.

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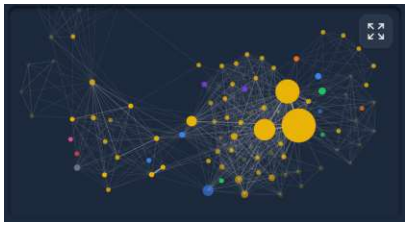
(a) Statistics at the top of the dashboard of a keyword.

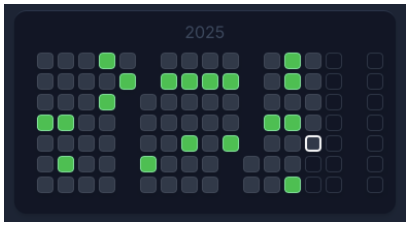


(b) Ego network visualization of a person.

Figure 5.19

Ego Network

The ego network visualization (see 5.19b) received limited engagement, with several users finding it too complex. P7 had not seen it before and found it very complex, noting that it requires tapping to see everyone's names, though they acknowledged it is good and asked whether it can be presented in a less cluttered way. P6 looked at it a few times initially, but found it became too complex, particularly with many people. They explained that for some people, who had seen them in clearly separated settings, the ego network was beautiful to look at because it clearly showed different clusters. However, once connections formed between these clusters, it became a single large clump, making patterns harder to recognize. They mentioned that viewing it on a computer rather than a phone might help, and noted that they could see the large nodes but mostly already knew that information from other statistics. P3 rarely looks at the ego network, stating it takes too long to load on their phone.



(a) A heatmap with a square for each day that highlights a keyword used on that day.

(b) Line Chart with three options to switch the timeframe.

Figure 5.20

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Heatmap

The year grid heatmap (see 5.20a) emerged as one of the most frequently used and appreciated visualizations across participants. P10 reported that they always look at the squares and manually count them, particularly to track streaks, expressing joy when seeing streaks of over a week. They described this as their favorite chart. P1 has developed an intuitive sense of frequency patterns from the heatmap, explaining that, after extended use, they get a good understanding of how high the frequency is relative to other things. They primarily interact with the data by scrolling over the last months or year to see temporal patterns without needing to count. P9 finds the monthly view of the heatmap most useful, stating it is what they look at most often when viewing person dashboards. They also mentioned that they primarily look at the heatmap in the preview and often do not click through to the whole tag dashboard.

P8 particularly appreciated the calendar-style heatmap (comparing it to GitHub's contribution graph), stating that it allows them to estimate very well whether they have done something more or less frequently in recent weeks because it provides an immediate look back into the near past. They described it as the first thing they look at when opening a tag dashboard and find it very pleasant to use. P5 also mentioned frequently looking at the heatmap grid to identify patterns with the naked eye. P2 noted that they look at the heatmap but consider it essentially the same information as the line chart, just in a different visualization format. They also find it valuable to compare the same month across different years, since they have been using the app for more than a year.

Line Chart

The line chart showing active days per month (see Figure 5.20b) received generally positive feedback, though some users found it redundant with the heatmap. P1 finds the line chart very interesting and describes it as the most exciting way to see how often they did something in which month over time. P2 similarly finds the line chart most valuable, especially for understanding temporal patterns and trends. P8 also appreciates the line chart, finding it pleasant because it clearly shows the absolute numbers for each month. P5 admitted that they have not seen the charts below the heatmap and described the active days per month as a nice overview, noting that they often try to recognize patterns with the naked eye in the heatmap, which the line chart makes clearer and easier.

P9 found the line chart showing active days per month similar to the heatmap, but less satisfying and more cluttered, stating they prefer the heatmap because it is not as overwhelming. They mentioned seeing peaks in the data, such as when they were on vacation, but overall found the heatmap more appealing.

Bubble Chart

The bubble chart (see 5.21a) generated the polarized responses among participants, with strong preferences both for and against it. P4 found it very cool and highly informative, explaining that it is very easy to follow and extract a lot of information because they can

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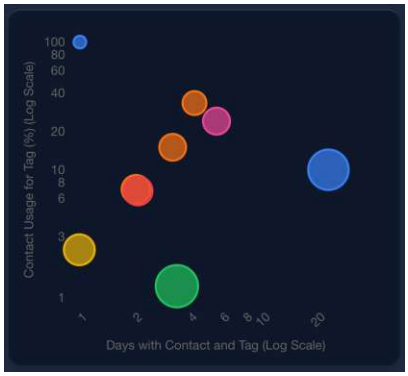
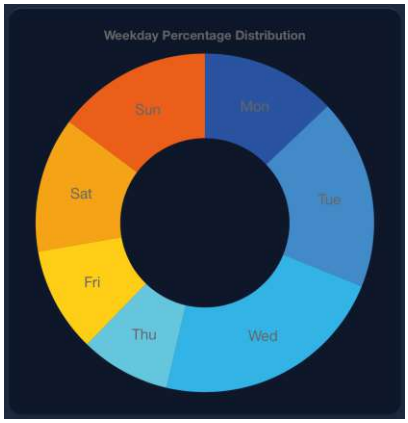


(a) A bubble chart in a person's dashboard showing tags as bubbles. The x-axis shows the number of co-occurrences with the person. The y-axis shows the conditional probability of the person given the tag.



(b) A pie chart showing the distribution of entries across weekdays for a tag.

Figure 5.21

see which tags or people co-occur most frequently with a person. They appreciated that they could identify positive or negative influences by looking at which positive or negative tags appeared in the upper-right corner of the chart. P4 went so far as to suggest that the bubble chart be placed above the network graph because they find it so valuable. They emphasized that it allows them to consume many pieces of information simultaneously in a visual format, an advantage over other visualizations such as cosine similarity.

However, several participants found the bubble chart overwhelming or confusing. P3 does not use it, stating it is too much for them and they would only engage with it if they really wanted to find something specific. They noted that, unlike other charts where you can see information at a glance, the bubble chart requires you to click each bubble to understand what it represents. P7 stated that they had not seen it before, but found it initially difficult to understand, and that if they had to remove one graph, it would be the bubble chart, but they believe it has its justification when wanting to go into detail. P6 admitted they had not looked at it closely because they needed to think more about what it meant. However, they suggested that color-coding the tags thematically might help with quick pattern recognition, since their tags and, therefore, bubbles currently look the same.

P8 did not intuitively understand the bubble chart without explanation, stating that it contains too much information in a single graphic. Even with an explanation, they understood it but felt it wasn't intuitive enough to use without guidance. P9 had no idea what it represented and did not check it. P10 found it totally unclear and could not understand it at first glance. P1 noted that for tags with many entries, such as alcohol, the bubbles overlap so much that they cannot be clicked on anymore, making

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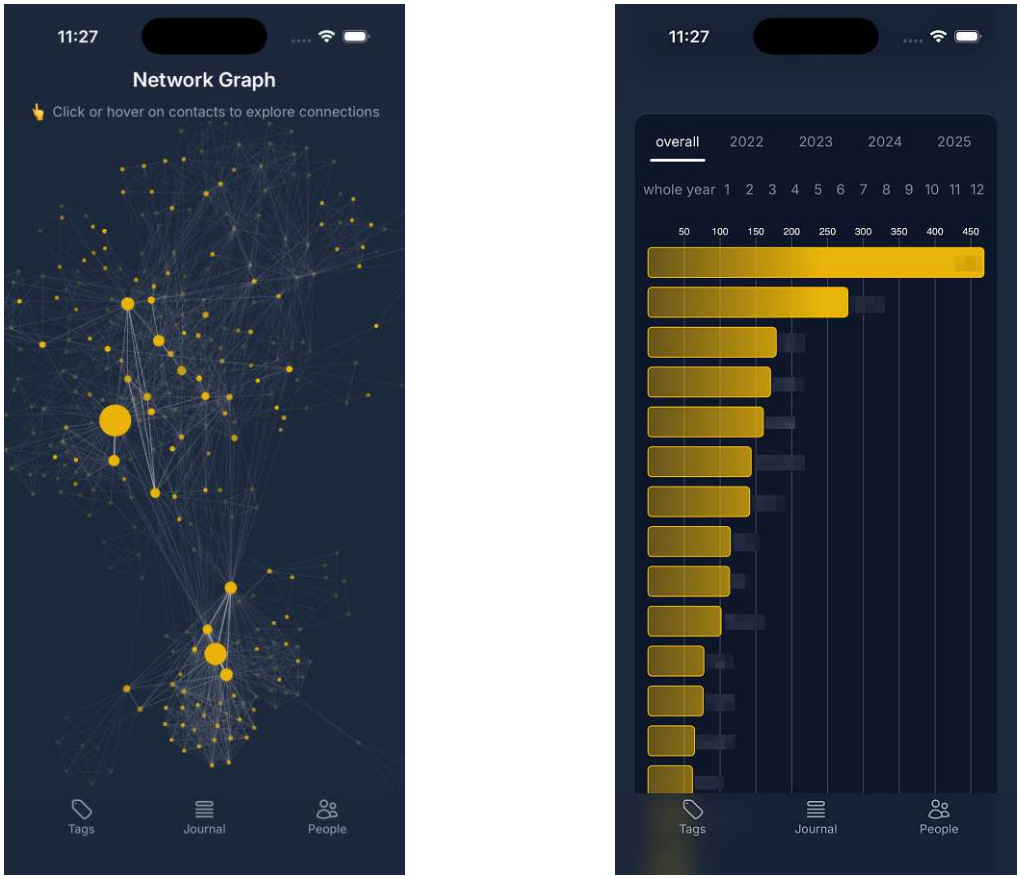
the visualization too dense to be useful.



(a) Network chart showing connections between people.

(b) Overall people chart showing how often people are seen.

Figure 5.22

Network Chart

P4 stated that if they were to show something from the app to another person, it would be the network graph (see Figure 5.22a) because it was rare to find an application that visualizes social circles in connection with habits. P8 opens the app to look at or show the network to friends, describing it as quite remarkable. They also occasionally open the app specifically to watch how the network has developed over time, not to enter data but to look at the graph.

P7 finds it interesting to see the connections and how most of the people they are connected to are somehow connected to others. P3 does not use the network chart much because their graph is relatively unimpressive, consisting mainly of a single large cluster with a few outliers. P1 looks at it occasionally and finds it quite interesting, particularly

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noting which nodes are large (indicating many connections) and how many connections a person has. They emphasized that it is essential to keep in mind that this is not a friend circle but rather people they have encountered, as people far out on the graph might be one-time meets rather than friends.

P9 looks at it sometimes but finds it depressing when comparing their network to others' much larger networks. They also mentioned having free-floating people who do not know anyone.

Overall People Chart

The visualization showing how often people were mentioned (see Figure 5.22b) received positive feedback. P9 finds the display showing how often a person is seen per year, as well as the monthly view, interesting. They noted that it makes clear how rarely some people are seen versus how often others are, and appreciate having an exact ranking of who is seen most and least often. P7 scrolls through the list of people they see most often, finding it really interesting to see how often they actually see certain people, though they only check it about every tenth time they open the app. P8 finds the ranking of who they have spent the most time with in the last month quite interesting and shows it to others frequently, describing it as quite cool.

P6 uses the chart to understand their relationship network, to track which person they see how often, and whether this corresponds to seeing good friends more often or if there is a discrepancy. They look at statistics most frequently when the app is open, and they are bored or find it interesting, usually starting with the question of which people they have seen most often. They find it exciting to see the overall top 10 people they have seen most frequently, because there are phases when one person catches up, and others fall behind. They also analyze monthly variations, asking themselves what triggered the increase in people in certain months, such as 80 people in June versus 60 in May. They noted that it is often not a single event but rather multiple rare events happening in the same month, such as class reunions or other gatherings where they see people they usually do not see. P3 regularly checks who they have met most recently because it interests them. One user used the data for a New Year's game in which friends had to guess how often they had met each other and rank themselves relative to each other, finding it interesting to compare people's beliefs with the actual data.

P1 does not use it actively because they feel it would lead to social evaluation of friendships and social structures. While they find it interesting occasionally, they had the impression that regularly looking at it could distort what they enter or influence them in some way.

Absolute Similarity

The bar graphs (see 5.23a) showing absolute co-occurrence of tags and people were among the visualizations that most users reported looking at regularly. Multiple participants (P3, P6, P9) mentioned that one of the first things they check is which people and tags frequently co-occur with a person. P6 noted that they are generally not interested in

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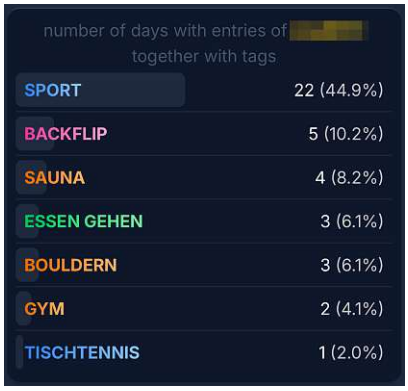
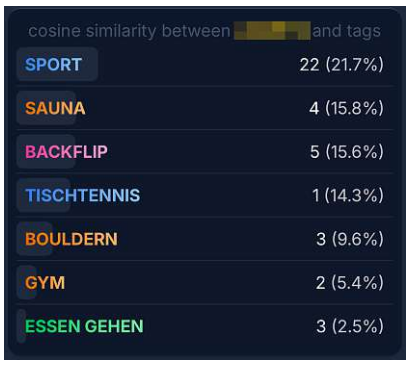


(a) Bar chart of the number of co-occurrences of tags with a person and the conditional probability of the tag given the person.



(b) Bar chart of the cosine similarity of tags with a person.

Figure 5.23

what they do most with a person because they usually already know this intuitively, such as playing volleyball with certain people. Instead, they find smaller patterns more interesting, such as realizing they have been to a restaurant with a person four times. P6 emphasized that these connection charts showing people and specific tags are what they look at most frequently.

P8 finds the connection charts showing the number of days spent together with people very pleasant and exactly what they want to see, noting they can see patterns at a glance, such as always seeing their mother together with their father. They also appreciate the tag connections displayed below the people connections. P1 similarly expressed interest in this visualization, particularly in seeing what activities they do most often with certain people.

Cosine Similarity

The cosine similarity (see Figure 3.3.2) visualization was poorly understood by most participants and received limited engagement. P4 found it more meaningful than absolute similarity because of its different weighting mechanism, which is proportional rather than absolute. They noted that when looking at a profile, what someone does most with a contact is weighted more strongly in cosine similarity compared to absolute similarity, which is just a pure percentage listing. They acknowledged that it might be difficult for laypeople or those unfamiliar with cosine similarity to understand the difference.

Most other participants did not understand or engage with the cosine similarity chart. P8 did not really understand it when first seeing it, only recognizing that it shows how frequently other tags or people co-occur with a person. Both P8 and P7 were confused because the percentages and numbers did not seem to match, as both absolute numbers

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and cosine similarity values were displayed together. P9 and P7 stated they would not have understood it without the explanation provided.

P9 also mentioned that the cosine similarity chart is too far down on the page, and users do not scroll that far, suggesting that people look at the first thing, are happy they did something, and then leave the dashboard. P3 finds it not very meaningful at a glance because they have to think about what it really says each time, which they find too cumbersome to engage with regularly.

P6 understood the visualization and noted that the information mainly consists of confirmations of what they already suspected, such as seeing a baby only when they see the parent. They find it reinforces what they already assume, seeing things confirmed again, but it usually adds no new value unless they are surprised in another direction. For this reason, they do not look at it very frequently.

Weekday

The weekday pie chart (see 5.21b) showing distribution across days of the week received positive feedback from users who found patterns in their data. P1 finds the pie chart with weekday distribution interesting. P2 finds the weekly distribution very interesting, particularly for negative activities. They explained that it helps them identify which days they are often triggered or should be more careful, allowing them to proactively plan and exercise more willpower on those days to prevent the activity. For example, with alcohol consumption, they can see that almost half of all instances occur on Saturday, which allows them to analyze why this always happens on Saturday and think about how to prevent it and what the triggers are.

P3 found the weekday distribution very interesting, discovering unexpected patterns, such as the fact that board games happen much less on Sunday than on other days, which they found somewhat funny and surprising. P5 mentioned they like looking at the weekday percentage distribution, noting it shows them patterns they try to identify with the naked eye in the heatmap. P6 stated that the weekday chart is what occasionally motivates them to look at tag dashboards. They would then check which weekday they ran most often, for instance.

P9 understood the visualization but was not particularly excited about it because their data did not show clear patterns. They explained that it is more valuable when one has a routine with activities planned for specific days, like running every Tuesday and swimming every Thursday, but since their schedule is irregular and they do things when they feel like it rather than on set days, the distribution is relatively similar across all days. They noted that it would be most exciting when there are considerable differences, like one day being huge and another very small, but their data does not show this.

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5.4.3 RQ3

Impact on Overall Phone Usage

When asked what apps users would use instead of *Habits Network* if they did not have it, users mentioned a variety of alternatives. They were also asked if using the app changed their overall phone usage time.

Participant	Effect on Phone Usage	Replaced Activities / Remarks
P1	Replaces other app time	Replaces social media or news consumption while on the go.
P2	No increase	Used shortly as part of evening routine, puts phone away afterwards.
P3	Slight increase	Uses phone slightly more, but app usage short (1-5 minutes).
P4	No relevant change	Very short daily use (approx. 4 minutes), clear benefit.
P5	Slight increase	Longer due to animations and viewing stats; finds analyzes motivating.
P6	Minimal change	Sometimes replaces news reading (e.g., on subway).
P7	Decrease in unnecessary use	Replaces time spent on unproductive activities.
P8	Replaces social media	Likely replaces "*doomscrolling*" time.
P9	No change	Uses app when bored; not a significant time factor.
P10	Replaces social media	Likely reduces time on Instagram or TikTok.

Table 5.4: Perceived impact of the app on phone usage.

Most participants reported that the app does *not* noticeably increase their overall phone usage. Instead, several users mentioned that *Habits Network* often replaces time previously spent on social media or news apps, particularly during idle moments (e.g., commuting or before bed). A few participants noted a slight increase in screen time due to animations or curiosity about statistics. However, this increase was perceived as minor and balanced by the app's meaningful use. In some cases, it even helps reduce unproductive smartphone behavior. A summary of the responses is provided in Table 5.4.

Being Aware Because of the App

Multiple participants reported that the app increases their awareness of their behaviors, providing an overview and sense of control over their lives. P10 explained that they want an overview of their drinking frequency, comparing the app to a scale that provides

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reassurance about weight. They noted that sometimes they feel like they have been drinking too much alcohol recently, but when they check the app, they realize it was not actually too much. This gives them a bit of control over their life and their habits, helping them distinguish between feelings and reality.

P2 stated that the app provides a good overview of both negative and positive activities, which enabled them to consciously decide to do certain activities less often or positive activities more often. P1 described awareness as a decisive first step in behavior change. For negative activities like alcohol consumption, making themselves aware of how often they consume these things is already a crucial step that shows them the patterns and motivates them to do it less frequently.

P7 provided a concrete example of awareness leading to action. At the beginning of the year, they tracked caffeine consumption and distinguished between coffee and energy drinks. They noticed that every day was marked red with energy drinks during university periods. Seeing it in black and white, knowing how bad certain things are, but then really seeing how often it actually happens, motivated them to stop and take a break from energy drinks for a few weeks.

When Users Add Entries

Timing of entries Participants showed varying patterns in when they create entries, ranging from structured evening routines to event-based immediate logging. P7 always enters everything in the evening rather than directly after activities. P6 mainly enters data in the evening or on the subway after being out all day. They try to enter everything on the same day because they notice they forget things otherwise. They enter specific activities, like planks, immediately after completing them because of the sense of accomplishment, but general entries about meeting people are always entered in the evening.

P8 mainly enters data before going to bed, reflecting on what they did that day. For certain activities that they might forget, like drinking coffee, they try to enter them directly. P5 states that they naturally enter at the end of the day, mainly for the same day, rarely in the morning for the previous day. Sometimes they forget and then enter two to three days at once when they get to it. P1 found it difficult at first to get into a rhythm. Still, the more regularly they integrated entering data into their evening routine, the easier it became to maintain an overview and be honest when entering. They estimate entering between ten and fourteen times per week, one to two times per day. When they have it under control, they enter in the evening or at the latest the next morning. When not in routine, they enter things on the go, which replaces time they would otherwise spend on social media or news.

P2 uses the app as part of their evening routine, letting the day pass in review and then making entries. They find this easiest and note that they use their phone less frequently this way and have a better overview. They tried entering directly after activities, but found it impractical because they do not always have their phone with them.

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Participant	Entry Timing	Daily Usage	Stats Viewing
P1	Evening routine or next morning; on-the-go when not in routine	10-14x/week (1-2x/day)	Not specified
P2	Evening routine, reviews day	Max 5 min/day	2-3 min after entering
P3	Evening/afternoon; sometimes random; soon after meeting people	2 min/day	Only when interested
P4	Event-based, immediately after activities	4 min/day	After entering, especially after gaps
P5	End of day, same day; sometimes 2-3 days at once	Slightly more due to animations	Always for entry, then maybe stats; regularly sport/alcohol
P6	Evening (shower/subway); specific activities immediate	Not specified	Mostly people profiles after entering
P7	Always evening, everything at once	Not specified	Sometimes out of curiosity after entering
P8	Before bed; some activities immediate	Not specified	Not specified
P9	Prefers when has more/cooler things; sometimes weekly gaps then catches up	1 min/day	Only with entering, after gaps, when meeting someone
P10	Variable; whenever occurs; immediate for medication	Variable by phase	Not specified

Table 5.5: Self-reported usage patterns by participant.

P4 enters data event-based, always after completing something they want to track, rather than sitting down at a fixed time to record everything. They do this to capture as much information as possible directly after an event, such as meeting friends, so that they can make notes. If they were to enter everything at a fixed time, they would forget small details they want to note.

P3 usually enters in the evening or later in the afternoon, though they sometimes enter randomly during activities when it occurs to them. Because they mainly track meeting people, they enter relatively soon after meeting many people at once, because otherwise they would not remember everyone. Sometimes they do not enter for a day and then

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enter retroactively.

P10 states that variable usage patterns exhibit phases of more or less frequent use, depending on what they were tracking. They enter whenever it occurs to them, sometimes forgetting days and then catching up retroactively because they still remember activities. For important things like medication, they enter immediately after taking it, so they have a record that they already took it that day. They sometimes look at the app when bored, like on the subway. P9 prefers entering when they have more or cooler things to track. They find it less appealing when they only have one person or nothing to enter. They compared it to replying to messages, noting that sometimes they lack motivation for the task, even though it takes little time. This can lead to gaps of up to a week without entering, after which they catch up by trying to enter everything retroactively.

Time spent on the app Most participants reported spending very little time on the app daily. P4 estimates using the app about four minutes per day, finding the added value clearly given despite the marginal time investment. P9 described it as taking one minute per day, stating it does not change their overall phone usage. P2 states they spend a maximum of five minutes per day on the app. In their evening routine, entering data is simply a task they check off before putting their phone aside, so they do not use their phone more because of it. After entering, they spend another two to three minutes randomly analyzing the tags they are currently interested in or focused on. P3 noted they are usually on the app for about two minutes to enter data and then leave, unless something interests them and they want to look something up. P5 mentioned spending slightly more time (compared to their own spreadsheet solution they used before) due to animations that deliberately include fade-in and fade-out effects, adding two to three seconds to the entry process. However, they also have more incentive than before to look at analyses because they do not have to do them themselves and can click around, so they spend more time looking at stats than they would have before.

When users look at statistics Three participants reported looking at statistics specifically after entering data, with P9 opening the app exclusively to create a new entry and only then viewing stats. P9 explicitly stated they would not open the app just to look at statistics. P5 opens the app, always intending to add a new entry, and might then look at statistics. They look at sports statistics relatively often because it is going well recently, maybe every three days, but otherwise look at tags almost exclusively only when entering them. Exercise and possibly alcohol are the only tags they regularly check. P6 mostly looks at people's profiles after entering data. P4 looks at dashboards after events or when they have time and have just entered something, particularly for tags they have not used for a long time or used very frequently recently, when they have not looked at the detailed charts in a while. P7 sometimes checks data out of interest; for example, when on a phone call with a colleague, they randomly check when they last spoke.

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Feed

It should be noted that only 8 of the 10 users had friends in the observation period who shared tags with them and are therefore able to use and judge the feed feature. The success of the feature depends on whether users' friends actively use the app and on which tags they share with them.

Among those who could access the feature, engagement varied. P3 very rarely looks at what friends do, only seeing their activities by accident when clicking the people button to access statistics. They are not particularly interested, even though they have at least one friend sharing tags with them. The same applies to P9, who also showed no interest in the feed feature even though they have at least one friend sharing tags with them.

P10 currently does not have friends sharing tags, but expressed interest in seeing what others do. They praised the feature as really good functionality, noting that seeing what friends do could inspire new ideas for activities to try. They gave the example of discovering journaling through seeing a friend's entries and being motivated to try it themselves.

Participants with active friends in their feed showed positive engagement. P2 finds the feed function very interesting and always checks it after making an entry to see what others have done. They compared it to a small dopamine hit, like when you see what someone posted on Facebook, describing it as gaining insight into people's lives, but focused on activities. P1 perceives the feed as inspiring, particularly when people make many entries. They appreciate insight into what happened during someone's day, usually positive things, when people share positive tags. They contrasted it favorably with platforms like Strava or Instagram, noting that entries are not about self-promotion but rather about mutual encouragement in a productive sense.

Goals

Participants showed mixed responses to the goals feature, with several users not actively engaging with it. P9 did not find it appealing and stated that seeing a failed goal would not affect them, expressing indifference to the feature. P1 occasionally looks at the goal system but currently does not use it actively because they do not place specific value on it at the moment, and it does not fully appeal to them yet. However, they could not explicitly explain why. P2 tried the goals feature once with Instagram but has not used it since. They wanted to test whether the tool would be helpful, but they failed to achieve their goal. They concluded that the tool does not help them because they check the app in the evening as part of their routine, but use Instagram in the morning, creating a significant time gap between the reminder and the behavior. However, they acknowledged that the tool could be handy if one changes their usage pattern.

P10 tested the goals feature but found themselves confused by the wording "behind schedule" when they were actually performing better than their set goal. P5 thinks the feature would be funny and cool, but has not used it yet.

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Two participants found the goals feature motivating and valuable. P8 emphasized that goals are among their top priorities when tracking, particularly when they forget to journal for a couple of days. They stated that having active goals to track, along with tracking people, are their main priorities. P8 finds that goals help them to be more motivated to reduce negative behaviors. They appreciate the progress bar showing whether they are ahead or behind schedule, which helps them decide whether they can allow themselves a cheat day or need to adjust their behavior. They explained that while one can look at statistics without a goal, having a concrete goal for a specific time period makes it more pleasant to see progress and helps them adapt their behavior accordingly. They noted that goals provide an extra piece of motivation that might lead them to do something they otherwise would not have done.

P4 stated that the goals feature was definitely a motivator for them, helping them understand how far they are from achieving their goal.

Even though P4 and P8 expressed their appreciation for the goals feature, both stopped using it after a while (see 2.4.1).

Achievements

Participants showed varying levels of engagement with the achievements feature, ranging from indifference to active pursuit. P9 expressed that they do not get anything out of achievements and finds them unimportant, responding sarcastically when asked about celebrating them. P4 considers achievements more of a side thing and does not actively change their activity or add extra tags to unlock them.

In contrast, some participants actively pursued achievements and modified their behavior to unlock them. P6 and P3 remembered that when achievements were introduced, they went through the list and completed many actions they had already done before, specifically to unlock the achievements. For example, they sent friend requests again or created and quickly fulfilled goals. P10 added a subtag to their *SPORT* tag specifically because it was required for an achievement, demonstrating that the feature motivated them to use a feature.

Streaks

Participants showed varying levels of engagement with streak tracking, with some finding it highly motivating while others remained indifferent. P10 regularly checks the lock icon to see how many days they have gone without mentioning something, and they try to do less, expressing happiness when they reach over a week. While they find the streak motivating for doing certain things more regularly, they clarified that most activities are done because they want to, not because they feel pushed by the app or need to maintain a streak.

P2 considers the longest streak without a negative activity to be the most important number for them and finds it very motivating. The longer the streak, the more motivated

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they become. They also mentioned that seeing a large number of days displayed on the lock icon sometimes triggers them to open a tag dashboard and review the overview and past entries to draw conclusions.

P5 generally does not care much about streaks, but notes exceptions when they have exceptionally long streaks. They remembered a two-month alcohol-free period during an exam phase, where they were aware they would break their record in a few days, which felt nice. Similarly, during a week when they exercised every day, they kept in mind that completing that day's exercise would mean achieving the whole week, which would be nice.

P1 finds the streak interesting, particularly for negatively associated tags. They use nicotine as an example, appreciating both the icon showing positive or negative streaks and the year view for scrolling through periods.

P8 finds the streak always nice to look at.

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CHAPTER 6

Discussion

6.1 Answering the Research Questions

6.1.1 How does Considering the Hook Model and Fogg Behavior Model Enhance User Satisfaction, Engagement, and Retention in a Self-Tracking App for Students?

The Hook Model (Section 2.2) and the Fogg Behavior Model (Section 2.3) provided the theoretical foundation for designing features intended to increase user satisfaction, engagement, and retention in *Habits Network*. Features such as achievements, goals, streaks, visualizations, and social functionality were designed with explicit consideration of principles from both frameworks.

Evaluating the effectiveness of these theoretical strategies presents methodological challenges. The app implements multiple features simultaneously, making it difficult to isolate which specific model principles drove observed outcomes. Furthermore, user behavior is influenced by numerous confounding variables, including life circumstances, academic schedules, and individual differences in motivation and technology use patterns. The findings presented in Chapter 5 combine quantitative behavioral measures of feature usage over time with qualitative interview data on user motivations, experiences, and perceptions. Although these data do not permit definitive causal claims, we interpret the observed patterns in relation to the Hook Model and the Fogg Behavior Model to assess how the empirical evidence supports or challenges the theoretical concepts.

In the following subsections, we will discuss the 4 phases of the Hook Model as well as ability and motivation from the Fogg Behavior Model. Since the two models overlap in the trigger phase, we will discuss triggers only once in the Hook Model section.

Trigger Phase: External Triggers Absent, Internal Triggers Developed

Habits Network currently implements minimal external triggers. The only external trigger in use is the app's presence on the user's home screen, while push notifications and email reminders remain unimplemented. Expert perspectives diverged on this limitation. Exp2 emphasized the potential value of notifications from a developer's perspective, particularly their potential to implement variable rewards rather than serving merely as reminders. However, Exp1 challenged the assumption that notifications are a primary driver of engagement, particularly in activity-tracking applications. Based on their research experience, they argued that for physical activity apps, notifications may have limited motivational impact due to users' established routines, contending that data visualizations hold greater motivational potential.

This apparent divergence highlights an essential distinction between two types of behaviors that *Habits Network* could attempt to promote through external triggers. The first is motivating users to engage in desired activities in their lives (e.g., exercising, socializing, pursuing personal goals). The second is encouraging users to engage with the app itself by creating journal entries and documenting their experiences. Viewed through the Fogg Behavior Model, these two behaviors differ significantly in their feasibility at the moment a notification is received. When a notification prompts a user to engage in a physical activity, such as going for a run, the user may possess the motivation and the external trigger, but often cannot act immediately due to contextual constraints, such as work schedules, location, or other commitments. In contrast, when a notification prompts the user to journal about an activity that has already occurred, the ability barrier is substantially lower - journaling can be performed in nearly any context where the user has their phone available.

This distinction suggests that Exp1's skepticism about notifications in physical activity apps addresses the first type of behavior, while Exp2's emphasis on variable reward notifications aligns more closely with the second. Notifications designed to prompt journaling behavior can serve as effective external triggers because they target an action with a higher inherent ability. When users repeatedly engage with the app in response to these prompts - opening the app, creating entries, and receiving variable rewards through statistics and insights - they progress through the *hook cycle*, gradually internalizing the behavior of documenting their lives.

This internalization process connects to Exp2's broader perspective on the ethical justification of engagement mechanisms in self-tracking applications. As discussed in the ethical considerations chapter (see 6.2), they articulated a causal chain: continuous app use leads to constant tracking, which in turn requires users to engage in tracked behaviors to generate meaningful data, ultimately helping them achieve their goals. From this perspective, notifications that successfully drive app engagement and journaling habits can indirectly promote the first type of behavior, actual life activities, by establishing a tracking routine that creates accountability and awareness.

Both expert perspectives offer valid insights when understood as addressing different

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stages of behavior change. External triggers through notifications may indeed have limited direct impact on motivating complex behaviors like physical exercise, but they can effectively target more accessible behaviors, such as app engagement and journaling, creating a feedback loop that indirectly supports users in pursuing and maintaining their real-world goals.

Despite the absence of deliberate external triggers, user interviews revealed that many participants developed internal triggers. Users reported emotional associations between completing positive activities and the immediate creation of an entry, with P10 even describing a "short dopamine rush" after creating an entry. This indicates that feeling proud or accomplished serves as an internal trigger after certain activities, prompting app engagement. The integration of app usage into evening routines and after certain activities suggests that the app was successful in creating habitual internal triggers.

A particularly powerful internal trigger emerged in P7's experience: thinking about a person prompted the question "when was the last time I saw that person?", which led them to open the app to find the answer. This demonstrates a cognitive association where the app becomes the answer mechanism for spontaneous personal questions. Critically, this trigger depends on prior investment; the user must have tracked enough interactions for the app to provide meaningful answers. Unlike social media, where investment loads the next external trigger, here investment enables the internal trigger itself.

Action Phase: Conceptual Tension Between Opening and Entering

The variable reward (random statistic) was initially designed to update after entry creation, as entries generate the data needed for statistics. However, Exp2's reconceptualization reveals a fundamental misalignment with the Hook Model: the action phase should target the simplest possible behavior (Section 2.2.2), yet creating an entry requires substantial brain cycles (Section 2.3.2) as users must reflect on their activities, select appropriate tags, and potentially recall specific people. Opening the app, by contrast, represents the minimal action that should follow a trigger. Exp2 argued that once users open the app, an entry will be created automatically 5.2.3. This misalignment has cascading implications for variable reward design: the random statistic should update when the app opens, rather than after entry creation. This shift reduces dependence on entries for engagement, a critical consideration given that three users reported decreased motivation during low-activity periods when they had little meaningful content to track (see 5.4.1). By decoupling the reward from data entry, the app could maintain engagement even when users' lives generate fewer trackable events, addressing a key retention vulnerability.

Variable Reward: Implementation Gaps and Unexpected Successes

The variable reward phase revealed a divergence between the intended implementation and its actual effectiveness. The random statistic feature (see 3.7) has room for improvement based on expert assessments, behavioral data, and user feedback. Exp2 argued it provided no praise and should occupy the entire screen with celebratory elements. At the same time,

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Exp1 questioned whether it functions as a reward at all since it is random and therefore mostly unrelated to user input. Behavioral data support these concerns, showing that the keywords in the random statistic did not facilitate exploration of a shown keyword. While this is not an indicator of failure per se, since the absence of a click on a keyword does not mean the statistic was not interesting, the user interviews revealed that the random statistic, although sometimes interesting, was often out of context or irrelevant to users. However, P9's anticipation of a new statistic before creating a journal entry indicates a strong connection between the action and the reward, underscoring the feature's potential.

Despite the limitations of the random statistic, variable rewards functioned effectively through unintended mechanisms embedded in the app's visualizations. Participants regularly viewed statistics immediately after creating entries, with P9, P6, and P2 consistently checking dashboards, profiles, or the feed post-entry despite stating they would not open the app solely for analysis (see 5.4.1). This post-entry viewing behavior creates an implicit variable reward mechanism: users enter data, anticipating their new entry might reveal interesting patterns, transforming visualizations into rewards of discovery.

Multiple participants explicitly described seeking and finding patterns. P5 frequently examined the heatmap to visually identify patterns, while P3 discovered unexpected patterns, such as board games occurring less on Sundays. P2 used weekday distributions to identify behavioral triggers, recognizing that half their alcohol consumption occurred on Saturdays, enabling proactive planning. P10 described experiencing a brief dopamine hit when seeing their activity appear in the heatmap, demonstrating immediate visual feedback as a reward.

This pattern-seeking behavior aligns with the Hook Model's three reward categories. Visualizations provided rewards of the hunt through temporal pattern discovery, rewards of the tribe through the feed (P2 also described a small dopamine hit when viewing the feed), and rewards of the self through visible evidence of competence and achievement.

These visualization-based rewards succeeded where the random statistic struggled because they were directly connected to user input and personally meaningful. When users entered a workout and immediately saw their streak extend or heatmap pattern strengthen, the reward felt earned and contextually relevant rather than arbitrary. This suggests that effective variable reward implementation requires not merely unpredictability, but tight coupling between user actions and personally significant outcomes that users can discover through exploration.

Investment Phase: Evidence of Increasing Commitment

Habits Network offers numerous investment opportunities throughout the user experience. Every interaction that contributes data to the system represents an investment: writing journal entries, creating tags, adding people to track social interactions, and connecting tags from others to one's own. Each of these actions requires effort from the user but

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increases the app's value by building a richer personal database that enables more meaningful insights and visualizations.

Sharing a tag with another person or connecting someone else's tag to one's own exemplifies collaborative investment, where the effort creates value for both the individual user and their social network within the app. These social investment mechanisms strengthen retention by creating mutual dependencies and shared context that would be lost if either party stopped using the app.

One form of investment, however, showed surprisingly low engagement despite its theoretical potential. There were only around 10 goals created by the observed users in total, and after 1.5 months after the release, no user was creating new goals anymore (see Figure 5.4). Despite this limited overall adoption, the single user who actively used goals over an extended period demonstrated exactly the commitment mechanism predicted by the investment phase.

This user explicitly described how creating goals transformed their tracking priorities, stating that when goals are active, they prioritize tracking those specific tags, followed by people (see user interview findings 5.4.3). The investment in creating a goal successfully served as a commitment device, motivating both the pursuit of the goal itself and the consistent tracking required to monitor progress toward it.

This demonstrates a powerful but underutilized investment mechanism: when users invest effort in setting explicit goals, they create a forward-looking commitment that shapes their subsequent behavior in the app.

The user's statement about prioritizing goals and people reveals an important insight: they explicitly mentioned tracking people as a top priority alongside goals. This emphasis on people tracking connects to a broader theme that emerged across multiple data sources. Both expert interviews identified the social dimension as what distinguishes *Habits Network* from conventional self-tracking applications, with the ability to track social interactions representing a unique value proposition that sets the app apart from competitors focused primarily on individual metrics (see 5.2.1). Furthermore, tracking people creates additional investment through an interdependency with tag tracking: one user reported they would not stop tracking certain tags because doing so would mean losing data about the people they track (see 5.4.1).

People Tracking vs Tag-Only Tracking: Engagement Trade-offs The distinction between users who track people versus those who use only tags presents an interesting tension in the Hook Model's action phase. Tracking people theoretically creates more triggers for entry creation, as every social interaction becomes a potential prompt to document the experience, leading to more frequent passes through the hook cycle. However, this increased trigger frequency comes at the cost of substantially higher cognitive effort per entry, as users must recall and select specific individuals rather than simply applying categorical tags.

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The limited sample size prevents drawing definitive conclusions about whether people tracking increases overall engagement. Among the three users who did not track people, entries per day varied dramatically: 0.98, 1.56, and 2.9, respectively. This substantial variation within the tag-only group suggests that the tracking approach alone does not determine engagement levels, and that individual differences in tracking motivation and life circumstances play significant roles.

A proper investigation of whether people tracking increases entry frequency would require a larger user base with controlled comparisons between tracking approaches. The current data can only establish that both approaches are viable, but cannot determine whether the additional triggers from social tracking outweigh the increased effort barrier.

Motivation in the Fogg Behavior Model

In the context of a self-tracking app, the motivation that makes users use the app is what they get out of it, i.e., the information about their life, the statistics, and the visualizations. The statistics and all the features the app offers that use the entered data are what motivate people to use the app. To use those features, they have to track their lives, i.e., enter data. This means, for a developer, that the ability to enter information must be very high, so the barrier to entry must be reduced to a minimum.

The Anticipation Paradox: Motivation Through Hope Rather Than Consumption An apparent contradiction arises when comparing users' reported primary motivation with their actual usage patterns. When asked what motivates them to use the app, multiple participants explicitly cited visualizations and statistics, with some specifically valuing how visible progress motivates continued activity (see 5.4.1). However, five participants explicitly reported that they primarily open the app to create entries rather than to analyze data, with P9 confirming they would not open the app solely to view statistics, and P1 explaining that they have to actively think about viewing charts for this to happen. This creates a puzzle: if users claim statistics motivate them, why do they report rarely viewing them independently of data entry?

The behavioral data provide a more nuanced picture than the self-reported usage patterns suggest. Figures 5.8, 5.9, and 5.10 show that users view tag overviews, individual tag dashboards, and person dashboards regularly and frequently throughout the observation period. This discrepancy between reported and observed behavior suggests that users may not be fully aware of how often they actually engage with visualizations, particularly when this occurs immediately after entering data within a single app session.

One possible explanation for this paradox lies in reframing motivation from consumptive to anticipatory, aligning with the Fogg Behavior Model's Hope/Fear dimension (see Section 2.3.1). Rather than being motivated by the repeated consumption of insights, users may be motivated by the anticipation of discovering patterns. The sequence would operate as follows: when something meaningful happens in their life (the trigger), users enter it into the app (the action) in anticipation of potentially seeing that event reflected

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in their statistics or discovering a new pattern (the reward). This interpretation suggests that users are not primarily motivated by viewing statistics per se, but by the possibility that their new entry might reveal something interesting when they check the visualizations immediately afterwards.

This anticipatory mechanism may also explain why users are not consciously aware of their frequent chart viewing. P1's statement that they actively have to think about viewing charts suggests that lightweight visualizations, such as the heatmap in the tag overview, require minimal brain cycles (one of Fogg's six elements of simplicity, see Section 2.3.2). When viewing charts is cognitively effortless and tightly coupled with the entry process, users may perceive it as part of the entry action itself rather than as a distinct analytical activity.

An alternative explanation warrants consideration: users may find intrinsic value in documenting their activities, independent of the analytical insights they gain. Notably, when asked what motivates them to use the app, no participant mentioned simply wanting to record their activities, yet this documentation impulse may operate as an implicit motivator that users themselves do not consciously recognize. P9's comparison of entering data to sending a chat message (see 5.4.1) provides a suggestive parallel, framing data entry as a form of communication rather than pure information storage. This analogy aligns with the design decision to make the input interface resemble a chat box, explicitly aiming to create a sense of social conversation. The parallel to telling someone about an accomplishment suggests that documentation itself may function as a form of acknowledgment or validation, independent of subsequent analysis.

However, the available data are insufficient to determine which of these mechanisms predominates or how they interact. The contradiction between stated motivations and reported behavior patterns, the discrepancy between reported and observed viewing frequencies, and the absence of explicit documentation-focused motivations in user responses all point to complex psychological processes that require further investigation to understand fully.

Pleasure as Entry Motivator: Asymmetric Effects on Tracking Behavior The Fogg Behavior Model identifies Pleasure and Pain as a primitive motivational dimension characterized by immediate responses to what is happening in the moment (see Section 2.3.1) [Fog09]. User interviews revealed that this motivator operates asymmetrically in *Habits Network*. Multiple participants reported entering positive activities immediately due to a fulfilling or accomplished feeling, with P10 describing a dopamine rush and P1 noting that this reinforces the behavior itself by extending the positive feeling beyond the activity into its documentation. In contrast, P1 explicitly delayed entering negative tags like alcohol until evening or the next day, revealing that pain avoidance creates temporal displacement in tracking.

This asymmetry has critical implications for app design. While pleasure drives a double reinforcement mechanism (motivating both the behavior and its immediate documentation), pain avoidance threatens data completeness through delayed or forgotten entries.

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Although awareness of negative behaviors can drive change (P2 quit nicotine through tracking), the psychological pain of acknowledgment creates friction in the documentation process itself.

These findings suggest that self-tracking apps should leverage pleasure while reducing the pain of logging negative behaviors. Making positive tracking immediately rewarding through visual feedback or streak visualization capitalizes on the power of pleasure to reinforce both behaviors and documentation. For negative behaviors, apps might reframe entries as progress toward goals rather than admissions of failure, provide positive feedback for the act of tracking itself regardless of content, or automate detection to eliminate deliberate acknowledgment.

Environmental Dependency: A Limitation in the Fogg Behavior Model's Motivation Framework The Fogg Behavior Model's three core motivators (Pleasure/Pain, Hope/Fear, Social Acceptance/Rejection) assume that motivation can be optimized through design choices independent of external circumstances (see Section 2.3.1). However, user interviews revealed a critical limitation for life-tracking applications: three participants reported decreased motivation during periods with little to track, with one describing busy work periods where there was nothing meaningful to enter, another finding that entering a *WORK* tag felt unsatisfying as it did not provide a sense of accomplishment, and P7 not opening the app during one and a half weeks when sick at home. This reveals that all three core motivators become inoperative when users' lives do not generate trackable events. The pleasure of logging achievements, the hope for visible progress, and the social acceptance that comes from sharing all require meaningful activities to track.

Since phases of low activity are inevitable in anyone's life, maintaining engagement during these periods is critical for long-term retention. If users disengage completely during such phases, they may not return when their life circumstances change and generate new trackable events, resulting in permanent churn. Social features could address this environmental dependency by providing alternative engagement pathways that operate independently of the user's personal activity levels. The feed feature demonstrates this potential, allowing users to consume others' content even when they have little to contribute themselves. Participants with active friends showed consistent feed engagement, with P2 always checking after entries and P1 finding it inspiring. This suggests that observing friends' activities might sustain app usage during personal low-activity periods by providing borrowed motivation through social connection.

Beyond passive consumption, additional social features could further bridge these gaps. Commenting on or reacting to friends' entries could maintain engagement without requiring personal trackable events. P10's example of discovering journaling through seeing a friend's entries illustrates how social exposure during low-activity phases might even generate new tracking behaviors, transforming the motivation gap into an opportunity for behavioral expansion. These social mechanisms could function as a bridge, keeping

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users connected to the app during inevitable life fluctuations so they remain positioned to resume personal tracking when circumstances improve.

Ability in the Fogg Behavior Model

Ability, in the context of a self-tracking app, can be seen as the ease with which new data is entered, which is then used to generate statistics and visualizations. However, ability extends beyond data entry to encompass users' capacity to discover, understand, and utilize the app's features. According to Fogg's framework, ability is determined by the scarcest resource at the moment a behavior is triggered, with brain cycles representing the cognitive effort required to perform a behavior (Section 2.3.2).

Complexity as a Brain Cycles Barrier All empirical methods revealed that complexity is a significant issue in the app. The System Usability Scale showed that question 3, which asks whether users think the system was easy to use, had the lowest mean score across all questions. User interviews and behavioral data revealed that participants discovered features for the first time during the interview itself, with P5 only seeing the charts below the heatmap when shown during the session. Exp2 found the app overwhelming due to the simultaneous presentation of numerous features scattered across the interface, making it unclear what users should prioritize. This complexity prevents feature discovery, meaning users cannot access all the analytical tools available to them. This has implications for motivation and behavior change. When complexity prevents users from discovering analytical features, they cannot extract the insights that motivated them to track in the first place. This affects both ability (high brain cycles required for discovery) and motivation (unmet expectations of insights).

6.1.2 Which Visualization Methods Best Convey Insights from Tracked Life Domains, Measured Through User Comprehension and Engagement?

User Preference for Low-Level Metrics: The Heatmap and Line Chart Success

A fundamental tension emerges between expert recommendations for visualization sophistication and actual user preferences revealed through interviews. Exp1's framework of metric abstraction levels characterizes the app's visualizations as predominantly low-level, with simple representations like heatmaps and line charts that users generally understand well. While acknowledging that higher-level abstractions typically convey more concrete behavioral insights, Exp1 observed that the app presents information at a relatively raw level and suggested incorporating additional abstraction layers, particularly through reference points.

User interview data, however, reveals a strong preference for precisely these low-level metrics that Exp1 characterized as basic. The heatmap and line chart emerged as the most appreciated visualizations, with users describing intuitive pattern recognition

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and immediate comprehension. This creates a paradox: experts recommend moving toward higher abstraction to provide more meaningful behavioral insights, while users demonstrate the greatest satisfaction and engagement with the simplest, most direct data presentations.

This divergence suggests that comprehensibility may trump analytical sophistication for most users in self-tracking contexts. Also, Exp2, despite being an expert in the field, identified the low-level heatmap as personally the most attractive feature, prioritizing it over more sophisticated visualizations. This expert-as-user perspective validates the preference for simple metrics, extending beyond technical literacy limitations.

There were some individual differences in visualization preferences among users. Some users preferred the heatmap, while others favored the line chart; several noted that both convey the same information in different formats. These individual differences in visualization preference suggest that providing multiple low-level representations of the same data serves diverse user needs and cognitive styles rather than representing redundancy.

Convergent Evidence of Complexity Failure: Ego Network and Bubble Chart

In contrast to the successful low-level visualizations, both the ego network and bubble chart demonstrated rare complete convergence across user feedback and expert assessment regarding excessive complexity. Multiple users reported the ego network as too complex and overwhelming, with specific complaints about loading times, visual clutter, and difficulty extracting meaningful patterns. Expert evaluation confirmed these user experiences through cognitive science frameworks, with Exp3 identifying the fundamental problem of cognitive overload when presenting 40 nodes and 100 edges simultaneously without adequate filtering mechanisms. Notably, Exp3 praised the ego network's aesthetic quality, describing it as beautiful and like artwork, yet this visual appeal could not overcome the usability barriers that prevented users from extracting actionable insights. The behavioral data validated both perspectives, showing that ego network viewing declined substantially after approximately 1.5 months, indicating insufficient perceived value despite the feature's easy accessibility through a single button click from person profiles.

The bubble chart generated remarkably similar patterns of user comprehension failure and expert criticism, though without corresponding behavioral tracking data. User interviews revealed that most participants could not understand the visualization without explanation, with several describing it as overwhelming and unclear. An important exception emerged with P4, who found the bubble chart highly valuable and informative, suggesting that certain visualizations may serve power users effectively even if they fail to achieve broad comprehension. Expert assessment converged with the majority user perspective, with both Exp1 and Exp3 questioning the visualization's value for users without technical backgrounds. Exp3 specifically identified mobile usability challenges, including overlapping bubbles, creating interaction difficulties, and memory limitations when users must click individual unlabeled elements.

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This triangulation of evidence reveals that complexity represents a critical failure point where user experience, expert evaluation, and behavioral patterns align. The contrast between these complex visualizations and the successful simple visualizations reinforces the conclusion that comprehensibility constitutes a primary design constraint, outweighing the potential analytical value that users cannot readily access.

6.1.3 To What Extent Can Well-Being Metrics, Such as Productivity and Motivation, Be Improved Through Persuasive Design Strategies in a Self-Tracking App?

The interviews suggest (see Findings 5.4.3) that using the app does not lead to an overall increase in smartphone usage but rather to a qualitative shift in how users spend their time. Several participants described using the app in moments that would otherwise be filled with passive activities, such as scrolling through social media or reading the news. Instead of engaging in habitual or distracting content consumption, users reported spending these short periods reflecting on their day, recalling meaningful activities, or documenting positive experiences. This reflective use appears to replace time previously associated with unproductive smartphone behavior, transforming idle moments into opportunities for self-awareness and intentional action.

Participants described this process as brief yet focused, typically lasting only a few minutes per day. For many, it became an integrated part of their daily routine, particularly in the evening or during commutes. Even those who mentioned a slight increase in overall phone usage emphasized that the additional time felt purposeful and beneficial. Thus, the app not only supports self-tracking but also subtly redirects attention from external stimulation toward internal reflection, fostering mindful engagement and reinforcing positive habits.

Self-Awareness as the Primary Mechanism for Well-Being Improvement

While self-awareness may appear to be a consequence of tracking rather than an intentional persuasive mechanism, self-monitoring constitutes an established persuasive design strategy within persuasive system design frameworks (see Section 2.6) [OH09; Fog03]. The design choices regarding which behavioral data to surface, how to visualize patterns, and when to present insights represent persuasive interventions that facilitate self-discovery rather than external prescription. Interview data revealed that these self-monitoring features functioned as a primary mechanism through which the app influenced well-being metrics, with users describing multiple distinct pathways from awareness to behavior change.

Multiple Pathways from Awareness to Well-Being Users demonstrated diverse routes through which awareness translated into improvements in well-being, each reflecting different aspects of the Fogg Behavior Model's motivation and ability components. P10's experience illustrates awareness, reducing perceived difficulty by correcting distorted

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perceptions about their alcohol consumption frequency. When feeling they had been drinking more than their internal goal, checking the app revealed their consumption was actually within acceptable limits, providing reassurance and a sense of control. This pathway represents a direct improvement in well-being through reduced anxiety, independent of any behavior change.

In contrast, P7's experience demonstrates awareness increasing motivation through pattern revelation. After distinguishing between coffee and energy drinks in their tracking, they discovered that energy drinks appeared every single day during university periods. Despite already knowing energy drinks were unhealthy, seeing the concentrated visual pattern provided concrete evidence that motivated them to take a break from consumption for several weeks. This shows the gap between knowing something in theory and actually feeling it: the visualization made the information feel real and motivated users.

P2 described a more deliberative pathway in which awareness enabled conscious decision-making about which activities to increase or decrease, based on the overview provided by the app. P1 conceptualized awareness as foundational rather than instrumental, positioning it as a decisive first step in behavior change that must occur before other mechanisms can take effect. Both perspectives support the Fogg Behavior Model's prediction that behavior change requires sufficient levels of both motivation and ability, with awareness simultaneously enhancing both components.

Awareness as Dual Influence on Motivation and Ability These diverse pathways extend the Fogg Behavior Model's framework in an important direction. While Fogg positions ability and motivation as independent variables that must simultaneously exceed thresholds, the awareness pathways demonstrate that self-monitoring through data visualization can simultaneously influence both components through different mechanisms.

Awareness increases ability by making problems feel more manageable, by providing concrete targets for intervention, and by externalizing cognitive work that would otherwise require mental reconstruction of behavior patterns. Simultaneously, awareness increased motivation through shock value when seeing concentrated patterns, through satisfaction of curiosity about personal behavior, and through cognitive dissonance when behavior patterns conflicted with self-image or goals.

Critically, these motivational and ability effects emerged from users' perception of neutral information provision rather than from persuasive tactics. Users achieved behavior change by seeing their own data honestly represented, not through app-generated suggestions or warnings. This suggests an essential distinction within persuasive design between coercive persuasion (external pressure) and facilitated self-persuasion (creating conditions for users to persuade themselves).

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6.2 Ethical Considerations, User Trust, and Life Improvement

The primary objective of the application is to improve the user's life by providing data in a format that yields actionable insights. By visualizing and organizing personal data, the app enables users to understand their behavior better and make informed decisions about change. James Baldwin's observation well captures this idea:

> *Not everything that is faced can be changed, but nothing can be changed until it is faced.*[Rat17]

Similarly, a quote often attributed to Peter Drucker underlines the same principle:

> *What gets measured gets managed.*

In this sense, the app's value lies in helping users face and measure aspects of their lives to manage and improve them.

However, the main challenge is not in confronting insights once they are visible - most individuals are naturally interested in their own progress - but in the consistent creation of entries. The act of journaling, which underlies these insights, requires effort and discipline. This is where persuasive technology plays a crucial role: by making the process of returning to the app enjoyable and rewarding, the system lowers the barrier to consistent data entry. In this way, the difficult part - maintaining regular input - is made easier through design that encourages engagement.

The ethical justification for using persuasive design was also emphasized by one of the experts interviewed (Exp2), who argued that even addictive aspects of an app can be legitimate if they serve a meaningful purpose:

> *I think these addictive aspects are important for you to continue to use the app because if you continue to use the app, it means you continue to track your habits, and if you continue to track your habits, it means you have to do these habits; otherwise, you have nothing to track. It helps you to achieve your goals, and only in this connection does it make sense. Otherwise, we would just be developing an addictive app, in which case a game might do a better job.*

In other words, the ethical boundary lies in the *incentive structure*. As Johann Hari discusses in *Stolen Focus: Why You Can't Pay Attention* [Har22], the persuasive mechanisms of large technology companies are primarily optimized for profit rather than user well-being. Their business models depend on maximizing the amount of time users spend on their platforms, since attention directly translates into advertising revenue. Hari illustrates this dynamic with the example of Google:

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> *The longer you make people look at their phones, the more advertising they see - and therefore the more money Google gets. [...] the incentives seemed to lead only one way: you should always design products that "engage" the maximum number of people, because engagement equals more dollars, and disengagement equals fewer dollars.*

In contrast, *Habits Network* currently has no paying customers, and this is intentional for two reasons. First, the data users enter is highly sensitive. Users must be able to trust that their information is protected and remains entirely under their control. Data privacy and user autonomy are therefore core requirements of the system, akin to the confidentiality expected in a personal diary.

Second, introducing a paywall at this stage could have hindered user adoption. To gain sufficient participation during the development and evaluation period, accessibility was prioritized over monetization. While it is possible that offering the app for free may cause some users to undervalue it or assume that they themselves are the product, the opposite risk - of excluding potential early users - was considered more significant at this stage. As Shoshana Zuboff describes in *The Age of Surveillance Capitalism* [Zub18],

> *We are the sources of surveillance capitalism's crucial surplus: the objects of a technologically advanced and increasingly inescapable raw-material-extraction operation.*

Habits Network explicitly rejects this model. Its design is grounded in user empowerment rather than data extraction, aiming to create a system that motivates reflection and growth while preserving privacy and trust.

6.3 Future Work

LLM-Powered Conversational Life Assistant for Pattern Discovery and Proactive Guidance An LLM-based conversational assistant could address multiple limitations identified in this thesis by serving as an intelligent intermediary between users and their complex life data. Rather than requiring users to navigate visualizations and mentally synthesize patterns, the assistant could proactively surface insights through natural language (e.g., "You tend to exercise more on days when you see Sarah - consider scheduling workouts together"), answer spontaneous questions conversationally (addressing P7's use case: "When was the last time I saw that person?"), and reduce cognitive load by translating complex metrics like cosine similarity into actionable recommendations. During low-activity periods, the assistant could maintain engagement by offering reflective prompts ("You haven't journaled in a week - would you like to reflect on why?"), suggesting social reconnections based on interaction patterns ("It's been 3 weeks since you saw Alex, who you usually meet monthly"), or generating personalized weekly summaries that highlight meaningful trends without requiring active data exploration.

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This approach transforms the app from a passive tracking tool into an active life coach that understands temporal context, social dynamics, and individual behavioral patterns, potentially bridging the gap between the high-level insights experts recommend and the simple interfaces users prefer.

Temporal Abstraction Metrics for Life Phase Detection and Social Network Evolution Future research could develop higher-level metrics that reveal long-term life changes beyond raw activity counts, particularly focusing on social network evolution and activity-relationship patterns. Potential metrics include social circle fragmentation (detecting when tight friend groups diverge over time), activity-person coupling strength (identifying which relationships are tied to specific activities and how these bonds shift), life phase transitions (algorithmically detecting periods of major change based on simultaneous shifts in both social interactions and activity patterns), and temporal relationship entropy (measuring whether someone's social life is becoming more concentrated around few people or more distributed). These abstraction layers would address expert feedback about the need for reference points and context by automatically highlighting meaningful changes (e.g., "Your social interactions have become 40% more concentrated around five people compared to last year" or "Board games used to involve 12 different people; now it's always the same 3") rather than requiring users to mentally synthesize patterns from heatmaps and line charts, making long-term tracking valuable for understanding life trajectory rather than just day-to-day habits.

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APPENDIX A

Appendix

A.1 Expert Interview Guides

Due to the varying expertise of the interviewed experts, specialized interview guides were developed for each expert. The guides were adapted to focus on each expert's domain knowledge while maintaining alignment with the research questions.

A.1.1 Common Sections Across All Interviews

The following sections are identical across all expert interview guides and are therefore presented here only once to avoid repetition.

Introduction & Background

- Can you tell me a bit about your background?
  - How long have you been active in the field?
  - What is your specialization?
  - Are you focused on any particular technology or platform?
- What kind of psychological or domain-specific education or training have you had?

Closing

- Do you have any other observations or suggestions?
- Are there any other questions I should be asking but haven't?

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A.1.2 Expert Interview Guide 1 (Exp1) - Data Visualization, Sports Technologies, Well-being Technologies, User Perception Studies, UI/UX Design Expert

Main Part: Visualizations & Data Understanding

A. Network Graph Visualization

- Looking at this network graph showing people and activities, what are your initial thoughts?
  - How do you think users will understand this?
- [Show bubble chart] What do you think of this visualization?
  - Follow-up: Do you think users will find this useful?
- What about comparison visualizations between users and activities?
  - What impact do you expect these comparison visualizations to have on users?

C. Derived Metrics vs Raw Data

- [Show cosine similarity] What do you think of this similarity measure?
  - Follow-up: Will users understand what this means?
  - Follow-up: How would you explain this to users?
- Would temporal visualizations (different time frames, rolling averages) be beneficial?

D. Behavioral Impact

- What effect do you think this app will have on users over time?
  - Behaviorally?
  - In terms of what visualizations they actually use?
- What happens when users achieve a habit and stop tracking it?
  - How can we keep them engaged?

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Input Methods & Interaction

- What do you think about the text-based input method?
  - Follow-up: Could it be made easier or more frictionless?
  - Follow-up: Would buttons or alternative input methods be better?
- What about global tags or leaderboards with strangers?
  - Follow-up: How does this affect the privacy perception of the app?

Notifications & Rewards

- What do you think about push notifications for this type of app?
- [Show variable reward on home page] What do you think about this reward system?
  - Follow-up: What would make rewards more effective?

A.1.3 Expert Interview Guide 2 (Exp2) - Human-Computer Interaction, Attention Management, Productivity Systems, Digital Self-control Tools, Intelligent User Interfaces Expert

Main Part: Motivation & Habit Formation

A. Intrinsic vs. Extrinsic Motivation

- How do you get people from extrinsic to intrinsic motivation?
- What role does visualization play in motivation?

C. App Features & Complexity

- [After watching video] What do you think of the app in general?
- What do you think about the social connection feature?

D. Visualization & Understanding

- What's the most important feature that should be visualized?
- What do you think about the variable rewards shown on the home page?
  - Follow-up: Are they prominent enough?
  - Follow-up: Should rewards be bigger and more celebratory?

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Input Methods & Tracking

- What do you think of the free-text input?
  - Follow-up: How do you think LLMs could help here?
- What do you think about binary tracking (only yes/no)?
  - Follow-up: How does binary thinking affect users psychologically?

Rewards & Notifications

- What do you think about notifications?

A.1.4 Expert Interview Guide 3 (Exp3) - Educational Technology, Usability Evaluation, Cognitive Science, Cognitive Biases, Cognitive Flow Theory, Digital Game-based Learning Expert

Hauptteil: Visualisierungen & Cognitive Load

A. Netzwerk-Graph

- [Zeige Netzwerk-Graph] Was ist dein erster Eindruck?
  - Follow-up: Verstehen die User das? Ist es zu komplex?

B. Bubble Chart & Komplexität

- [Zeige Bubble Chart] Was hältst du von dieser Visualisierung?

C. Cosine Similarity

- [Zeige Cosine Similarity] Verstehen die User das?
  - Follow-up: Ist das zu mathematisch/technisch?
- Wie sollte man Cosine Similarity erklären?

D. Alternative Visualisierungen

- Was hältst du von Line Charts mit Rolling Averages?
- Was hältst du von der *GitHub*-Heatmap?

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E. Ego-Netzwerk Visualisierung

- [Zeige Ego-Netzwerk] Was hältst du von dem Ego-Netzwerk?
- Wie könnte man die Cognitive Load reduzieren?

Input & Perceived Behavioral Control

- Wie schwer ist es für User, einen Eintrag zu erstellen?
  - Follow-up: Könnte man es noch reibungsloser machen?

A.1.5 Expert Interview Guide 4 (Exp4) - Visual Analytics, Visualization, Similarity Search, 3D Object Retrieval, Image Processing, Archaeological Applications of Visualization Expert

Hauptteil: Netzwerk-Graph-Evaluation

A. Erste Eindrücke & Struktur

- [Zeige Netzwerk-Graph] Was ist dein erster Eindruck?
- Was ist das Problem mit diesem Graph?

Alternative Visualisierungen

G. Adjacency Matrix / Heatmap

- Wäre eine Adjacency Matrix besser?
- Ist eine Heatmap für normale User zugänglicher?

H. Similarity Measures

- [Zeige Cosine Similarity] Was hältst du von diesem Ansatz?

A.2 User Interview Guide

This guide was used to conduct the semi-structured expert interviews with users of the journaling app *Habits Network*. The questions are organized according to the research questions (RQ1-RQ3) of this study. Follow-up questions were used flexibly to encourage elaboration and clarification.

RQ1: Nutzungsmotive und Interaktion mit der App

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- Warum benutzt du die App?
- Wie schwer fällt dir der Input für einen neuen Eintrag?
- Hat dich die Komplexität der Eingabe davon abgehalten, einen Eintrag zu machen?
- Was motiviert dich, einen neuen Eintrag zu machen?
- Zu welchem Zeitpunkt machst du häufig einen Eintrag?
- Gibt es Auslöser, die dich dazu bringen, einen Eintrag zu machen?
  - Gibt es Unterschiede zwischen Einträgen mit *Tags* und *Personen*?
- Wie findest du die zufällige Statistik über den Einträgen?
- Wie bist du in der Vergangenheit damit umgegangen, wenn du ein paar Tage nichts eingetragen hast?
- Warum benutzt du manche Features der App nicht? Hast du sie bereits gesehen?

RQ2: Wahrnehmung und Nutzung der Visualisierungen

- Welche Charts siehst du dir gerne an?
- Welche Charts schaust du dir nie an?
- Welche Charts verstehst du nicht?
  - Verstehst du den *Bubble Chart*?
- Warum schaust du dir die Charts an? Was willst du herausfinden?
- Wann schaust du dir die Charts an? Gibt es Auslöser oder bestimmte Situationen?
- Zeigst du die Charts anderen Personen?

RQ3: Wahrgenommene Wirkung der App-Nutzung

- Wie wirkt sich die Nutzung der App auf dein Leben aus?
- Wie nutzt du die App?
  - Mehr, um positive Dinge häufiger zu machen oder um negative Dinge zu reduzieren?
  - Welche Features nutzt du besonders?

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- Hast du Unterschiede bemerkt, seit du die App nutzt?
- Hast du, bevor du die App genutzt hast, weniger Zeit am Handy verbracht oder ersetzt *Habits Network* nun eine andere App?
- Wie denkst du, wie sich deine Nutzung auf andere Menschen auswirkt?
- Hast du dich durch die App verbessert?
  - Hat sich etwas verschlechtert?
- Konntest du bereits einen Tag zu einem Habit machen, den du jetzt nicht mehr einträgst?
- Was motiviert dich dann weiterzumachen?

At the end, I asked the users if there was something I should have asked but did not. I also asked them about their answers in the SUS questionnaire, if any stood out.

A.3 System Usability Score Questionnaire

The complete System Usability Scale (SUS) questionnaire is provided on the following page.

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Fragebogen zur System-Gebrauchstauglichkeit
(System Usability Scale)

1. Ich denke, dass ich das System gerne häufig benutzen würde.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
2. Ich finde, dass das System unnötig komplex war.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
3. Ich finde, dass das System leicht zu benutzen war.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
4. Ich denke, dass ich die Unterstützung einer fachkundigen Person benötigen würde, um das System benutzen zu können.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
5. Ich finde, dass die verschiedenen Funktionen des Systems gut integriert waren.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
6. Ich denke, dass das System sehr inkonsistent ist.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
7. Ich glaube, dass die meisten Menschen den Umgang mit diesem System sehr schnell lernen würden.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
8. Ich finde, dass das System sehr umständlich zu benutzen ist.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
9. Ich fühlte mich bei der Benutzung des Systems sehr sicher.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5
10. Ich musste eine Menge lernen, bevor ich mit dem System arbeiten konnte.	Stimme überhaupt nicht zu				Stimme voll zu
	1	2	3	4	5

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A.4 User Interview Consent Form

The complete consent form used for user interviews is provided on the following pages.

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Zustimmungserklärung
Informed Consent Form

für das Diplomarbeitsprojekt:
for the Master Thesis Project:

Increasing User Satisfaction, Engagement and Retention in Life-Tracking Apps for Students: A Case Study of Methods on the App Habits.Network

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1. Zweck der Studie / Purpose of This Study

Als Teil meines Diplomarbeitsprojekts *'Increasing User Satisfaction, Engagement and Retention in Life-Tracking Apps for Students: A Case Study of Methods on the App Habits.Network'* werden Userinterviews durchgeführt. Die gesammelten Daten werden später verwendet, um die Effektivität der App in Bezug auf Nutzungsmetriken zu bewerten, die Forschungsfragen der Arbeit zu beantworten und die App zu verbessern.

Die Userinterviews zielen darauf ab, persuasive Designstrategien zu identifizieren, die geeignet sind, das Engagement, die Zufriedenheit und die Bindung der Nutzer in Self-Tracking-Apps zu erhöhen, mit besonderem Fokus auf die App "habits.network". Des Weiteren ist es das Ziel, Mängel oder Erfolge bei der Implementierung von persuasiven Designstrategien zu identifizieren, die durch die Literaturrecherche ausgewählt wurden. Die Bewertung der Visualisierungen von Beziehungen zwischen den Aktivitäten eines Nutzers und sozialen Interaktionen ist ebenfalls ein Ziel der Userinterviews.

As part of my master thesis project *'Increasing User Satisfaction, Engagement and Retention in Life-Tracking Apps for Students: A Case Study of Methods on the App Habits.Network'*, user interviews are being conducted. The data collected will later be used to evaluate the effectiveness of the app with respect to usage metrics, to answer the research questions of the thesis, and to improve the app.

The user interviews aim to identify persuasive design strategies suitable for increasing user engagement, satisfaction, and retention in self-tracking apps, with a special focus on the app "habits.network". Furthermore, the goal is to identify shortcomings or successes in the implementation of persuasive design strategies chosen through the literature review. Evaluating the visualizations of relationships between a user's activities and social interactions is also a goal of the user interviews.

2. Ort und Dauer der Studie / Where and How Long Does It Last?

Die Interviews werden primär persönlich durchgeführt, können aber auch online über Zoom stattfinden. Jedes Interview wird voraussichtlich etwa 30 Minuten dauern. Zeit und

Datum der Interviews werden individuell mit den teilnehmenden Personen vereinbart.

The interviews are primarily conducted in person but can also be held online via Zoom. Each interview is expected to last approximately 30 minutes. The time and date of the interviews will be individually arranged with participating individuals.

3. Erhobene Daten und deren Verarbeitung und Speicherung / What Data Is Collected and How Is It Processed and Stored?

Generell ist zu beachten, dass personenbezogene Daten stets anonymisiert werden, sofern nicht anders vereinbart. Die Teilnehmer sind nicht verpflichtet, persönliche oder sensible Informationen gegen ihren Willen preiszugeben.

Das Audio der persönlichen Interviews wird aufgezeichnet; bei Online-Interviews wird auch das Video mit der integrierten Aufnahmefunktion von Zoom zu Auswertungszwecken aufgezeichnet (einschließlich Transkription durch Drittanbieter-Software). Während der Interviews werden zusätzliche individuelle Notizen gemacht.

Das aufgezeichnete Video wird nicht veröffentlicht. Wichtige Teile der Interviews werden transkribiert und in anonymisierter Form in die schriftliche Arbeit aufgenommen (die Namen der Interviewteilnehmer werden anonymisiert).

Die aufgezeichneten Interviews und die daraus resultierenden Transkriptionen werden ausschließlich auf den persönlichen Rechnern des Forschenden gespeichert. Es wird sichergestellt, dass keinem Dritten Zugang zu diesen Daten gewährt wird. Die Daten werden unter strenger Einhaltung der Datenschutzbestimmungen verarbeitet und durch geeignete technische und organisatorische Maßnahmen vor unbefugtem Zugriff, Verlust oder Missbrauch geschützt.

In general, please note that personal data will always be anonymized unless otherwise agreed upon. Participants are not required to share any personal or sensitive information against their will.

The audio of in-person interviews will be recorded; for online interviews, the video will also be recorded using Zoom's built-in recording function for evaluation purposes (including transcription using third-party software). Additional individual notes will be taken during the interviews.

The recorded video will not be published. Key parts of the interviews will be transcribed and included in the written thesis in anonymized form (the names of interview participants will be anonymized).

The recorded interviews and their resulting transcriptions will be stored exclusively on the researcher's personal machines. It will be ensured that no third party is granted access to this data. The data will be processed in strict compliance with data protection regulations and protected against unauthorized access, loss, or misuse through appropriate technical and organizational measures.

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4. Ihre Rechte / Your Rights

Die Teilnahme ist freiwillig. Sie können die Studie jederzeit ohne Angabe von Gründen verlassen. Wenn Sie Ihre Teilnahme zurückziehen, werden alle gesammelten Daten ausnahmslos gelöscht.

Auf Anfrage können Sie Ihre Interviewaufzeichnungen oder Transkriptionen einsehen.

Sollten Sie Bedenken bezüglich der Nutzung von Transkriptionssoftware von Drittanbietern haben, teilen Sie mir dies bitte im Voraus mit. In solchen Fällen wird eine alternative Lösung gesucht.

Participation is voluntary. You may withdraw from the study at any time without stating a reason. If you withdraw, all collected data will be deleted without exception.

Upon request, you may review your interview recordings or transcriptions.

If you have any concerns regarding the use of third-party transcription software, please let me know in advance. An alternative solution will be sought in such cases.

5. Zustimmungserklärung / Declaration of Consent

Bitte kreuzen Sie die entsprechenden Kästchen an:
Please tick the appropriate boxes:

- ☐ Ich habe die Informationen zu diesem Projekt gelesen und verstanden.
  I have read and understood the information provided for this project.
- ☐ Ich habe die Möglichkeit gehabt, Fragen zu stellen, und diese wurden zu meiner Zufriedenheit beantwortet.
  I have had the opportunity to ask questions, and these have been answered to my satisfaction.
- ☐ Ich stimme der Audio- und/oder Videoaufzeichnung meines Interviews zu.
  I consent to the audio and/or video recording of my interview.
- ☐ Ich stimme der Transkription meines Interviews, auch durch Drittanbieter-Software, zu.
  I consent to the transcription of my interview, including by third-party software.
- ☐ Ich bin damit einverstanden, dass Zitate aus meinem Interview in der Masterarbeit und zugehörigen akademischen Publikationen in anonymisierter Form verwendet werden.
  I agree that quotes from my interview may be used in the master thesis and associated academic publications in anonymized form.

Ich bestätige, dass ich freiwillig an dieser Studie teilnehme und jederzeit ohne Angabe von Gründen meine Teilnahme zurückziehen kann. Ich verstehe, dass meine Daten in diesem Fall vollständig gelöscht werden.

I confirm that I participate in this study voluntarily and can withdraw my participation at any time without stating a reason. I understand that in this case, all my data will be completely deleted.

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6. Kontakt / Contact

Sollten Sie Fragen oder Bedenken zu diesem Projekt haben, können Sie mich jederzeit über die folgenden Informationen kontaktieren:

If you have questions or concerns regarding this project, you can contact me at any time via the following information:

Cederic Demoulin, BSc
E-Mail: cederic.demoulin@student.tuwien.ac.at

Sollten Sie weitere Fragen oder Bedenken bezüglich der Verarbeitung Ihrer Daten haben, wenden Sie sich bitte an die akademische Betreuerin an der TU Wien:

If you have further questions or concerns regarding the processing of your data, please contact the academic supervisor at TU Wien:

Dr. Astrid Weiss
TU Wien, Forschungsbereich Human-Computer Interaction
Argentinierstrasse 8, 2. Stock, E193/5, 1040 Wien
Tel: +43-1-58801-187-35
E-Mail: astrid.weiss@tuwien.ac.at

Allgemeine Informationen zum Datenschutz finden Sie bei der österreichischen Datenschutzbehörde unter: https://www.dsb.gv.at/

Interviewte Person / Interviewee	**Interviewer / Interviewer**
_____	_____
Name (Druckbuchstaben) / Name (Printed)	Name (Druckbuchstaben) / Name (Printed)
_____	_____
Unterschrift / Signature	Unterschrift / Signature
_____	_____
Datum / Date	Datum / Date

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Overview of Generative AI Tools Used

The following AI tools were used during the creation of this thesis:

- **Claude Code** was used to generate data visualizations, reformulate text passages for improved clarity, spell-check, and search for passages in interview transcripts related to specific themes.
- **Gemini CLI** was used for finding the missing DOIs, URLs, and ISBNs in the bibliography.
- **GitHub Copilot** was used to assist in writing LaTeX code for the thesis document.
- **Grammarly** was used for spell checking and improving the clarity of various sections of the thesis.
- **ChatGPT** was used to support literature research for the related work chapter.
- **Consensus** was used for discovering and analyzing academic papers.
- **PostHog AI** was used for querying and analyzing user behavior data.

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